
AP EAMCET-2018-Engineering-24 Apr-Shift-1
Mathematics

1. If $f(x)$ is a polynomial function satisfying

$$f(x) \cdot \frac{1}{f(x)} = f(x) + f\left(\frac{1}{x}\right) \text{ and } f(4) = 257, \text{ then } f(3) =$$

- (1) 28
- (2) 65
- (3) 82
- (4) 244

2. The set of all real values of x for which the real valued function

$$f(x) = \left(1 + \frac{1}{x}\right)^x \text{ is defined, is}$$

- (1) $(0, \infty)$
- (2) $\mathbb{R} - \{0\}$
- (3) $(-\infty, -1) \cup (0, \infty)$
- (4) $\mathbb{R} - \{0, -1\}$

3. Let $S_n = \sum_{k=1}^n (-1)^{k-1} \cdot k^2$ for $n \geq 1$. Given that $S_{2n} = -n(2n+1)$ for

$n = 1, 2, 3, \dots$ then $S_{77} =$

(1) -3003

(2) 3003

(3) -2926

(4) 2926

4.
$$\begin{vmatrix} 125 & 5 & 25 \\ 343 & 7 & 49 \\ 729 & 9 & 81 \end{vmatrix} =$$

(1) $9!$

(2) $6!$

(3) $8!$

(4) $7!$

5. Let matrix $A = \begin{bmatrix} 5 & -3 & 0 \\ -3 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, X be a non-zero matrix of order

3×1 and c be a real number. If $A^2 X = cAX$, then the number of distinct value of c is

(1) 3

(2) 2

(3) 1

(4) 0

6. A set of values of θ for which the system of equations

$$(\sin 3\theta)x - y + z = 0$$

$$(\cos 2\theta)x + 4y + 3z = 0$$

$$2x + 7y + 7z = 0$$

has non-trivial solutions, is

(1) $(n-1)\frac{\pi}{2} + (-1)^n \frac{\pi}{4}$
(here n is an integer)

(2) $(n-1)\frac{\pi}{2} + (-1)^n \frac{\pi}{3}$
(here n is an integer)

(3) $\frac{n\pi}{2} + (-1)^n \frac{\pi}{6}$
(here n is an integer)

(4) $n\pi + (-1)^n \frac{\pi}{6}$
(here n is an integer)

7. The maximum value of modulus of e^{z^2} on the set $\{z \in \mathbb{C} / 0 \leq \operatorname{Re}(z) \leq 1, 0 \leq \operatorname{Im}(z) \leq 1, \}$ is

(1) $\frac{2}{e}$

(2) e

(3) $e+1$

(4) e^2

8. If $z \neq \pm 1$ is a complex number and $\operatorname{Arg}\left(\frac{Z-1}{Z+1}\right) = \frac{\pi}{4}$ Then the locus of Z in the Argand plane is

(1) $x^2 + y^2 - 2y - 1 = 0$

(2) $x^2 + y^2 + 2y - 1 = 0$

(3) $x^2 + y^2 - 2x + 1 = 0$

(4) $x^2 + y^2 + 2x + 1 = 0$

9. If $C_r = {}^nC_r$, then $C_0 + C_4 + C_8 + C_{12} + \dots =$

(1) $\frac{2^{\frac{n}{2}} \left[\sin \frac{n\pi}{4} + 2^{\frac{n}{2}-1} \right]}{2}$

$$(2) 2^{\frac{n}{2}} \sin \frac{n\pi}{4}$$

$$(3) 2^{n-1} \cos \frac{n\pi}{4}$$

$$(4) \frac{2^{\frac{n}{2}} \left[\cos \frac{n\pi}{4} + 2^{\frac{n}{2}-1} \right]}{2}$$

10. One of the complex root of the equation $x^{11} - x^6 - x^5 + 1 = 0$ is

$$(1) \operatorname{cis} \frac{3\pi}{5}$$

$$(2) \operatorname{cis} \frac{\pi}{3}$$

$$(3) \operatorname{cis} \frac{5\pi}{6}$$

$$(4) \operatorname{cis} \frac{7\pi}{5}$$

11. If the quadratic equation $4\sec^2 \alpha \cdot x^2 + 2x + \left(\beta^2 - \beta + \frac{1}{2} \right) = 0$ has real roots, then the value of $\cos^2 \alpha + \cos^{-1} \beta$ is

$$(1) \frac{\pi}{3}$$

(2) $\frac{\pi}{3} + 1$

(3) $\frac{\pi}{2}$

(4) $\frac{\pi}{2} - 1$

12. The sum of non-real roots of $(p^2 + p - 3)(p^2 + p - 2) - 12 = 0$ is

(1) 1

(2) -1

(3) 6

(4) -6

13. The solution set of the inequation $\sqrt{x^2 + 6x + 5} > (8 - x)$ is

(1) $(8, \infty)$

(2) $\left(\frac{59}{22}, 8\right]$

(3) $\left(\frac{59}{22}, \infty\right)$

(4) $(-1, \infty)$

14. If α and β are the irrational roots of the equation

$$x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0$$

then the roots of the equation $(\alpha + \beta)x^2 + 2\alpha\beta x - \alpha\beta = 0$ are

(1) $-1, \frac{1}{3}$

(2) $\frac{3 \pm \sqrt{5}}{2}$

(3) $\frac{1 \pm i\sqrt{3}}{2}$

(4) $1, -\frac{1}{3}$

15. Let m be a natural number such that $20000 < m < 60000$ and let k be the sum of all the digits in m . Then the number of numbers m for which k is even, is

(1) 19909

(2) 19989

(3) 18999

(4) 19999

16. There are three sections in a Question paper, each containing 4 Questions. If a candidate has to answer only 5 Questions from

this paper without leaving any section, then number of ways the candidate can make the choice of questions is

- (1) 624
- (2) 704
- (3) 384
- (4) 432

17. The coefficient of x^4 in the expansion of $(1 + x - x^2 - x^3)^{11}$ is

- (1) 990
- (2) 220
- (3) -220
- (4) -385

18. If $|x|$ is so small that x^2 and higher power of x may be neglected

then the approximate value of $\frac{\sqrt{4+x} + \sqrt[3]{8-x}}{\left(1 - \frac{2x}{3}\right)^{\frac{3}{2}}}$ when $x = \frac{6}{25}$ is

- (1) 6
- (2) 5
- (3) $\frac{2}{3}$

(4) $\frac{5}{6}$

19. The coefficient of x^4 in the power series expansion of

$\frac{x^2 - 1}{(x^2 + 1)(x^2 + 2)}$ is

(1) $\frac{15}{16}$

(2) $\frac{15}{4}$

(3) $-\frac{13}{8}$

(4) $\frac{77}{324}$

20. The smallest positive root of the equation $\tan x - x = 0$ lies in the interval

(1) $\left(0, \frac{\pi}{2}\right)$

(2) $\left(\frac{\pi}{2}, \pi\right)$

(3) $\left(\pi, \frac{3\pi}{2}\right)$

$$(4) \left(\frac{3\pi}{2}, 2\pi \right)$$

$$21. \left[\left(1 + \cos \frac{\pi}{8} \right) \left(1 + \cos \frac{2\pi}{8} \right) \left(1 + \cos \frac{3\pi}{8} \right) \left(1 + \cos \frac{4\pi}{8} \right) \right] =$$
$$\left[\left(1 + \cos \frac{5\pi}{8} \right) \left(1 + \cos \frac{6\pi}{8} \right) \left(1 + \cos \frac{7\pi}{8} \right) \right]$$

$$(1) \frac{1}{8}$$

$$(2) \frac{1}{16}$$

$$(3) \frac{1}{32}$$

$$(4) \frac{1}{64}$$

22. If A is in third quadrant and $\tan A = \frac{\sqrt{7}}{3}$, then

$$18 - 16 \sin^2 \frac{A}{2} - 32 \sin \frac{A}{2} \sin \frac{5A}{2} =$$

$$(1) -6$$

$$(2) 11$$

$$(3) 5$$

$$(4) 10$$

23. Let $P(\alpha, \beta)$ and $Q(\gamma, \delta)$ be two points that lie on the curve $\tan^2(x+y) + \cos^2(x+y) + y^2 + 2y = 0$ in the XY -plane. If the distance between P and Q is d , then $\cos d =$

(1) 0

(2) $(-1)^n, n \in \mathbb{N}$

(3) $\pm\pi$

(4) $\pm 2n\pi, n \in \mathbb{N}$

24. $\tan^{-1} 2 + \cot^{-1}(-3) + \cot^{-1} \frac{1}{3} + \tan^{-1} \left(-\frac{1}{2}\right) =$

(1) $-\frac{3\pi}{2}$

(2) $\frac{\pi}{2}$

(3) $-\frac{\pi}{2}$

(4) $\frac{3\pi}{2}$

25. If $x = -\frac{1}{2}$, $\sinh^{-1} x + \operatorname{cosech}^{-1} x =$

$$(1) \log_e \left(\frac{7-3\sqrt{5}}{2} \right)$$

$$(2) \log_e \left(\frac{3+\sqrt{5}}{2} \right)$$

$$(3) \log_e \left[\frac{(\sqrt{5}-1)(2+\sqrt{3})}{2} \right]$$

$$(4) \log_e \left[\frac{(\sqrt{5}+1)(2+\sqrt{3})}{2} \right]$$

26. In $\triangle ABC$, if the $\frac{s-a}{11} = \frac{s-b}{12} = \frac{s-c}{13}$ then $\tan^2 \left(\frac{A}{2} \right) + \tan^2 \left(\frac{C}{2} \right) =$

$$(1) \frac{290}{429}$$

$$(2) \frac{290}{143}$$

$$(3) \frac{143}{33}$$

$$(4) \frac{113}{33}$$

27. In a $\triangle ABC$ $(a-b)^2 \sin^2 \left(\frac{A+B}{2} \right) + (a+b)^2 \sin^2 \left(\frac{C}{2} \right) =$

- (1) b^2
- (2) a^2
- (3) c^2
- (4) $a^2 + b^2 - c^2$

28. In a triangle, if the ex-radii r_1, r_2, r_3 are in the ratio 1:2:3 then its sides are in the ratio

- (1) 5:8:9
- (2) 5:4:3
- (3) 7:9:11
- (4) 1:2:3

29. In $\triangle ABC$ if D and E are the mid points of the sides BC and CA respectively, then $2(\overline{AD} + \overline{EB})$

- (1) $3\overline{AB}$
- (2) $\frac{3}{2}\overline{AB}$
- (3) $2\overline{AB}$
- (4) $3\overline{BC}$

30. If $\vec{n} = 2\vec{i} - 3\vec{j} + 4\vec{k}$, $\vec{m} = \vec{i} - \vec{j}$, $\vec{l} = 2\vec{i} - \vec{j} + \vec{k}$, then the Cartesian equation of the plane passing through the line of intersection of two planes $\vec{r} \cdot \vec{n} = 1$ and $\vec{r} \cdot \vec{m} = -4$ and perpendicular to the plane $\vec{r} \cdot \vec{l} = -8$ is

(1) $5x - 20y - 12z - 44 = 0$

(2) $x - 2y - 12z - 45 = 0$

(3) $5x - 20y - 12z - 47 = 0$

(4) $5x - 2y - 12z + 47 = 0$

31. If $\vec{a}, \vec{b}, \vec{c}$ are vectors of equal magnitude such that $(\vec{a}, \vec{b}) = \alpha$, $(\vec{b}, \vec{c}) = \beta$, $(\vec{c}, \vec{a}) = \gamma$ then the minimum value of $\cos \alpha + \cos \beta + \cos \gamma$ is

(1) $\frac{3}{2}$

(2) $-\frac{3}{2}$

(3) $\frac{1}{2}$

(4) $-\frac{1}{2}$

32. In ΔABC if $A(\vec{\alpha})$, $B(\vec{\beta})$ and $C(\vec{\gamma})$ are the position vectors of the vertices then the length of the perpendicular from A to BC is

(1) $|\vec{\alpha} \times \vec{\beta}| + |\vec{\beta} \times \vec{\gamma}| + |\vec{\gamma} \times \vec{\alpha}|$

(2) $|\vec{\alpha} \times \vec{\beta} + \vec{\beta} \times \vec{\gamma} + \vec{\gamma} \times \vec{\alpha}|$

(3) $\frac{|\vec{\alpha} \times \vec{\beta} + \vec{\beta} \times \vec{\gamma} + \vec{\gamma} \times \vec{\alpha}|}{|\vec{\alpha} - \vec{\beta}|}$

(4) $\frac{|\vec{\alpha} \times \vec{\beta} + \vec{\beta} \times \vec{\gamma} + \vec{\gamma} \times \vec{\alpha}|}{|\vec{\gamma} - \vec{\beta}|}$

33. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors, then match the items of List-I with those of List-II.

List-I

List-II

(A) $[\vec{b} \times \vec{c} \vec{c} \times \vec{a} \vec{a} \times \vec{b}] =$

(I) $[\vec{a} \vec{b} \vec{c}]^2$

(B) $[\vec{a} \times \vec{b} \vec{a} \times \vec{c} \vec{b}] =$

(II) $2[\vec{a} \vec{b} \vec{c}]$

(C) $[\vec{a} + \vec{b} \vec{b} + \vec{c} \vec{c} + \vec{a}] =$

(III) $[\vec{a} \vec{b} \vec{c}]$

For three mutually perpendicular unit

(D) vectors $\vec{a}, \vec{b}, \vec{c}$:

(IV) $[\vec{a} \vec{b} \vec{c}] [\vec{a} \cdot \vec{b}]$

$[(\vec{a} + \vec{b} + \vec{c}) \vec{b} \times \vec{c} \vec{c} \times \vec{a}] =$

(V) 0

The correct answer is

A	B	C	D
(1) III	IV	V	II

	A	B	C	D
(2)	IV	V	II	III

	A	B	C	D
(3)	I	IV	V	III

	A	B	C	D
(4)	I	IV	II	III

34. $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three unit vectors such that no two of them are collinear. If $\vec{b} = 2\{\vec{a} \times (\vec{b} \times \vec{c})\}$ and α is the angle between $\vec{a}$, $\vec{c}$ and β is the angle between $\vec{a}$, $\vec{b}$ then $\cos(\alpha + \beta) =$

(1) $\frac{\sqrt{3}}{2}$

(2) $-\frac{\sqrt{3}}{2}$

(3) $\frac{1}{2}$

(4) $-\frac{1}{2}$

35. If the variance of 6, 7, 8, 9, 10, 11 is σ^2 , then the variance of 112, 114, 116, 118, 120, 122 is

(1) $2\sigma^2$

(2) $4\sigma^2$

(3) $100 + 2\sigma^2$

(4) $100 + 4\sigma^2$

36. Consider the frequency distribution

C.I.	75-175	175-275	275-375	375-475	475-575	575-675	675-775
f_i	3	2	1	0	1	2	3

If the variance of this distribution is 60000, then the coefficient of variation of the distribution is

(1) 60

(2) $\frac{400\sqrt{6}}{17}$

(3) $\frac{400\sqrt{6}}{9}$

(4) 595.75

37. If two number a and b are chosen from the set of integers 1 to 39, then the probability that those numbers satisfy the equation $7a - 9b = 0$ is

(1) $\frac{1}{742}$

(2) $\frac{4}{743}$

(3) $\frac{4}{741}$

(4) $\frac{5}{741}$

38. Two balls are drawn from an urn containing 7 white, 6 red and 8 black balls one after the other without replacement. Then the probability that atleast one of them is white is

(1) $\frac{4}{9}$

(2) $\frac{13}{30}$

(3) $\frac{11}{30}$

(4) $\frac{17}{30}$

39. In a manufacturing company, three machines A, B and C respectively produce 20%, 30% and 50% of the total product. The defective products from A, B and C are respectively 5%, 3% and 2%. If an article produced by the company is selected at

random and is found to be defective, then the probability that it is produced by machine B is

(1) $\frac{10}{29}$

(2) $\frac{8}{29}$

(3) $\frac{9}{29}$

(4) $\frac{11}{29}$

40. The variance of the random variable X having the following distribution

$$X = k \quad : \quad -2 \quad -1 \quad 0 \quad 1 \quad 2$$

$$P(X = k): \quad \frac{1}{6} \quad \frac{1}{6} \quad \frac{1}{3} \quad \frac{1}{6} \quad \frac{1}{6}$$

(1) $\frac{1}{3}$

(2) $\frac{2}{3}$

(3) $\frac{4}{3}$

(4) $\frac{5}{3}$

41. Give that the probability of a man hitting a target with a gun is $\frac{1}{3}$

. If the fires 8 time, then the probability of his hitting the target atleast twice is

(1) $5\left(\frac{2}{3}\right)^8$

(2) $1 - 5\left(\frac{2}{3}\right)^8$

(3) $\left(\frac{2}{3}\right)^8$

(4) $\left(\frac{3}{8}\right)^4$

42. A variable line 'L' passing through the origin cuts two parallel line $x - y + 10 = 0$ and $x - y + 20 = 0$ at two points A and B respectively. If P is a point on line 'L' such that OA, OP, OB are in harmonic progression then the locus of P is

(1) $3x + 3y + 40 = 0$

(2) $3x + 3y + 20 = 0$

(3) $3x - 3y + 40 = 0$

(4) $3x - 3y + 20 = 0$

43. When the coordinate axes are rotated through an angle of 45° about the origin in the positive direction, if the transformed equation of a curve is $17x^2 - 16xy + 17y^2 = 225$, then the original equation of that curve is

- (1) $25x^2 + 9y^2 = 225$
- (2) $9x^2 - 25y^2 = 225$
- (3) $25x^2 - 16xy + 9y^2 = 225$
- (4) $9x^2 + 25y^2 = 225$

44. The vertex A of a triangle lies on the lines $x + y = 1$ and $2x + 3y = 6$. If the orthocenter of the triangle is $O\left(\frac{3}{7}, \frac{22}{7}\right)$ then the equation of OA in the normal form is

- (1) $x \cos \alpha + y \sin \alpha = 7, \alpha = \tan^{-1} \frac{1}{7}$
- (2) $x \cos \alpha + y \sin \alpha = \frac{13}{\sqrt{17}}, \alpha = \tan^{-1} \left(\frac{1}{4} \right)$
- (3) $x \cos \alpha + y \sin \alpha = \frac{13}{4}, \alpha = \tan^{-1} \left(\frac{13}{\sqrt{17}} \right)$
- (4) $x \cos \alpha + y \sin \alpha = \frac{13}{\sqrt{17}}, \alpha = \tan^{-1} (4)$

45. The equation of the line passing through the point of intersection of the lines $2x + 3y + 6 = 0$, $3x - y - 13 = 0$ and parallel to the line $3x - 4y + 5 = 0$ is

(1) $3x - 4y + 75 = 0$

(2) $3x - 4y + 15 = 0$

(3) $3x - 4y + 25 = 0$

(4) $3x - 4y - 25 = 0$

46. If O, G, S are respectively the orthocenter, centroid and circumcentre of a triangle whose vertices are $A(2, 3)$, $B(2, 4)$, $C(4, 3)$, then $AO^2 + 9BG^2 + 4CS^2 =$

(1) $\frac{77}{36}$

(2) 13

(3) $\frac{8}{9}$

(4) $\frac{5}{4}$

47. If two sides of a triangle are given by $3x^2 - 5xy + 2y^2 = 0$ and its orthocenter $(2, 1)$ then the equation of the third side of the triangle is

(1) $5x - 10y + 1 = 0$

(2) $10x + 5y - 1 = 0$

(3) $5x - 10y = 21$

(4) $10x + 5y = 21$

48. The equation of the pair of lines joining the origin to the points of intersection of two circles $x^2 + y^2 - 4x + 8y + 5 = 0$ and $x^2 + y^2 + 2x + 4y - 3 = 0$ is

(1) $13x^2 + 6xy - 28y^2 = 0$

(2) $xy - 28y^2 = 0$

(3) $(x + 4)(x - 5) = 0$

(4) $13x^2 + 68xy - 28y^2 = 0$

49. If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ ($c > 0$) touches both the coordinate axes and lies in the third quadrant, then the length of the chord intercepted by the circle on the line $x + y + \sqrt{c} = 0$ is

(1) $\sqrt{2c}$

(2) c

(3) $\sqrt{c}$

(4) $\sqrt{\frac{c}{2}}$

50. If the shortest distance from $(2, -14)$ to the circles $x^2 + y^2 + 6x + 4y - 12 = 0$ is d and the length of the tangent drawn from the same point to the circle is l , then $\sqrt{d+l} =$

- (1) 13
- (2) $2\sqrt{5}$
- (3) 12
- (4) 5

51. The number of common tangents to the circles $x^2 + y^2 - 4x - 2y + k = 0$ and $x^2 + y^2 - 6x - 4y + l = 0$ having radii 2 and 3 respectively, is

- (1) 4
- (2) 2
- (3) 3
- (4) 1

52. The center of the circle which intersects the circle $x^2 + y^2 - 2x - 2y - 2 = 0$ orthogonally and passes through the point $(2, 0)$ and touches the X-axis is

- (1) $(4, 1)$

(2) $(-1, 2)$

(3) $(1, 4)$

(4) $(2, -1)$

53. If from any point on the circle $x^2 + y^2 + 2gx + 2fy + c = 0$, tangents are drawn to the circle

$$x^2 + y^2 + 2gx + 2fy + c \sin^2 \alpha + (g^2 + f^2) \cos^2 \alpha = 0, \left(0 < \alpha < \frac{\pi}{2} \right)$$

then the angle between those tangents is

(1) $\frac{\pi}{4}$

(2) $\frac{\pi}{3}$

(3) 2α

(4) α

54. If the double ordinate of the parabola $y^2 = 8x$ is of the length 16, then the angle subtended by it at the vertex of the parabola is

(1) $\frac{\pi}{2}$

(2) $\frac{\pi}{3}$

(3) $\frac{3\pi}{4}$

(4) $\frac{\pi}{4}$

55. If P and the origin are the points of intersection of the parabolas $y^2 = 32x$ and $2x^2 = 27y$, and if θ is the acute angle between these curves at P then $5\sqrt{\tan \theta} =$

(1) 2

(2) $2\sqrt{3}$

(3) $3\sqrt{2}$

(4) 3

56. If the latus rectum of an ellipse subtends a right angle at the center of that ellipse, then the eccentricity of that ellipse is

(1) $\frac{\sqrt{5}+1}{4}$

(2) $\frac{\sqrt{5}-1}{2}$

(3) $\frac{\sqrt{10-2\sqrt{5}}}{5}$

(4) $\frac{\sqrt{10+2\sqrt{5}}}{5}$

57. If the tangent at the point $(1, 2)$ on the ellipse $3x^2 + 4y^2 = 19$ is also a tangent to the parabola $y^2 - kx = 0$, then $k =$

(1) $\frac{57}{16}$

(2) $-\frac{57}{64}$

(3) $\frac{57}{64}$

(4) $-\frac{57}{16}$

58. If the equation of one asymptote of the hyperbola $14x^2 + 38xy + 20y^2 + x - 7y - 91 = 0$ is $7x + 5y - 3 = 0$, then the other asymptote is

(1) $2x - 4y + 1 = 0$

(2) $2x + 4y + 1 = 0$

(3) $2x - 4y - 1 = 0$

(4) $2x + 4y - 1 = 0$

59. If the midpoints of the side AB, BC, CA of a triangle are $(1, 5, -1)$, $(0, 4, -2)$, $(2, 3, 4)$ respectively, then the length of the median drawn from C to AB is

(1) $\sqrt{29}$

(2) $\sqrt{27}$

(3) $\sqrt{5}$

(4) 5

60. If a line makes angles $\frac{\pi}{4}$ and $\frac{\pi}{3}$ with Y-axis and Z-axis respectively, then the obtuse angle made by that line with X-axis is

(1) $\frac{\pi}{3}$

(2) $\frac{2\pi}{3}$

(3) $\frac{\pi}{6}$

(4) $\frac{5\pi}{6}$

61. The equation of the plane bisecting the line segment joining the points $(2, 0, 6)$ and $(-6, 2, 4)$ and perpendicular to it, is

(1) $2x - y + 4z - 15 = 0$

(2) $4x - y + 3z - 6 = 0$

(3) $4x - y + z + 4 = 0$

(4) $x - 2y + 3z - 11 = 0$

62. $\lim_{x \rightarrow 0} \frac{\sqrt{1+x \sin x} - \sqrt{\cos x}}{\tan^2 2x} =$

(1) 3

(2) $\frac{3}{2}$

(3) $\frac{3}{4}$

(4) $\frac{3}{16}$

63. If the function f defined by $f(x) = \begin{cases} \cos x & \text{if } x \leq 0 \\ 3x + \alpha & \text{if } 0 < x < 2 \\ \beta x + 3 & \text{if } 2 \leq x \leq 4 \\ 11 & \text{if } x > 4 \end{cases}$

Where α and β are real constants, is continuous on $\mathbb{R}$, then

$$\alpha^2 + \beta^2 =$$

- (1) 3
- (2) 9
- (3) 5
- (4) 1

64. If $f : [0, 3] \rightarrow [0, 3]$ is defined by $f(x) = \begin{cases} 1+x & 0 \leq x \leq 2 \\ 3-x & 2 < x \leq 3 \end{cases}$ then

$f \circ f$ is

- (1) Continuous at $x = 1$
- (2) Continuous at $x = 2$
- (3) Discontinuous at $x = 1$ and $x = 2$
- (4) Continuous on $[0, 3]$

65. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that

$f(x+y) = f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$ and if $f'(4) = 24$ and

$f'(0) = 3$, then $f(4) =$

- (1) 72
- (2) 5
- (3) 11

(4) 8

$$66. \frac{d}{dx} \left[x^{\sin x} + (\sin x)^x \right] =$$

$$(1) \ x^{\sin x} \left[\frac{\sin x}{x} + \cos x \cdot \log x \right] + (\sin x)^x \left[\frac{x \tan x}{x} + \log(\sin x) \right]$$

$$(2) \ x^{\sin x} [x \tan x + \cos x \log x] + (\sin x)^x \left[\frac{\sin x}{x} + \log(\sin x) \right]$$

$$(3) \ x^{\sin x} \left[\frac{x}{\sin x} + \cos x \log x \right] + (\sin x)^x [x \cot x + \log(\sin x)]$$

$$(4) \ x^{\sin x} \left[\frac{\sin x}{x} + \sin x \log x \right] + (\sin x)^x [x \cot x + \log(\cos x)]$$

$$67. \text{ If } x = a(t + \sin t) \text{ and } y = a(1 - \cos t), \text{ then } \frac{d^2 y}{dx^2} =$$

$$(1) \ \frac{1}{4a \sin^4 \left(\frac{t}{2} \right)}$$

$$(2) \ \frac{1}{4a \cos^4 \left(\frac{t}{2} \right)}$$

$$(3) \ 4a \operatorname{cosec}^4 \left(\frac{t}{2} \right)$$

(4) $4a\sec^4\left(\frac{t}{2}\right)$

68. If θ is an angle between the curves $x^2 + 4y = 0$ and $xy = 2$, then

$\tan \theta =$

(1) -1

(2) $\frac{1}{3}$

(3) $\frac{1}{2}$

(4) 3

69. If the function $f(x) = x^3 + 2px^2 + 27x + 16$ is strictly increasing for all $x \in \mathbb{R}$, then the range of p is

(1) $\left(-\infty, \frac{-9}{2}\right) \cup \left(\frac{9}{2}, \infty\right)$

(2) $(-\infty, -9) \cup (9, \infty)$

(3) $\left(\frac{-9}{2}, \frac{9}{2}\right)$

(4) $(-9, 9)$

70. If $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{\sqrt{x}}$ for $x \in [3, 12]$, then the value of

$c \in (3, 12)$ for which

$\frac{f'(c)}{g'(c)} = \frac{f(12) - f(3)}{g(12) - g(3)}$ holds, is

(1) 7.5

(2) 4.8

(3) 6

(4) 9

71. If the distance s described in time ' t ' by a particle moving on a straight line is given by $s = t^5 - 40t^3 + 30t^2 + 80t - 250$, then its minimum acceleration is

(1) 260

(2) -260

(3) 130

(4) -130

72. If $g\left(\frac{t+1}{2t+1}\right) = t+1$, then $\int g(x) dx =$

(1) $\frac{x^2}{2} + c$

$$(2) \log_e(2x-1) + \frac{1}{2} \log_e |(x+1)| + c$$

$$(3) \frac{1}{2} \log_e \left| \left(\frac{x+1}{2x+1} \right) \right| + c$$

$$(4) \frac{x}{2} + \frac{1}{4} \log_e |(2x-1)| + c$$

$$73. \int \frac{x^8 - 9x^2 + 18}{x^4 - 3x^2 + 3} dx =$$

$$(1) \frac{x^5}{4} + x^3 + 6x^2 + c$$

$$(2) \frac{x^5}{5} + \frac{x^4}{4} + 6x + c$$

$$(3) \frac{x^5}{5} + x^3 + 6x + c$$

$$(4) \frac{x^5}{5} - \frac{x^3}{2} + 6x^2 + c$$

$$74. \int \frac{x^4 + 1}{1 + x^6} dx =$$

$$(1) \tan^{-1}(x^3) + \tan^{-1} x + c$$

$$(2) \frac{1}{3} \tan^{-1} x + \tan^{-1} x^3 + c$$

$$(3) 3 \tan^{-1} x^3 + \tan^{-1} x + c$$

$$(4) \tan^{-1} x + \frac{1}{3} \tan^{-1} x^3 + c$$

$$75. \int (\log x)^3 x^5 dx =$$

$$(1) x^6 \left[\frac{(\log x)^3}{12} - \frac{1}{6} (\log x)^2 + \frac{1}{6} \log x - \frac{1}{36} \right] + c$$

$$(2) x^6 \left[\frac{(\log x)^3}{6} - \frac{1}{18} (\log x)^2 + \frac{\log x}{12} - \frac{1}{36} \right] + c$$

$$(3) x^6 \left[\frac{(\log x)^3}{6} + \frac{1}{12} (\log x)^2 - \frac{\log x}{12} + \frac{1}{36} \right] + c$$

$$(4) x^6 \left[\frac{(\log x)^3}{6} - \frac{(\log x)^2}{12} + \frac{\log x}{36} - \frac{1}{216} \right] + c$$

$$76. \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2} \right) \left(1 + \frac{2^2}{n^2} \right) \dots \left(1 + \frac{n^2}{n^2} \right) \right]^{\frac{1}{n}} =$$

$$(1) 3e^{\frac{\pi-4}{6}}$$

$$(2) 2e^{\frac{\pi-2}{4}}$$

$$(3) 2e^{\frac{\pi-4}{2}}$$

$$(4) 4e^{\frac{\pi-4}{4}}$$

$$77. \int_0^{\pi/2} \frac{\cos x}{\sqrt{1 + \cos x \sin x}} dx =$$

$$(1) \sqrt{2} \cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$(2) \frac{1}{\sqrt{2}} \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$(3) \sqrt{2} \sin^{-1} \left(\frac{1}{\sqrt{3}} \right)$$

$$(4) \sqrt{2} \sin^{-1} (\sqrt{3})$$

78. The area (in square units) bounded by the curves $y = 2x^2$ and $y = \text{Max} \{x - [x], x + |x|\}$ in between the lines $x = 0$ and $x = 2$ is

$$(1) \frac{4}{3}$$

$$(2) \frac{1}{2}$$

$$(3) 1$$

$$(4) 2$$

79. If a and b are arbitrary constants, then the differential equation

having $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ as its general solution is

(1) $\left(\frac{d^2y}{dx^2}\right)^2 = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^3$

(2) $(x^2 - y^2)\frac{d^2y}{dx^2} - 2xy\frac{dy}{dx} - y = 0$

(3) $xy\frac{d^2y}{dx^2} + x\left(\frac{dy}{dx}\right)^2 - y\frac{dy}{dx} = 0$

(4) $x^2\frac{d^2y}{dx^2} + 2x\frac{dy}{dx} - 2y = 0$

80. The general solution of the differential equation $\frac{dx}{dy} + \frac{x}{y} = x^2$ is

(1) $\frac{1}{y} = cx - y \log x$

(2) $\frac{1}{x} = cy + x \log x$

(3) $\frac{1}{x} = cy - y \log y$

(4) $\frac{1}{y} = cx + y \log x$

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81. If 'A' represents Boltzmann constant, 'B' represents Planck's constant and 'C' represents speed of light in vacuum, then the quantity having the dimensions of $A^4B^{-3}C^{-2}$ is
- (1) Universal gas constant
 - (2) Specific heat capacity
 - (3) Stefan's constant
 - (4) Heat energy
82. The motion of a particle along a straight line is described by the function $x = (2t - 3)^2$ where x is in metres and t is in seconds. The acceleration of the particle at $t = 2$ s is

- (1) 1 ms^{-2}
- (2) 4 ms^{-2}
- (3) 8 ms^{-2}
- (4) 7 ms^{-2}

83. A particle moves in the x-y plane with velocity $\vec{V} = x\hat{i} + yt\hat{j}$; At $t = \frac{x\sqrt{3}}{y}$, the magnitudes of tangential and normal accelerations,

respectively are

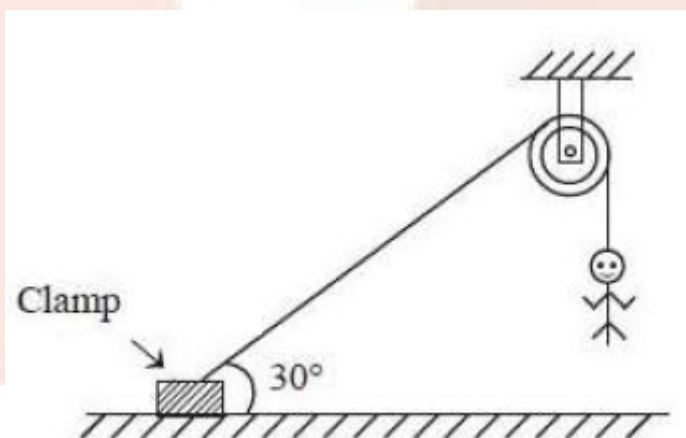
- (1) $\frac{\sqrt{3}y}{2}, \frac{y}{2}$
- (2) $\frac{\sqrt{2}y}{3}, \frac{\sqrt{3}y}{2}$
- (3) $\frac{\sqrt{3}y}{2}, \frac{5y}{2}$
- (4) $2\sqrt{3}y, \frac{11y}{\sqrt{3}}$

84. Assertion (A) : The speed of a body in uniform circular motion is constant.

Reason (R) : In uniform circular motion, the acceleration of the body is constant.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true and (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false.
- (4) (A) is false but (R) is true.

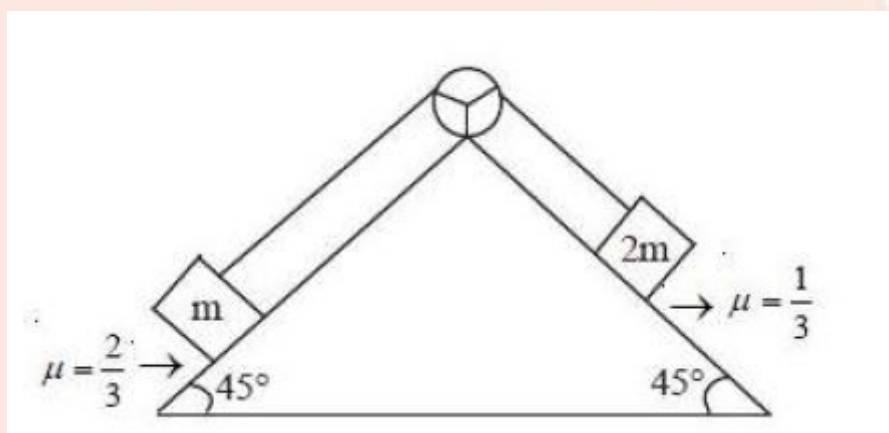
85. One end of a light string is fixed to a clamp on the ground and the other end passes over a fixed frictionless pulley as shown in the figure. It makes an angle of 30° with the ground. The clamp can tolerate a vertical force of 40 N. If a monkey of mass 5 kg were to climb up the rope then the maximum acceleration in the upward direction with which it can climb safely is ($g = 10 \text{ ms}^{-2}$)



- (1) 2 ms^{-2}
- (2) 4 ms^{-2}
- (3) 6 ms^{-2}

(4) 8 ms^{-2}

86. In the arrangement shown in the figure, if the blocks of masses ' m ' and ' $2m$ ' are released from the state of rest, tension in the strings is (μ = coefficient of friction, string is massless and inextensible, pulley is frictionless)



(1) mg

(2) $\sqrt{2}mg$

(3) $\frac{2\sqrt{2}mg}{3}$

(4) $\frac{\sqrt{2}mg}{3}$

87. A stone of mass 2 kg tied to a light inextensible string of length $\frac{5}{3} \text{ m}$ is whirling in a circular path in a vertical plane. If the ratio

of maximum tension to the minimum tension in the string is 4, then the speed of the stone at the highest point of the circle is _____ ($g = 10 \text{ ms}^{-2}$)

(1) 20 ms^{-1}

(2) $10\sqrt{3} \text{ ms}^{-1}$

(3) $\sqrt{50} \text{ ms}^{-1}$

(4) 10 ms^{-1}

88. A ball falls freely from a height of 180 m on to a hard horizontal floor and repeatedly bounces. If the coefficient of restitution is 0.5, the average speed and average velocity of the ball before it ceases to rebound are respectively

(Acceleration due to gravity = 10 ms^{-2})

(1) $10 \text{ ms}^{-1}, 10 \text{ ms}^{-1}$

(2) $50 \text{ ms}^{-1}, \frac{50}{3} \text{ ms}^{-1}$

(3) $\frac{50}{3} \text{ ms}^{-1}, 10 \text{ ms}^{-1}$

(4) $\frac{20}{3} \text{ ms}^{-1}, \frac{50}{3} \text{ ms}^{-1}$

89. A circular disc of radius 'R' is removed from one end of a bigger circular disc of radius '2R'. The centre of mass of the new disc is at a distance ' αR ' from the centre of the bigger disc.

The value of ' α ' is

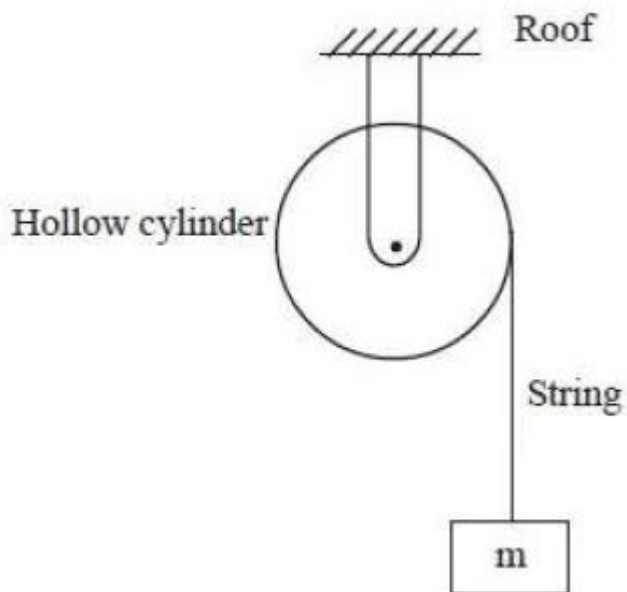
(1) $\frac{1}{2}$

(2) $\frac{1}{3}$

(3) $\frac{1}{4}$

(4) $\frac{1}{6}$

90. In the figure shown, acceleration with which the mass 'm' falls down when released is (consider the string to be massless, g-acceleration due to gravity)



(1) $\frac{2g}{3}$

(2) $\frac{g}{2}$

(3) $\frac{5g}{6}$

(4) g

91. At $t = 0$, a particle executing SHM with a time period 3 s is in phase with another particle executing SHM. The time period of the second particle is T (less than 3 s). If they are again in the same phase for the third time after 45 s, then the value of T is ____

(1) 1 s

(2) 1.5 s

- (3) 2 s
- (4) 2.5 s

92. A body is projected vertically upwards from the surface of Earth with a velocity sufficient to carry it to infinity. The time taken by it to reach a height of three times the radius of the Earth is (Acceleration due to gravity = 9.8 ms^{-2} and radius of the Earth = 6400 km)

- (1) 44.44 minutes
- (2) 22.22 minutes
- (3) 18.76 minutes
- (4) 37.52 minutes

93. A copper wire of cross-sectional area 0.01 cm^2 is under a tension of 22 N. The decrease in the cross-sectional area is (Young modulus = $1.1 \times 10^{11} \text{ Nm}^{-2}$, Poisson's ratio = 0.32)

- (1) $0.128 \times 10^{-6} \text{ cm}^2$
- (2) $128 \times 10^{-6} \text{ cm}^2$
- (3) $12.8 \times 10^{-6} \text{ cm}^2$
- (4) $1.28 \times 10^{-6} \text{ cm}^2$

94. When a soap bubble of radius 0.2 mm is charged, it experiences

an outward electrostatic pressure of magnitude $\frac{\sigma^2}{2\epsilon_0}$ where

$\sigma = 20 \mu\text{Cm}^{-2}$ is the surface charge density. If the excess pressure inside the soap bubble due to the surface tension is same as this electrostatic pressure, then the surface tension of the soap solution is

$$(\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2})$$

(1) $8.85 \times 10^{-4} \text{ Nm}^{-1}$

(2) $12.4 \times 10^{-4} \text{ Nm}^{-1}$

(3) $11.3 \times 10^{-4} \text{ Nm}^{-1}$

(4) $90 \times 10^{-4} \text{ Nm}^{-1}$

95. The temperature of a spherical black body is inversely proportional to its radius. If its radius is doubled then the power radiating from it will be

(1) Doubled

(2) $\frac{1}{4}$ times of initial value

(3) Halved

(4) 4 times of initial value

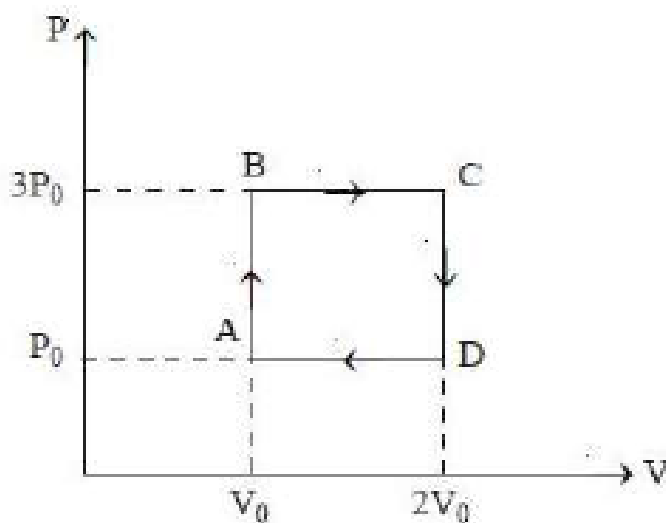
96. A metal sphere immersed in water weights W_1 at 0°C and W_2 at 50°C . The coefficient of cubical expansion of the metal is less than that of water. Then

- (1) $W_1 > W_2$
- (2) $W_1 < W_2$
- (3) $W_1 = W_2$
- (4) $W_1 = 2W_2$

97. A reversible Carnot heat engine converts $\frac{1}{4}$ th of its input heat into work. When the temperature of the sink is reduced by 50 K, its efficiency becomes $33\frac{1}{3}\%$. The initial temperature of the source and the sink respectively are

- (1) 600 K, 550 K
- (2) 600 K, 450 K
- (3) 300 K, 150 K
- (4) 450 K, 350 K

98. An ideal monatomic gas is carried along the cycle ABCDA as shown in the figure. The total heat absorbed during this process is



- (1) $10.5P_0V_0$
- (2) $7.5P_0V_0$
- (3) $2.5P_0V_0$
- (4) $1.5P_0V_0$

99. The ratio of the speed of sound in a monatomic gas at 27°C and rms speed of the molecules of the same gas at a temperature of 127°C is _____

- (1) 1:2
- (2) $\sqrt{5}:\sqrt{12}$
- (3) 3:4

(4) $\sqrt{13} : \sqrt{17}$

100. Two uniform stretched steel strings A and B are vibrating under the same tension. The first overtone of A is equal to the second overtone of B. If the radius of A is twice that of B, then the ratio of the lengths of the strings is _____

- (1) 2:3
- (2) 1:2
- (3) 1:3
- (4) 1:4

101. An open pipe of length ' l ' is vibrating in 3rd overtone with maximum amplitude ' A '. The amplitude at a distance of $\frac{l}{16}$ from any open end is

- (1) A
- (2) 0
- (3) $\frac{A}{\sqrt{2}}$
- (4) $\frac{\sqrt{3}A}{2}$

102. Magnifying power of an astronomical telescope for normal adjustment is 10 and length of the telescope is 110 cm. Magnifying power of the same telescope when the image is formed at the near point is

- (1) 14
- (2) 18
- (3) 23
- (4) 26

103. In Young's double slit experiment, the two slits are illuminated by a light beam consisting of wavelengths $4200 \text{ \AA}$ and $5040 \text{ \AA}$. If the distance between the slits is 2.4 mm and the distance between the slits and the screen is 200 cm, the minimum distance from the central bright fringe to the point where the bright fringe due to both the wavelengths coincide is _____

- (1) 0.7 mm
- (2) 1.4 mm
- (3) 2.1 mm
- (4) 2.8 mm

104. Flux coming out from a positive charge of 8 C, placed in a medium of dielectric constant 4 is

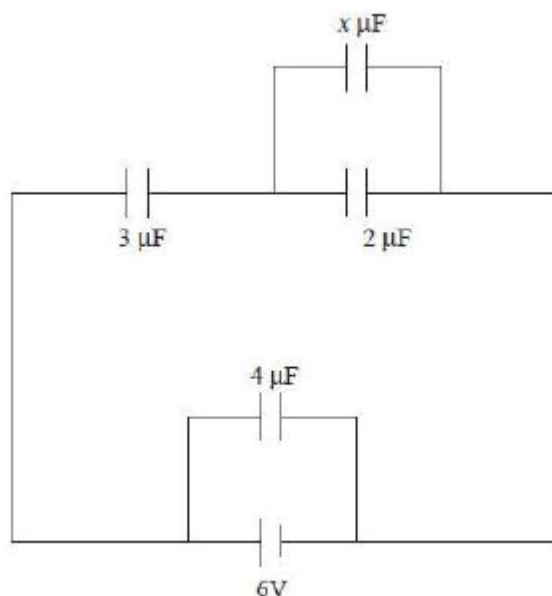
- (1) $\frac{1}{2\epsilon_0}$
- (2) $\frac{2}{\epsilon_0}$
- (3) $8\epsilon_0$
- (4) $32\epsilon_0$

105. Two charged particles each of mass 9.8 g and charges $+20\ \mu\text{C}$ and $-20\ \mu\text{C}$ are attached to the two ends of a massless and rigid uniform non-conducting rod of length 50 cm. This arrangement is held in a uniform electric field of $12.1\ \text{NC}^{-1}$ such that the rod makes a very small angle with the field direction. If the rod is set free, the minimum time needed for the rod to become parallel to the direction of the electric field is _____ seconds.

- (1) 5
- (2) 8
- (3) 12
- (4) 17

106. Four capacitors of capacitances $2\ \mu\text{F}$, $3\ \mu\text{F}$, $4\ \mu\text{F}$ and $x\ \mu\text{F}$ are connected to a battery of emf 6 V and of negligible internal

resistance, as shown in the figure. If the ratio of the charges on $x \mu\text{F}$ and $4 \mu\text{F}$ capacitances is $\frac{3}{8}$, then the value of x is _____



- (1) 2
- (2) 5
- (3) 3
- (4) 8

107. The plates of a parallel plate capacitor are charged upto 100 V. A 2 mm thick insulator sheet is inserted between the plates. Then to maintain the same potential difference the distance between the plates is increased by 1.6 mm. The dielectric constant of the insulator is

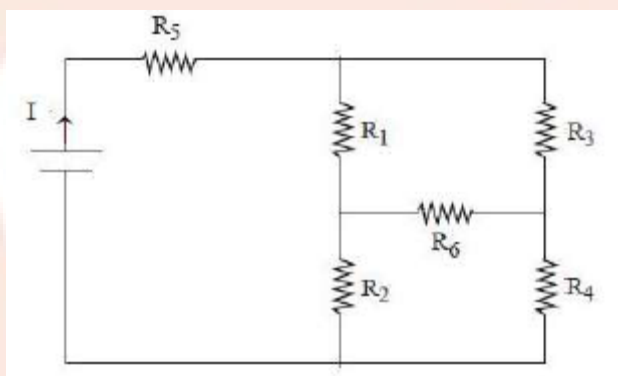
- (1) 6

(2) 8

(3) 5

(4) 4

108. In the given circuit, current I is independent of the resistance R_6 . Then



(1) $R_1 R_2 R_5 = R_3 R_4 R_6$

(2) $\frac{1}{R_5} + \frac{1}{R_6} = \frac{1}{R_1 + R_2} + \frac{1}{R_3 + R_4}$

(3) $R_1 R_4 = R_2 R_3$

(4) $R_1 R_3 = R_2 R_4$

109. A fuse wire of radius 0.2 mm blows off with a current of 5 A. The fuse wire of same material but of radius 0.3 mm will blow off with a current of _____

(1) $\frac{15}{2}$ A

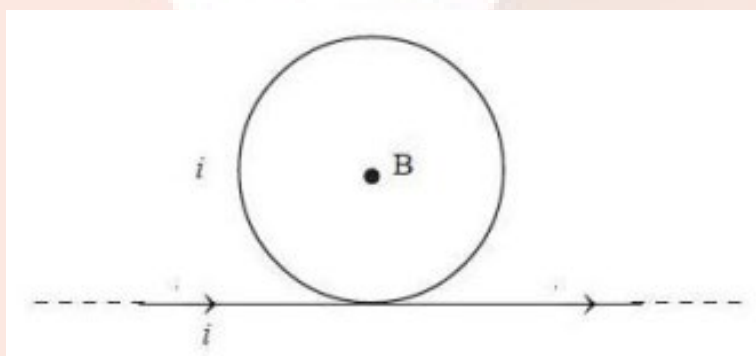
(2) $\frac{5\sqrt{3}}{2} \text{ A}$

(3) $5\sqrt{\frac{27}{8}} \text{ A}$

(4) 5 A

110. A circular loop and an infinitely long straight conductor carry equal currents, as shown in the figure. The net magnetic field at the centre of the loop is B_1 when the current in the loop is clock-wise and B_2 when the current in the loop is anti-clock-wise.

Then $\frac{B_1}{B_2}$ is



(1) $\frac{15}{19}$

(2) $\frac{13}{15}$

(3) $\frac{13}{17}$

$$(4) \frac{17}{19}$$

111. Two circular loops of diameters 0.6 cm and 40 cm are kept coaxially with a separation of 15 cm between their centres. If a current 2 A flows through the smaller loop, the flux linked with the bigger loop is (approximately)

- (1) 9.0×10^{-11} Wb
- (2) 0.9×10^{-11} Wb
- (3) 1.8×10^{-11} Wb
- (4) 2.7×10^{-11} Wb

112. In the magnetic meridian of a certain place, the vertical component of the earth's magnetic field is 0.3464 G and the dip angle is 30° . The horizontal component of the earth's magnetic field at this location is _____

- (1) 0.4 G
- (2) 0.6 G
- (3) 0.7 G
- (4) 0.8 G

113. A current carrying circular loop is perpendicular to a magnetic field of induction 10^{-4} T. If the radius of the loop starts shrinking at a uniform rate of 2 mms^{-1} , then the emf induced in the loop at the instant when its radius is 20 cm will be _____

- (1) $0.02\pi \mu\text{V}$
- (2) $0.08\pi \mu\text{V}$
- (3) $0.03\pi \mu\text{V}$
- (4) $0.05\pi \mu\text{V}$

114. A resistor and an inductor are connected in series to an ac source of voltage $150\sin(100\pi t + \pi)$ volt. If the current in the circuit is $5\sin\left(100\pi t + \frac{2\pi}{3}\right)$ ampere then the average power dissipated and the resistance of the resistor are respectively _____

- (1) 187.5 W, 30Ω
- (2) 187.5 W, 15Ω
- (3) 375 W, 30Ω
- (4) 375 W, 15Ω

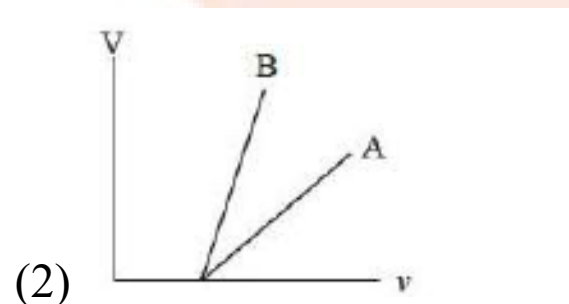
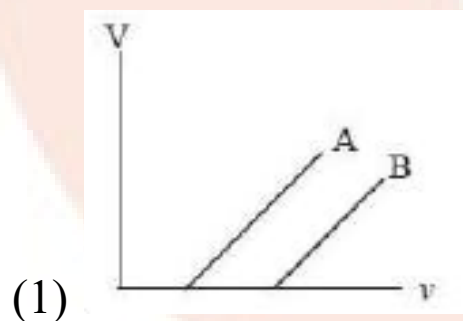
115. An electromagnetic wave of frequency 45 MHz travels in free space along X-axis. At some point and at some instant, the

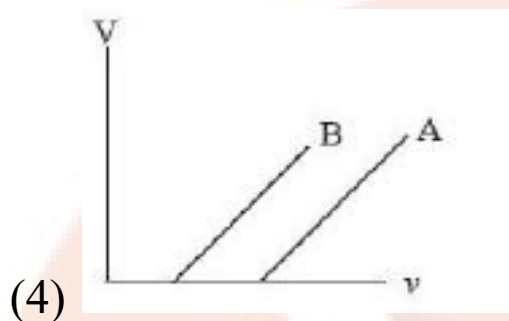
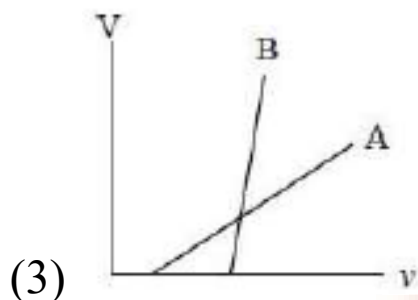
electric field has a maximum value of 750 NC^{-1} along Y-axis.

The magnetic field at this position and time is

- (1) $2.5 \times 10^{-6} \hat{j} \text{ T}$
- (2) $5 \times 10^{-6} \hat{k} \text{ T}$
- (3) $2.5 \times 10^{-6} \hat{k} \text{ T}$
- (4) $2.5 \times 10^{-6} \hat{i} \text{ T}$

116. When light of frequency ' ν ' incident on two metallic plates A and B, photo electrons are emitted. If the work function of A is more than that of B, the correct curve of the following curves drawn between stopping potential ' V ' and incident frequency ' ν ' is





117. The approximate value of principle quantum number for a circular orbit of hydrogen atom of radius 530 nm is

- (1) 26
- (2) 100
- (3) 200
- (4) 21

118. A radioactive element 'X' converts into another stable element 'Y'. Half life of X is 2 hours. Initially only X is present. After a time 't', if the ratio of atoms of X to Y is 1 : 4 then the value of 't' is

- (1) 2 hours
- (2) 4 hours

(3) Between 4 hours and 6 hours

(4) 6 hours

119. Match the following List-I and List-II

	<u>List-I</u>		<u>List-II</u>
(A)	Small Scale Integration (SSI)	(I)	Logic gates < 100
(B)	Medium Scale Integration (MSI)	(II)	Logic gates > 1000
(C)	Large Scale Integration (LSI)	(III)	Logic gates ≤ 10
(D)	Very Large Scale Integration (VLSI)	(IV)	Logic gates < 1000

The correct answer is

	A	B	C	D
(1)	II	III	I	IV
	A	B	C	D
(2)	IV	I	II	III
	A	B	C	D
(3)	I	IV	III	II
	A	B	C	D
(4)	III	I	IV	II

120. A modulation signal is given by

$$C_m(t) = A_c \sin \omega_c t + \mu A_c \sin \omega_m t \sin \omega_c t \quad \text{where } \mu \text{ is modulation}$$

index. To keep the signal without distortion, the value of μ should be

- (1) > 1
- (2) ≥ 1
- (3) $= 1$
- (4) ≤ 1

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121. The work functions (W_0) of K, Na, Li, Mg and Cu are 2.25, 2.30, 2.42, 3.70 and 4.80 eV respectively. How many of these metals do not undergo photoelectric effect when a radiation of wavelength 450 nm is allowed to fall on them ?

$$(1 \text{ eV} = 1.602 \times 10^{-19} \text{ J})$$

- (1) 2
- (2) 1
- (3) 3
- (4) 5

122. Number of completely filled orbitals in Xenon atom (Xe) is

- (1) 17
- (2) 18
- (3) 27
- (4) 28

123. The mass numbers of two elements X and Z are 52 and 75 respectively. X contains 16.6% more neutrons compared to protons. Z contains 27.3% more neutrons compared to protons. X and Z are respectively

- (1) ${}_{24}\text{Cr}$, ${}_{33}\text{As}$
- (2) ${}_{24}\text{Cr}$, ${}_{30}\text{Zn}$
- (3) ${}_{19}\text{K}$, ${}_{33}\text{As}$
- (4) ${}_{29}\text{Cu}$, ${}_{30}\text{Zn}$

124. An atom in a molecule has electrons in 1s, 2s, 2p, 3s, 3p, 3d and 4s orbitals. This atom can undergo hybridisations of type

- (1) sp^3d^2 , sp^3 , p^3ds
- (2) d^2sp^3 , p^2ds , dsp^2
- (3) sp^3 , dsp^2 , d^2sp^3
- (4) sp^3 , dsp^2 , dsp

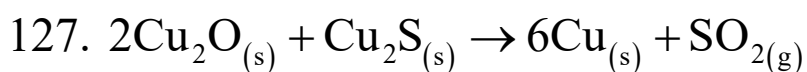
125. The bond dissociation energy (E) and bond length (R) of O_2 , N_2 , F_2 follow the order as

(E)	(R)
(1) $O_2 > N_2 > F_2$	$F_2 > O_2 > N_2$
(2) $N_2 > F_2 > O_2$	$N_2 > O_2 > F_2$
(3) $N_2 > O_2 > F_2$	$F_2 > O_2 > N_2$
(4) $N_2 > O_2 > F_2$	$N_2 > O_2 > F_2$

126. If the RMS speed of nitrogen at a certain temperature is 3000 ms^{-1} , the approximate kinetic energy of one mole of nitrogen at that temperature in kJ is

(assume nitrogen as ideal gas)

- (1) 9.0
- (2) 126.0
- (3) 90.0
- (4) 12.6



the oxidant and reductant respectively in the above reaction are

- (1) Oxide of Cu_2O and sulphide of Cu_2S

- (2) Sulphide of Cu_2S and Oxide of Cu_2O
- (3) Cu (I) of Cu_2O , Cu_2S and sulphide of Cu_2S
- (4) Cu (I) of Cu_2S , Cu (I) of Cu_2O

128. If 1.5 L of an ideal gas at a pressure of 20 atm expands isothermally and reversibly to a final volume of 15 L, the work done by the gas in L atm is

- (1) 69.9
- (2) 34.55
- (3) -34.55
- (4) -69.09

129. At T(K), the equilibrium constant of $\text{H}_{2(g)} + \text{I}_{2(g)} \rightleftharpoons 2\text{HI}_{(g)}$ is

49. If $[\text{H}_2]$, $[\text{I}_2]$ at equilibrium at the same temperature are 2.0×10^{-2} M and 8.0×10^{-2} M respectively, the $[\text{HI}]$ at equilibrium in mol L^{-1} is

- (1) 2.8
- (2) 0.28
- (3) 0.14
- (4) 1.4

130. If the pH of 0.10 M monobasic acid at 298 K is 5.0, the value of pK_a at the same temperature is

- (1) 5.0
- (2) 8.0
- (3) 9.0
- (4) 6.0

131. Identify the correct statements from the following.

- i. The number of hydrogen bonded water molecules in copper sulphate pentahydrate is one.
- ii. Lanthanum and zirconium form non-stoichiometric hydrides.
- iii. In solid form of H_2O , each oxygen is surrounded by six oxygens in octahedral positions at a distance of 276 pm.

- (1) i, ii, iii
- (2) i, ii
- (3) ii, iii
- (4) i, iii

132. Observe the following compounds.

$CaCO_3$ (i), $MgSO_4$ (ii), $BaCl_2$ (iii), $Sr(NO_3)_2$ (iv), $MgBr_2$ (v), $MgCl_2$ (vi)

The oxoacid salts of group II elements from the above list are

(1) i, ii, iii, iv, v, vi

(2) i, ii, iv

(3) iii, v, vi

(4) ii, v, vi

133. A few grams of borax is dissolved in distilled water. The pH range of resultant solution is

(1) 1-4

(2) 4-7

(3) 2-5

(4) 7-14

134. An element (X) when burnt in oxygen forms neutral XO and acidic XO_2 . The element X is-

(1) Sn

(2) C

(3) Ge

(4) Pb

135. Match the following

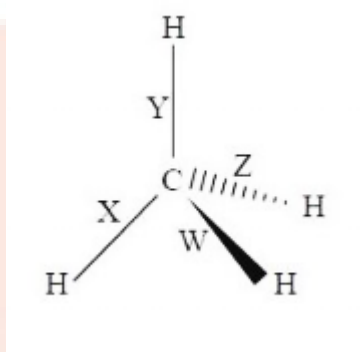
List-I	List-II
--------	---------

A) Methemoglobinemia	I) 1 ppm of dissolved oxygen in water
B) Kidney damage	II) 1000 ppb of lead in drinking water
C) Bones and teeth damage	III) BOD of drinking water is 2 ppm.
D) Growth of fish is stopped	IV) 2000 ppm of nitrates in drinking water.
	V) 50 ppm of fluoride in drinking water

The correct answer is

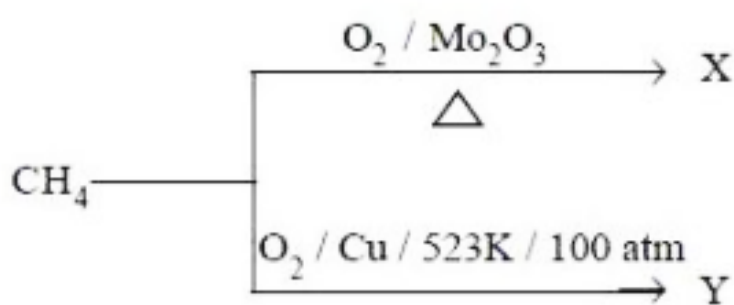
- (1) A B C D
 IV II V I
- (2) A B C D
 IV III V I
- (3) A B C D
 IV II I V
- (4) A B C D
 III II I V

136. In the following three dimensional structure of CH_4 , the bonds are labelled as W, X, Y and Z. The bonds projecting out of the plane are



- (1) X, Y
- (2) W, Z
- (3) X, Z
- (4) W, Y

137. Identify X and Y in the following reactions



- | | | |
|-----|------------------------|------------------------|
| | X | Y |
| (1) | CH_3OH | HCO_2H |
| | X | Y |
| (2) | HCO_2H | CH_2O |



138. Which one of the following has highest dipole moment?

- (1) cis-But-2-ene
- (2) trans-1, 2-Dichloroethane
- (3) cis-1, 2-Dichloroethene
- (4) trans-But-2-ene

139. If the side length of a face centered unit cell of a metal is 400 pm, approximate radius of the metal in pm is ($\sqrt{2} = 1.414$)

- (1) 14.14
- (2) 35.3
- (3) 176.7
- (4) 141.4

140. If CO₂ gas having a partial pressure of 1.67 bar is bubbled through 1 L water at 298 K, the amount of CO₂ dissolved in water in g L⁻¹ is approximately

(Henry's law constant of CO_2 is 1.67 k bar at 298 K)

- (1) 24.42
- (2) 12.21
- (3) 2.44
- (4) 1.22

141. 12.25 g of $\text{CH}_3\text{CH}_2\text{CHClCOOH}$ is added to 250 g of water to make a solution. If the dissociation constant of above acid is 1.44×10^{-3} , the depression in freezing point of water in $^\circ\text{C}$ is (K_f for water is $1.86 \text{ K kg mol}^{-1}$)

- 1. 0.789
- 2. 0.394
- 3. 1.183
- 4. 0.592

142. Assertion (A) : The charge on one mole of electrons is one Faraday.

Reason (R) : The quantity of current required to deposit one mole of Mg from Mg^{2+} electrolyte solution is two Faradays.

The correct answer is

1. Both (A) and (R) are correct and (R) is the correct explanation of (A)
 2. Both (A) and (R) are correct and (R) is not the correct explanation of (A)
 3. (A) is correct but (R) is not correct.
 4. (A) is not correct but (R) is correct.
143. If the half lives of the first order reaction at 350 K and 300 K are 2 and 20 seconds respectively, the activation energy of the reaction in kJ mol^{-1} is
1. 40.2
 2. 20.1
 3. 60.3
 4. 30.2
144. Which one of the following is present in gas mask?
1. Silica gel
 2. V_2O_5
 3. Activated charcoal
 4. Fluorescein

145. The pair of oxides, which can be leached out, when powdered bauxite containing ceratin impurities is digested with concentrated solution of NaOH at 473-573 K temperature and 35-36 bar pressure are

1. $\text{TiO}_2, \text{SiO}_2$
2. $\text{SiO}_2, \text{Al}_2\text{O}_3$
3. $\text{SiO}_2, \text{Fe}_2\text{O}_3$
4. $\text{Al}_2\text{O}_3, \text{Fe}_2\text{O}_3$

146. Nitrous acid was disproportionated to form water, HNO_3 and X. In another reaction, sodium nitrite was reacted with H_2SO_4 to form NaHSO_4 , HNO_3 , water an Y. What are X and Y respectively ?

- (1) $\text{NO}, \text{N}_2\text{O}_3$
- (2) NO, NO
- (3) $\text{N}_2\text{O}, \text{NO}_2$
- (4) $\text{NO}_2, \text{N}_2\text{O}_5$

147. Identify the correct statements from the follwing

- i Oxygen shows $-2, -1, +1, +2$ oxidation states.

ii The thermal stability of H_2O , H_2Se , H_2S follows the order



iii The reducing nature of H_2Se , H_2S , H_2Te follows the order



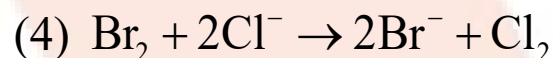
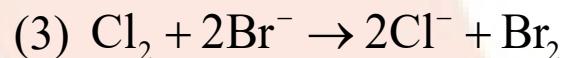
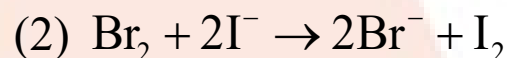
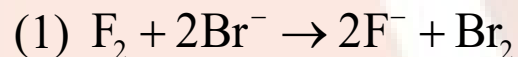
(1) i, ii, iii

(2) i, ii

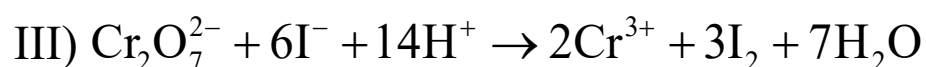
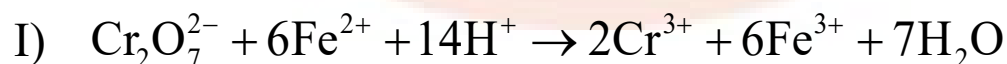
(3) i, iii

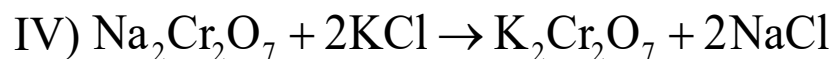
(4) ii, iii

148. Which one of the following reactions does not take place?



149. Identify the reactions in which dichromate acts as an oxidizing reagent.





- (1) I, IV
- (2) I, III
- (3) II, III
- (4) II, IV

150. The green coloured complex ion of nickel in its aqueous solution is

- (1) $[\text{Ni}(\text{en})_3]^{2+}$
- (2) $[\text{Ni}(\text{H}_2\text{O})_2(\text{en})_2]^{2+}$
- (3) $[\text{Ni}(\text{H}_2\text{O})_4(\text{en})]^{2+}$
- (4) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$

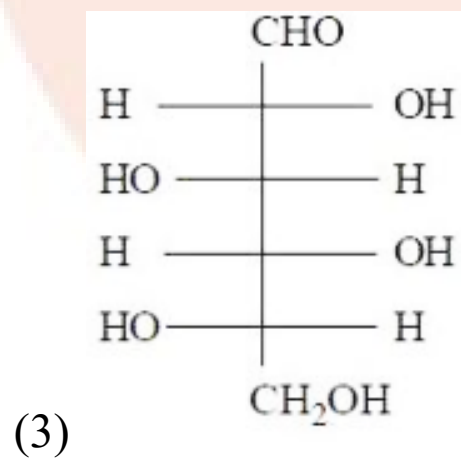
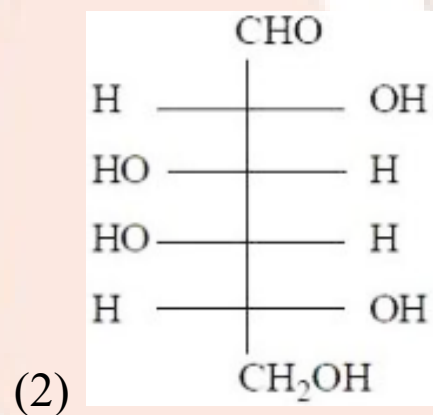
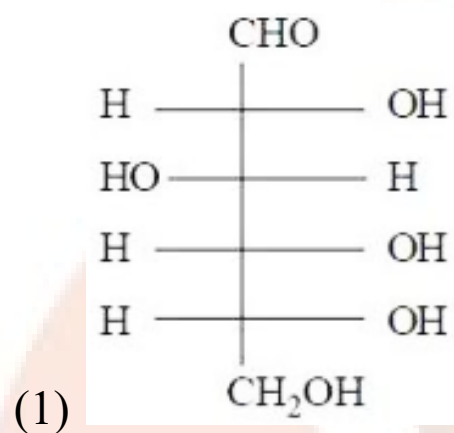
151. Examples of synthetic polymer (X) and semi-synthetic polymer (Y) are

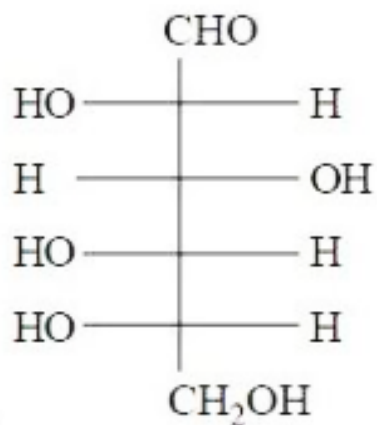
- | X | Y |
|----------------|------------|
| (1) polyethene | rayon |
| (2) rayon | nylon 6, 6 |
| (3) rubber | polythene |

(4) cellulose nitrate

PVC

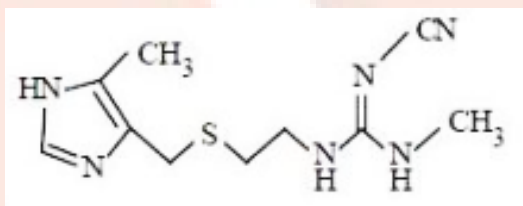
152. Fischer projection formula of L-(–)- glucose is



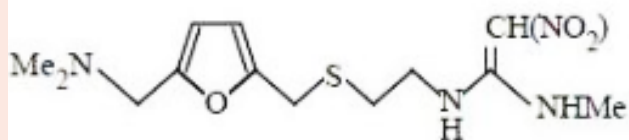


(4)

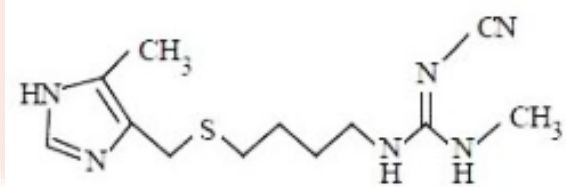
153. Structure of Ranitidine is



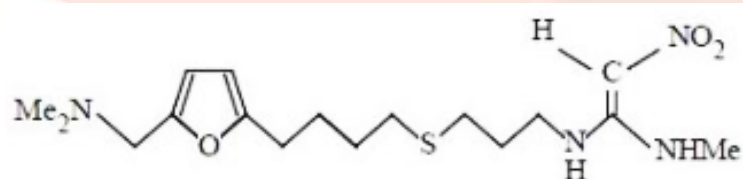
(1)



(2)

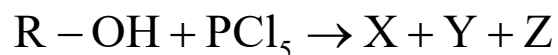


(3)



(4)

154. What are X, Y and Z in the following reaction



X

Y

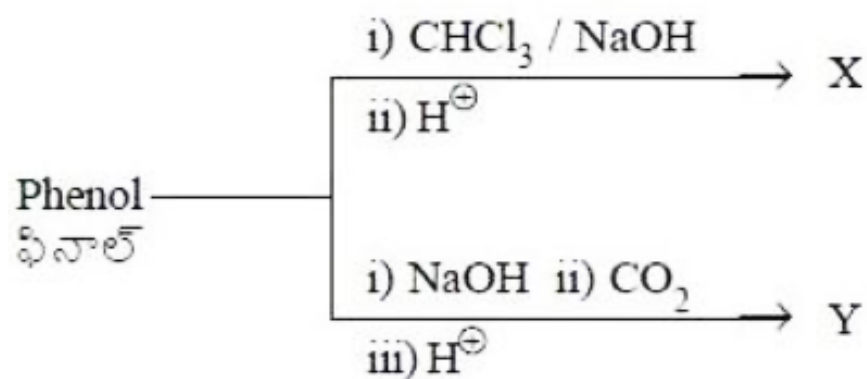
Z

- | | | | |
|-----|--------|-----------|----------|
| (1) | $R-Cl$ | HCl | $POCl_3$ |
| (2) | ROR | H_3PO_3 | H_2O |
| (3) | $R-Cl$ | H_2O | $POCl_3$ |
| (4) | ROR | H_2O | $POCl_3$ |

155. Butanone reacts with methyl magnesium bromide to form an addition product (Z). On hydrolysis Z gives

- (1) $CH_3CH_2C(OH)(CH_3)CH_3$
- (2) $(H_3C)_3CCH_2OH$
- (3) $CH_3CH_2CH_2CH(OH)CH_3$
- (4) $CH_3CH_2CH(CH_3)CH_2OH$

156. What are X and Y in the following reactions?

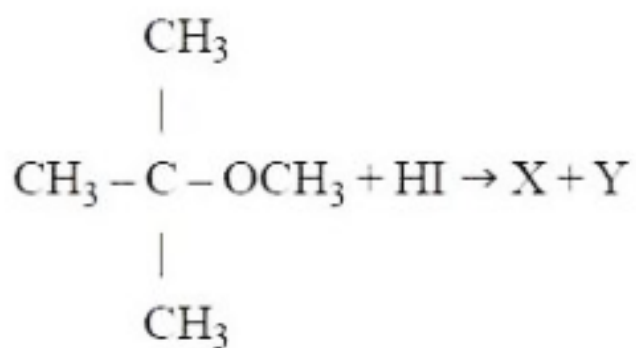
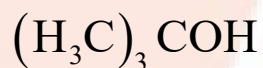
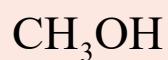
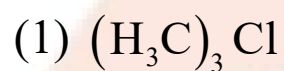


X

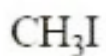
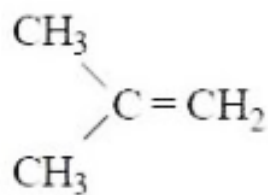
Y

- | | | |
|-----|--|--|
| (1) | | |
| (2) | | |
| (3) | | |
| (4) | | |

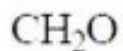
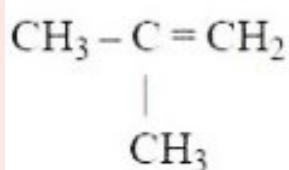
157. What are X and Y in the following reaction?

XY

(3)



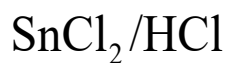
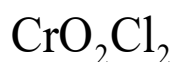
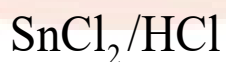
(4)



158. Reagents used in Etard reaction (I) and Stephen reaction (II) are

I

II





159. What are X and Y in the following reactions?



- | | |
|--------------------------------|-------------------------|
| (1) $\text{RCOONH}_4^{\oplus}$ | RCH_2CN |
| (2) RCN | RCONH_2 |
| (3) RNHCH_3 | RCH_2NC |
| (4) RCONH_2 | RCN |

160. The order of basic strength of methyl substituted amines and NH_3 in aqueous solution is

- (1) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N} > \text{NH}_3$
- (2) $(\text{CH}_3)_3\text{N} > (\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3$
- (3) $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3 > (\text{CH}_3)_3\text{N}$
- (4) $\text{NH}_3 > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$

AP EAMCET-2018-Engineering-24 Apr-Shift-1
Answer Key

1	3	33	4	65	4	97	2	129	2
2	3	34	2	66	*	98	1	130	3
3	2	35	2	67	2	99	2	131	2
4	4	36	2	68	4	100	3	132	2
5	2	37	3	69	3	101	3	133	4
6	4	38	4	70	3	102	1	134	2
7	2	39	3	71	2	103	3	135	1
8	1	40	4	72	4	104	2	136	2
9	4	41	2	73	3	105	1	137	4
10	2	42	3	74	4	106	2	138	3
11	2	43	1	75	4	107	3	139	4
12	2	44	4	76	3	108	3	140	3
13	3	45	4	77	3	109	3	141	1
14	1	46	2	78	4	110	*	142	1
15	4	47	4	79	3	111	*	143	1
16	1	48	4	80	3	112	2	144	3
17	3	49	1	81	3	113	2	145	2
18	2	50	2	82	3	114	2	146	2

19	3	51	2	83	1	115	3	147	3
20	3	52	4	84	3	116	4	148	4
21	2	53	3	85	3	117	2	149	2
22	2	54	1	86	3	118	3	150	4
23	2	55	3	87	3	119	4	151	1
24	4	56	2	88	3	120	4	152	4
25	1	57	4	89	2	121	3*	153	2
26	1	58	2	90	2	122	3	154	1
27	3	59	4	91	4	123	1	155	1
28	1	60	2	92	1	124	3	156	4
29	1	61	3	93	4	125	3	157	2
30	4	62	4	94	3	126	2	158	3
31	2	63	3	95	2	127	3	159	4
32	4	64	3	96	2	128	4	160	1

AP EAMCET-2018-Engineering-24 Apr-Shift-1

Solutions – Mathematics

1. Let the function $f(x)$ is given by

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n$$

The given relation is

$$f(x) \cdot f\left(\frac{1}{x}\right) = f(x) + f\left(\frac{1}{x}\right)$$

Substitute $f(x)$ in the given equation

$$\left(a_0x^n + a_1x^{n-1} + \dots + a_n\right) \left(\frac{a_0}{x^n} + \frac{a_1}{x^{n-1}} + \dots + a_n\right)$$

Compare the coefficient of x^n ,

$$a_0a_n = a_0 \Rightarrow a_n = 1$$

Compare the coefficient of x^{n-1}

$$a_0a_{n-1} + a_na_1 = a_1$$

$$a_0a_{n-1} = 0$$

So,

$$a_{n-1} = a_{n-2} = \dots = a_1 = 0$$

$$a_0 = \pm 1$$

$$f(x) = 1 \pm x^n$$

$$f(4) = 1 \pm 4^n = 257$$

$$4^n = 256$$

$$n = 4$$

$$\text{Therefore, } f(3) = 1 + 3^4 = 82$$

2. The $f(x)$ is given as

$$f(x) = \left(x + \frac{1}{x}\right)^x$$

For this function to be defined

$$\left(x + \frac{1}{x}\right) > 0$$

$$\frac{x+1}{x} > 0$$

Therefore the interval is

$$x \in (-\infty, -1) \cup (0, \infty)$$

3. Given $S_n = \sum_{k=1}^n (-1)^{k-1} \cdot k^2$

Expand the above expression,

$$S_n = 1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{n-1} n^2$$

$$\begin{aligned} S_{77} &= 1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{76} 77^2 \\ &= (1^2 + 2^2 + 3^2 + \dots + 77^2) - 2(2^2 + 4^2 + 6^2 + \dots + 76^2) \\ &= \sum_{r=1}^{77} r^2 - 2 \cdot 2^2 \sum_{k=1}^{58} k^2 \\ &= \frac{77 \cdot 78 \times 155}{6} - \frac{8 \cdot 38 \cdot 39 \cdot 77}{6} = 3003 \end{aligned}$$

4. Let,

$$A = \begin{vmatrix} 125 & 5 & 25 \\ 343 & 7 & 49 \\ 729 & 9 & 81 \end{vmatrix}$$

Apply row reduction as, $R_1 \rightarrow \frac{1}{5}R_1$, $R_2 \rightarrow \frac{1}{7}R_2$, $R_3 \rightarrow \frac{1}{9}R_3$

$$A = 5 \times 7 \times 9 \begin{vmatrix} 25 & 1 & 5 \\ 49 & 1 & 7 \\ 81 & 1 & 9 \end{vmatrix}$$

Use $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - R_1$

$$A = 5 \times 7 \times 9 \begin{vmatrix} 25 & 1 & 5 \\ 24 & 0 & 2 \\ 56 & 0 & 4 \end{vmatrix}$$

$$A = 5 \times 7 \times 9 [-1(96 - 112)] = 71$$

5. It is given that

$$A^2x = cAx \rightarrow A$$

$$Ax = c$$

Therefore $Ax \rightarrow c$ is the Eigen value of A .

And the Eigen values of A are 8, 2, 2.

Therefore, the number of values of c is 2.

6. Condition for the non-trivial solution is,

$$\begin{vmatrix} \sin 3\theta & -1 & 1 \\ \cos 2\theta & 4 & 3 \\ 2 & 7 & 7 \end{vmatrix} = 0$$

Solve further,

$$\sin 3\theta(28 - 21) + 1(7 \cos 2\theta - 6) + 1(7 \cos 2\theta - 8) = 0$$

$$\sin 3\theta + 2 \cos 2\theta - 2 = 0$$

$$-\sin \theta(4 \sin^2 \theta + 4 \sin \theta - 3) = 0$$

$$\sin \theta(2 \sin \theta - 1)(2 \sin \theta + 3) = 0$$

$$\theta = n\pi, \theta = n\pi + (-1)^n \frac{\pi}{6}$$

7. Let $z = x + iy$

$$e^{z^2} = e^{x^2 - y^2 + 2ixy}$$

$$= e^{x^2 - y^2} \times e^{2ixy}$$

$$|e^{z^2}| = e^{x^2 - y^2}$$

$$0 \leq x^2 \leq 1, 0 \leq y^2 \leq 1$$

$$-1 \leq x^2 - y^2 \leq 1$$

$$\text{Maximum of } |e^{z^2}| = e$$

8. Let $z = x + iy$

$$\frac{z-1}{z+1} = \frac{x+iy-1}{x+iy+1}$$

$$= \frac{x+iy-1}{x+iy+1} \times \frac{x+iy-1}{x+iy-1}$$

$$= \frac{x^2 + y^2 - 1 + 2iy}{(x+1)^2 + y^2}$$

Argument is given by

$$\arg\left(\frac{z-1}{z+1}\right) = \tan^{-1}\left(\frac{2y}{x^2 + y^2 - 1}\right) = \frac{\pi}{4}$$

$$\frac{2y}{x^2 + y^2 - 1} = 1$$

$$x^2 + y^2 - 2y - 1 = 0$$

9. From the given series,

$$\begin{aligned}(1)^{\frac{1}{4}} &= (\cos 0 + i \sin 0)^{\frac{1}{4}} \\ &= (\cos 2r\pi + i \sin 2r\pi)^{\frac{1}{4}} \\ &= \cos \frac{r\pi}{2} + i \sin \frac{r\pi}{2}\end{aligned}$$

The four roots of unity for the following equation is $1, i, -1, -i = 1, \alpha, \alpha^2, \alpha^3$

$$\text{And } (1+x)^n = \sum_{r=0}^n {}^nC_r x^r$$

$$\text{For } x=1, (2)^n = \sum_{r=0}^n {}^nC_r$$

$$\text{For } x=\alpha \quad (1+\alpha)^n = \sum_{r=0}^n {}^nC_r \alpha^r$$

$$\text{For } x=\alpha^2 \quad (1+\alpha^2)^n = \sum_{r=0}^n {}^nC_r \alpha^{2r}$$

$$\text{For } x=\alpha^3 \quad (1+\alpha^3)^n = \sum_{r=0}^n {}^nC_r \alpha^{3r}$$

Add the above equations,

$$\begin{aligned}2^n + (1+\alpha)^n + (1+\alpha^2)^n + (1+\alpha^3)^n \\ = \sum_{r=0}^n {}^nC_r (1 + \alpha^r + \alpha^{2r} + \alpha^{3r})\end{aligned}$$

Substitute the values in the above equation

$$\begin{aligned}
 4\left({}^nC_0 + {}^nC_4 + {}^nC_8 + \dots\right) &= 2^n + (1+i)^n + (1+i^2)^n + (1+i^3)^n \\
 &= 2 \cdot 2^{\frac{n}{2}} \left[\cos \frac{n\pi}{4} + 2^{\frac{n}{2}-1} \right] \\
 &= \frac{2^{\frac{n}{2}} \left[\cos \frac{n\pi}{4} + 2^{\frac{n}{2}-1} \right]}{2}
 \end{aligned}$$

10. $x^{11} - x^6 - x^5 + 1 = 0$

$$(x^6 - 1)(x^5 - 1) = 0$$

$$x = (1)^{\frac{1}{6}}, x = (1)^{\frac{1}{5}}$$

$$x = \cos \frac{2k\pi}{6} + i \sin \frac{2k\pi}{6} \text{ or } x = \cos \frac{2k\pi}{5} + i \sin \frac{2k\pi}{5}$$

For $k = 0, 1, 2, 3, 4$

$$x = \cos \frac{k\pi}{3} + i \sin \frac{k\pi}{3}$$

for $k = 1$

$$\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} = \text{cis} \frac{\pi}{3}$$

11. The given quadratic equation is

$$4^{\sec^2 \alpha} \left(\beta^2 - \beta + \frac{1}{2} \right) \leq 1$$

This equation has real roots.

$$D = 4 - 4 \times 4^{\sec^2 \alpha} \left(\beta^2 - \beta + \frac{1}{2} \right) \geq 0$$

From this we will get

$$4^{\sec^2 \alpha} = 4, \left(\beta - \frac{1}{2} \right)^2 = 0$$

$$\sec^2 \alpha = 1, \beta = \frac{1}{2}$$

Therefore the required value is

$$\cos^2 \alpha + \cos^{-1} \beta = 1 + \frac{\pi}{3}$$

12. The given expression is

$$(P^2 + P - 3)(P^2 + P - 2) - 12 = 0$$

$$\text{Let } y = P^2 + P - 2$$

$$y(y - 1) - 12 = 0$$

$$y^2 - y - 12 = 0$$

$$(y - 4)(y + 3) = 0$$

$$P^2 + P - 6 = 0 \text{ and } P^2 + P + 1 = 0$$

To imaginary roots

$$P^2 + P + 1 = 0$$

Sum of non-real roots = -1

13. Given expression is,

$$\sqrt{x^2 + 6x + 5} > (8 - x)$$

Solve further,

$$\left(\sqrt{x^2 + 6x + 5}\right)^2 > (8 - x)^2$$

$$6x + 16x > 64 - 5$$

$$x > \frac{59}{22}$$

$$x \in \left(\frac{59}{22}, \infty\right)$$

14. The given equation is

$$x^5 - 5x^4 + 9x^3 - 9x^2 + 5x - 1 = 0$$

$x = 1$ is one root of equation.

$$(x - 1)(x^4 - 4x^3 + 5x^2 - 4x + 1) = 0$$

$$\Rightarrow x^4 - 4x^3 + 5x^2 - 4x + 1 = 0$$

$$x^2 - 4x + 5 - \frac{4}{x} + \frac{1}{x^2} = 0$$

$$\left(x^2 + \frac{1}{x^2}\right) - 4\left(x + \frac{1}{x}\right) + 5 = 0$$

$$\left(x + \frac{1}{x}\right)^2 - 4\left(x + \frac{1}{x}\right) + 3 = 0$$

$$\text{Let } x + \frac{1}{x} = y$$

$$y^2 - 4y + 3 = 0$$

$$(y-1)(y-3) = 0$$

$$y = 1, 3$$

$$x + \frac{1}{x} = 1, x + \frac{1}{x} = 3$$

$$x^2 - x + 1 = 0, x^2 - 3x + 1 = 0$$

Here, $x^2 - 3x + 1 = 0$ have irrational roots. Which are,

$$x = -1, \frac{1}{3}$$

15. Since 20000 is not the part of the series.

The possible value of K is,

$$4 \times 10^3 \times 5 - 1 = 19999$$

16. The possible number of ways to select 5 questions

$$\begin{aligned} N &= {}^4C_2 \times {}^4C_2 \times {}^4C_1 \times 3 + {}^4C_3 \times {}^4C_1 \times {}^4C_1 \times 3 \\ &= 432 + 192 \\ &= 624 \end{aligned}$$

17. Given, $(1 + x - x^2 - x^3)^{11}$

Solve further,

$$= \left[(1-x^2)(1+x) \right]^{11}$$

$$= (1-x^2)^{11} (1+x)^{22}$$

For the above expression the coefficient of x^4 can be given as

$$= \sum_{r=0}^4 (-1)^r \cdot {}^{11}C_r \cdot {}^{22}C_{4-r}$$

$$= {}^{11}C_0 \times {}^{22}C_4 - {}^{11}C_1 \times {}^{22}C_3 + {}^{11}C_2 \times {}^{22}C_2 - {}^{11}C_3 \times {}^{22}C_1 + {}^{11}C_4 \times {}^{22}C_0$$

$$= 7315 - 16940 + 12705 - 3630 + 330$$

$$= 220$$

18. Expand this using binomial expression

$$\frac{\left[2\left(1+\frac{x}{4}\right)^{\frac{1}{2}} + 2\left(1-\frac{x}{8}\right)^{\frac{1}{3}} \right]}{\left(1-\frac{2x}{3}\right)^{\frac{3}{2}}} = 2 \left[1 + \frac{1}{2} \times \frac{x}{4} + 1 - \frac{1}{3} \times \frac{x}{8} \right] \left(1 - \frac{2x}{3} \right)^{-\frac{3}{2}}$$

$$= 2 \left[2 + \frac{x}{8} - \frac{x}{24} \right] \left[1 - \frac{3}{2} \left(\frac{-2x}{8} \right) \right]$$

$$= 2 \left[2 + \frac{2x}{24} \right] [1+x]$$

$$= 2 \left[2 + \frac{25x}{12} \right]$$

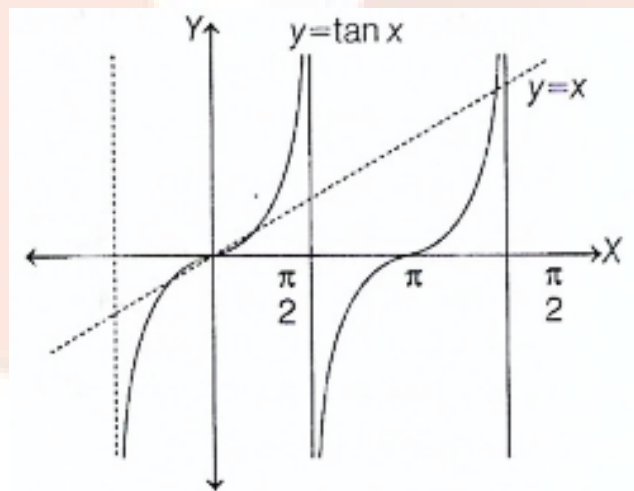
Substitute $x = \frac{6}{25}$

$$= 2 \left[2 + \frac{25}{12} \times \frac{6}{25} \right] = 5$$

19. By the use of binomial expansion,

$$\begin{aligned} \frac{(x^2 - 1)}{(x^2 + 1)(x^2 + 2)} &= \frac{1}{2} (x^2 - 1) (1 + x^2)^{-1} \left(1 + \frac{x^2}{2} \right)^{-1} \\ &= \frac{1}{2} \left[(x^2 - 1) (1 - x^2 + x^4 \dots) \left(1 - \frac{x^2}{2} + \frac{x^4}{4} \dots \right) \right] \\ &= \frac{1}{2} \left[(x^2 - 1) \left(1 - \frac{3}{2}x^2 + \frac{7}{4}x^4 \dots \right) \right] \\ &= \frac{1}{2} \left(-\frac{7}{4} - \frac{3}{2} \right) = -\frac{13}{8} \end{aligned}$$

20. The curve of function $\tan x = x$ is shown below,



From the graph it is clear that smallest root lies in the interval

$$\left(\pi, \frac{3\pi}{2}\right).$$

21. The given expression is

$$= \left[\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{2\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{4\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \right]$$

$$= \left[\left(1 + \cos \frac{6\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right) \right]$$

$$= \left[\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{\pi}{4}\right) \left(1 + \cos \frac{3\pi}{8}\right) (1) \left(1 - \cos \frac{3\pi}{8}\right) \right]$$

$$= \left[\left(1 + \cos \frac{3\pi}{4}\right) \left(1 - \cos \frac{\pi}{8}\right) \right]$$

Solve further,

$$= \left(1 - \cos^2 \frac{\pi}{8}\right) \left(1 - \cos^2 \frac{3\pi}{8}\right) \left(1 - \frac{1}{2}\right)$$

$$= \sin^2 \frac{\pi}{8} \times \sin^2 \frac{3\pi}{8} \times \frac{1}{2}$$

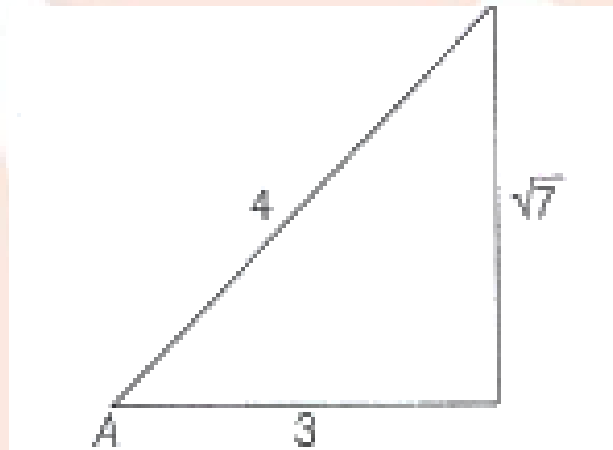
$$= \frac{1}{4} \left(1 - \cos \frac{\pi}{4}\right) \left(1 - \cos \frac{3\pi}{4}\right) \times \frac{1}{2}$$

$$= \frac{1}{4} \left(1 - \frac{1}{2}\right) \times \frac{1}{2} = \frac{1}{16}$$

22. Given expression is solved as,

$$\begin{aligned}
& 18 - 16 \sin^2 \frac{A}{2} - 32 \sin \frac{A}{2} \sin \frac{5A}{2} \\
&= 18 - 8 \sin^2 \frac{A}{2} - 16 \left(2 \sin \frac{A}{2} \sin \frac{5A}{2} \right) \\
&= 18 - 8(1 - \cos A) - 16 \left(\cos \left(\frac{5A}{2} - \frac{A}{2} \right) - \cos \left(\frac{5A}{2} + \frac{A}{2} \right) \right) \\
&= 18 - 8 + 8 \cos A - 16 \cos 2A + 16 \cos 3A
\end{aligned}$$

Consider the figure,



Substitute the values of ratios from the triangle.

$$\begin{aligned}
&= 10 + 8 \left(-\frac{3}{4} \right) - 16 \left[2 \times \left(\frac{9}{16} \right) - 1 \right] + 16 \left[4 \times \left(\frac{-27}{64} \right) - 3 \left(\frac{-3}{4} \right) \right] \\
&= 10 - 6 - 2 + 16 \times \frac{9}{16} = 11
\end{aligned}$$

23. The given expression is

$$\sec^2(x+y) - 1 + \cos^2(x+y) + y^2 + 2y = 0$$

$$\sec^2(x+y) + \cos^2(x+y) + y^2 + 2y + 1 = 2$$

For the above expression to be true

$$x + y = 0 \text{ and } y^2 + 2y + 1 = 0$$

$$x = -y \text{ and } (y+1)^2 = 0 \Rightarrow y = -1$$

Hence point P and Q coincides and so $d = 0$.

$$\cos d = 1$$

$$x + y = \pi \text{ and } y^2 + 2y + 1 = 0$$

$$x = \pi + 1, y = -1$$

So,

$$\cos d = -1$$

Therefore, $\cos d = (-1)^n, n \in N$

24. Given expression is solved as,

$$\tan^{-1} 2 + \cot^{-1}(-3) + \cot^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(-\frac{1}{2}\right)$$

$$= \tan^{-1}(2) - \tan^{-1}\left(\frac{1}{2}\right) + \pi - \cot^{-1}(3) + \cot^{-1}\left(\frac{1}{3}\right)$$

$$= \tan^{-1}(2) - \tan^{-1}\left(\frac{1}{2}\right) + \pi - \tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}(3)$$

$$= \tan^{-1}\left(\frac{2 - \frac{1}{2}}{1 + 2 \times \frac{1}{2}}\right) + \pi + \tan^{-1}\left(\frac{3 - \frac{1}{3}}{1 + 3 \times \frac{1}{3}}\right)$$

$$= \pi + \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{4}{3}\right)$$

$$= \pi + \frac{\pi}{2} = \frac{3\pi}{2}$$

25. Given that, $x = -\frac{1}{2}$

As,

$$\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right), \operatorname{cosech}^{-1} x = \ln\left(\frac{1}{x}\right)$$

So,

$$\begin{aligned}
 \sin^{-1} x + \operatorname{cosech}^{-1} x &= \ln \left(x + \sqrt{x^2 + 1} \right) + \ln \left(\frac{1}{x} + \sqrt{\frac{1}{x^2} + 1} \right) \\
 &= \ln \left(-\frac{1}{2} + \sqrt{\frac{1}{4} + 1} \right) + \ln \left(-2 + \sqrt{4 + 1} \right) \\
 &= \ln \left[\frac{\sqrt{5} - 1}{2} \times (\sqrt{5} - 2) \right] \\
 &= \ln \left(\frac{7 - 3\sqrt{5}}{2} \right)
 \end{aligned}$$

26. The given relation is

$$\frac{s-a}{11} = \frac{s-b}{12} = \frac{s-c}{13} = k$$

Using this equation

$$36 - (a + b + c) = 36k$$

$$36 - 2s = 36k$$

$$s = 36k$$

$$\tan^2 \frac{A}{2} + \tan^2 \frac{C}{2} = \frac{(s-b)(s-c)}{s(s-c)} + \frac{(s-a)(s-b)}{s(s-c)}$$

Substitute the values

$$\begin{aligned}
 &= \frac{12k \times 13k}{36k \times 11k} + \frac{11k \times 12k}{36k \times 13k} \\
 &= \frac{12}{36} \left[\frac{13}{11} + \frac{11}{13} \right] = \frac{290}{429}
 \end{aligned}$$

27. In the triangle ABC ,

$$A + B + C = \pi$$

Substitute the value in the given relation

$$\begin{aligned} &= (a-b)^2 \sin^2 \left(\frac{A+B}{2} \right) + (a+b)^2 \sin^2 \frac{C}{2} \\ &= [a^2 + b^2 - 2ab] \sin^2 \left(\frac{\pi - C}{2} \right) + [a^2 + b^2] \sin^2 \frac{C}{2} \\ &= (a^2 + b^2) \sin^2 \frac{C}{2} + 2ab \sin^2 \frac{C}{2} \\ &= a^2 + b^2 - 2ab \cos C \\ &= a^2 + b^2 - a^2 - b^2 + c^2 = c^2 \end{aligned}$$

28. The ratio of radii r_1, r_2, r_3

$$r_1 : r_2 : r_3 = 1 : 2 : 3$$

Let's assume

$$r_1 = x, r_2 = 2x, r_3 = 3x$$

$$s - a = \frac{\Delta}{r_1} = \frac{\Delta}{x}$$

$$s - b = \frac{\Delta}{r_1} = \frac{\Delta}{2x}$$

$$s - c = \frac{\Delta}{r_1} = \frac{\Delta}{3x}$$

Add the above three equations

$$3s - (a + b + c) = \frac{\Delta}{x} + \frac{\Delta}{2x} + \frac{\Delta}{3x}$$

$$s = \frac{11\Delta}{6x}$$

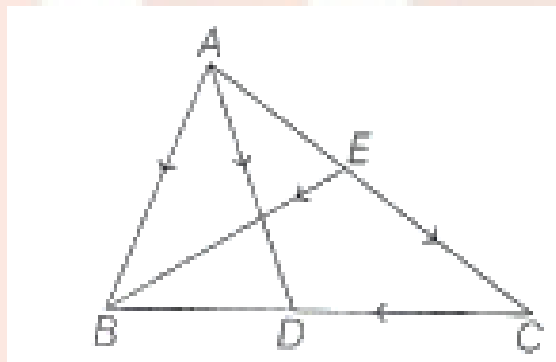
Therefore, the sides of the triangle is

$$a = \frac{5\Delta}{6x}, b = \frac{8\Delta}{6x}, c = \frac{9\Delta}{6x}$$

Therefore, the ratio of the sides of the triangle is

$$a : b : c = 5 : 8 : 9$$

29. Consider the triangle given in the figure below



$$AD = AC + CD = AC + \frac{1}{2}CB$$

$$EB = EC + CB = \frac{1}{2}AC + CB$$

Add above two equation

$$AD + EB = \frac{3}{2}AC + \frac{3}{2}CB$$

$$2(AD + EB) = 3AB$$

30. The equation of plane through line of intersection of planes

$$r \cdot n = 1 \text{ and } r \cdot m = -4 \text{ is}$$

$$r \cdot (n + \lambda m) = 1 - 4\lambda$$

Given that this plane is perpendicular to $r \cdot l = -8$, then

$$(n + \lambda m) \cdot l = 0$$

$$n \cdot l + \lambda m \cdot l = 0$$

$$(2\bar{i} - 3\bar{j} + 4\bar{k}) \cdot (2\bar{i} - \bar{j} + \bar{k}) + \lambda(\bar{i} - \bar{j}) \cdot (2\bar{i} - \bar{j} + \bar{k}) = 0$$

$$\lambda = \frac{-11}{3}$$

Therefore, the equation of the required plane is

$$r \cdot \left[2\bar{i} - 3\bar{j} + 4\bar{k} - \frac{11}{3}(\bar{i} - \bar{j}) \right] = 1 - 4\left(\frac{-11}{3}\right)$$

$$r \cdot \left(\frac{-5\bar{i}}{3} + \frac{2\bar{j}}{3} + 4\bar{k} \right) = \frac{47}{3}$$

$$(x\bar{i} + y\bar{j} + z\bar{k}) \cdot (-5\bar{i} + 2\bar{j} + 12\bar{k}) = 47$$

$$5x - 2y - 12z + 47 = 0$$

31. From the given equation,

$$\cos \alpha = \frac{\bar{a} \cdot \bar{b}}{\lambda^2}, \cos \beta = \frac{\bar{b} \cdot \bar{c}}{\lambda^2}, \cos \gamma = \frac{\bar{c} \cdot \bar{a}}{\lambda^2}$$

Then,

$$\cos \alpha + \cos \beta + \cos \gamma = \frac{1}{\lambda^2} (\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{c} + \bar{c} \cdot \bar{a})$$

Use the relation,

$$|\bar{a} + \bar{b} + \bar{c}|^2 = [|\bar{a}|^2 + |\bar{b}|^2 + |\bar{c}|^2 + 2(\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{c} + \bar{c} \cdot \bar{a})]$$

we can write

$$[\lambda^2 + \lambda^2 + \lambda^2 + 2(\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{c} + \bar{c} \cdot \bar{a})] \geq 0$$

$$2(\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{c} + \bar{c} \cdot \bar{a}) \geq -3\lambda^2$$

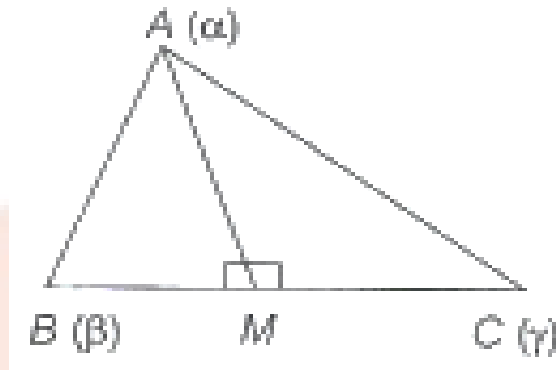
$$\bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{c} + \bar{c} \cdot \bar{a} > \frac{-3\lambda^2}{2}$$

Then

$$\cos \alpha + \cos \beta + \cos \gamma \geq \frac{1}{\lambda^2} \left(\frac{-3\lambda^2}{2} \right) = -\frac{3}{2}$$

Therefore, the minimum value of the required equation is $-\frac{3}{2}$

32. The area of the below triangle is given by the equation,



$$A = \frac{1}{2}(BC) \cdot (AM)$$

$$\frac{1}{2}|\overrightarrow{BC}||\overrightarrow{AM}| = \frac{1}{2}|\alpha \times \beta + \beta \times \gamma + \gamma \times \alpha|$$

$$|\overrightarrow{AM}| = \frac{|\alpha \times \beta + \beta \times \gamma + \gamma \times \alpha|}{|\gamma - \beta|}$$

33. Consider the solutions of the vector expression as,

(A)

$$[\mathbf{b} \times \mathbf{c} \mathbf{c} \times \mathbf{a} \mathbf{a} \times \mathbf{b}]$$

$$= (\mathbf{b} \times \mathbf{c}) \times (\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b})$$

$$\text{Let } \mathbf{d} = \mathbf{b} \times \mathbf{c}$$

$$\begin{aligned}
&= \{ \mathbf{d} \times (\mathbf{c} \times \mathbf{a}) \times (\mathbf{a} \times \mathbf{b}) \} \\
&= \{ (\mathbf{d} \cdot \mathbf{a})\mathbf{c} - (\mathbf{d} \cdot \mathbf{c})\mathbf{a} \} \cdot (\mathbf{a} \times \mathbf{b}) \\
&= \{ [\mathbf{a} \mathbf{b} \mathbf{c}]\mathbf{c} - 0 \} \cdot (\mathbf{a} \times \mathbf{b}) \\
&= [\mathbf{a} \mathbf{b} \mathbf{c}][\mathbf{c} \mathbf{b} \mathbf{a}] \\
&= [\mathbf{a} \mathbf{b} \mathbf{c}]^2
\end{aligned}$$

(B)

$$[\mathbf{a} \times \mathbf{b} \mathbf{a} \times \mathbf{c} \mathbf{b}] = \{ (\mathbf{a} \times \mathbf{b}) \times (\mathbf{a} \times \mathbf{c}) \} \cdot \mathbf{b}$$

Let $\mathbf{d} = \mathbf{a} \times \mathbf{b}$

$$\begin{aligned}
&= \{ \mathbf{d} \times (\mathbf{a} \times \mathbf{c}) \} \cdot \mathbf{b} \\
&= \{ (\mathbf{d} \cdot \mathbf{c})\mathbf{a} - (\mathbf{d} \cdot \mathbf{a})\mathbf{c} \} \cdot \mathbf{b} \\
&= \{ (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} \} \mathbf{a} \cdot \mathbf{b} - \{ (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} \} \mathbf{c} \cdot \mathbf{b} \\
&= [\mathbf{abc}][\mathbf{a} \cdot \mathbf{b}] - 0 \\
&= [\mathbf{abc}][\mathbf{a} \cdot \mathbf{b}]
\end{aligned}$$

(C)

$$\begin{aligned}
[\mathbf{a} + \mathbf{b} \mathbf{b} + \mathbf{c} \mathbf{c} + \mathbf{a}] &= (\mathbf{a} + \mathbf{b}) \{ (\mathbf{b} + \mathbf{c}) \times (\mathbf{c} + \mathbf{a}) \} \\
&= (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{b} \times \mathbf{c} + \mathbf{b} \times \mathbf{a} + \mathbf{c} \times \mathbf{c} + \mathbf{c} \times \mathbf{a}) \\
&= [\mathbf{abc}] + [\mathbf{bca}] \\
&= 2[\mathbf{abc}]
\end{aligned}$$

(D)

 $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are three mutually perpendicular unit vectors

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{c} = \mathbf{c} \cdot \mathbf{a} = 0$$

$$\begin{aligned}
 [(\mathbf{a} + \mathbf{b} + \mathbf{c}) \mathbf{b} \times \mathbf{c} \mathbf{c} \times \mathbf{a}] &= (\mathbf{a} + \mathbf{b} + \mathbf{c}) \cdot (\mathbf{b} \times \mathbf{c})(\mathbf{c} \times \mathbf{a}) \\
 &= [\mathbf{cab}] \\
 &= [\mathbf{abc}]
 \end{aligned}$$

Thus, option (4) is correct.

$$34. \bar{b} = 2\{\bar{a} \times (\bar{b} \times \bar{c})\}$$

$$\bar{b} = 2\{(\bar{a} \cdot \bar{c})\bar{b} - (\bar{a} \cdot \bar{b})\bar{c}\}$$

Compare the coefficient of $\bar{b}$ and $\bar{c}$

$$2(\bar{a} \cdot \bar{c}) = 1 \text{ and } \bar{a} \cdot \bar{b} = 0$$

$$|\bar{a}||\bar{c}|\cos\alpha = \frac{1}{2}, \bar{a} \cdot \bar{b} = 0$$

$$\cos\alpha = \cos\frac{\pi}{3}, \cos\beta = \cos\frac{\pi}{2}$$

Therefore,

$$\cos(\alpha + \beta) = -\sin\frac{\pi}{3} = -\frac{\sqrt{3}}{2}$$

35. The variance is proportional to the square of the data. So, when the data is doubled the variance will be four times.

Therefore, the new variance is $= 4\sigma^2$

36. By using the given data

CI	f_i	Mid-Value x_i	$f_i x_i$
75–175	3	125	375
175–275	2	225	450
275–375	1	325	325
375–475	0	425	0
475–575	1	525	525
575–675	2	652	1250
675–775	4	725	2175
$\Sigma f_i = 12$		$\Sigma f_i x_i = 5100$	

The mean of the data is

$$\bar{X} = \frac{\Sigma f_i x_i}{\Sigma f_i} = \frac{5100}{12} = 425$$

The coefficient of variation is

$$\begin{aligned}
 CV &= \frac{\sigma}{\bar{X}} \times 100 \\
 &= \frac{100\sqrt{6}}{425} \times 100 = \frac{400\sqrt{6}}{17}
 \end{aligned}$$

37. The ways to choose 2 numbers from 1 to 39 = ${}^{39}C_2$

Favourable cases are

$$(a, b) = (9, 7), (18, 14), (27, 21), (36, 28)$$

Therefore, the required probability = $\frac{4}{{}^{39}C_2} = \frac{4}{741}$

38. (I) First ball is white and second is non-white

$$P_1 = \frac{7}{21} \times \frac{14}{20}$$

(II) First is non-white and second is white

$$P_2 = \frac{14}{21} \times \frac{7}{20}$$

(III) Both balls are white

$$P_3 = \frac{7}{21} \times \frac{6}{20}$$

Therefore, the required probability is

$$\begin{aligned} P &= \frac{7}{21} \times \frac{14}{20} + \frac{14}{21} \times \frac{7}{20} + \frac{7}{21} \times \frac{6}{20} \\ &= \frac{17}{30} \end{aligned}$$

39. Let events that production is done by machine A is E_1 , the production done by machine B is E_2 , the production done by machine C is E_3 . The event that product being defective is E .

The probabilities are

$$P(E_1) = \frac{20}{100}, P(E_2) = \frac{30}{100}, P(E_3) = \frac{50}{100}$$

Probability,

Machine A has defective production $P\left(\frac{E}{E_1}\right) = \frac{5}{100}$

Machine B has defective production $P\left(\frac{E}{E_2}\right) = \frac{3}{100}$

Machine C has defective production $P\left(\frac{E}{E_3}\right) = \frac{2}{100}$

An item being selected and found defective, then the probability that it is produced by machine B is

$$\begin{aligned} P\left(\frac{E_2}{E}\right) &= \frac{P\left(\frac{E_2}{E}\right) \times P(E_2)}{P\left(\frac{E_1}{E}\right) \times P(E_1) + P\left(\frac{E_2}{E}\right) \times P(E_2) + P\left(\frac{E_3}{E}\right) \times P(E_3)} \\ &= \frac{\frac{30}{100} \times \frac{3}{100}}{\frac{20}{100} \times \frac{5}{100} + \frac{30}{100} \times \frac{3}{100} + \frac{50}{100} \times \frac{2}{100}} = \frac{9}{29} \end{aligned}$$

40. Variance of the distribution is given by

$$\text{Variance}(x) = E(x^2) - [E(x)^2]$$

Then,

$$\begin{aligned}
 E(x) &= \sum P_i x_i \\
 &= \frac{1}{6} \times (-2) + \frac{1}{6}(-1) + \frac{1}{3} \times 0 + \frac{1}{6}(1) + \frac{1}{6}(2) = 0
 \end{aligned}$$

And,

$$\begin{aligned}
 E(x^2) &= \sum P_i x_i^2 \\
 &= \frac{1}{6}(-2)^2 + \frac{1}{6}(-1)^2 + \frac{1}{3}(0)^2 + \frac{1}{6}(0)^2 + \frac{1}{6}(1)^2 + \frac{1}{6}(2)^2 \\
 &= \frac{5}{3}
 \end{aligned}$$

Substitute the values in the expression of variance

$$\text{Variance}(x) = \frac{5}{3} - (0)^2 = \frac{5}{3}$$

41. Probability to hit the target is $= \frac{1}{3}$

Probability to not hit the target is $= \frac{2}{3}$

The probability to hit the target atleast twice is

$$\begin{aligned}
 P &= 1 - [P(0) + P(1)] \\
 &= 1 - \left[\left(\frac{2}{3}\right)^8 + {}^8C_1 \left(\frac{2}{3}\right)^7 \cdot \frac{1}{3} \right] \\
 &= 1 - \left(\frac{2}{3}\right)^7 \cdot \frac{10}{3} = 1 - 5 \left(\frac{2}{3}\right)^8
 \end{aligned}$$

42. Solve the lines $y = mx$ and $x - y + 10 = 0$ for coordinate of point A and B.

$$A\left(\frac{10}{m-1}, \frac{10m}{m-1}\right), B\left(\frac{20}{m-1}, \frac{20m}{m-1}\right)$$

$$\text{The distance } OA = \sqrt{\frac{100}{(m-1)^2} + \frac{100m^2}{(m-1)^2}} = \frac{10\sqrt{1+m^2}}{(m-1)}$$

$$\text{The distance } OB = \sqrt{\frac{400}{(m-1)^2} + \frac{400m^2}{(m-1)^2}} = \frac{20\sqrt{1+m^2}}{(m-1)}$$

Let a point $P(h, k)$ lies on the line $y = mx$

$$k = mh \Rightarrow m = \frac{k}{h}$$

$$\text{The distance } OP = \sqrt{h^2 + k^2}$$

Now, OA, OP, OB are in HP.

Therefore,

$$\frac{1}{OA}, \frac{1}{OP}, \frac{1}{OB} \text{ in AP.}$$

$$\frac{1}{OP} = \frac{1}{OA} + \frac{1}{OB}$$

$$\frac{2}{\sqrt{h^2 + k^2}} = \frac{m-1}{10\sqrt{1+m^2}} + \frac{m-1}{20\sqrt{1+m^2}}$$

$$\frac{2}{\sqrt{h^2 + k^2}} = \frac{m-1}{10\sqrt{1+m^2}} \left[1 + \frac{1}{2} \right]$$

$$\frac{2}{\sqrt{h^2 + k^2}} = \frac{\left(\frac{k}{h} - 1 \right)}{10\sqrt{1 + \left(\frac{k}{h} \right)^2}} \times \frac{3}{2}$$

$$\frac{2}{\sqrt{h^2 + k^2}} = \frac{3(k-h)}{20\sqrt{h^2 + k^2}}$$

$$3(k-h) = 40$$

Therefore, the locus of the point is

$$3x - 3y + 40 = 0$$

43. When the axes are rotated through 45° about the origin then,

$$(x, y) = \left(\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}, -\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}} \right)$$

Substitute the values in the equation of the curve

$$\left[17\left(\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right)^2 - 16\left(\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right)\left(\frac{-x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right) + 17\left(\frac{-x}{\sqrt{2}} + \frac{y}{\sqrt{2}}\right)^2 \right] = 225$$

$$17\frac{x^2}{2} + 17\frac{y^2}{2} + 17xy - 8y^2 + 8x^2 + 17\frac{x^2}{2} + 17\frac{y^2}{2} - 17xy = 225$$

$$17x^2 + 17y^2 - 8y^2 + 8x^2 = 225$$

$$25x^2 + 9y^2 = 225$$

44. Point of intersection of $x + y = 1$ and $2x + 3y = 6$ is $A(-3, 4)$

Therefore, equation of the line passing through

$A(-3, 4)$ and $O\left(\frac{3}{7}, \frac{22}{7}\right)$ is

$$y - 4 = \frac{\frac{22}{7} - 4}{\frac{3}{7} + 3}(x + 3)$$

$$x + 4y = 13$$

Normal form of the line is

$$x \cos \alpha + y \sin \alpha = P$$

On comparing,

$$P = \frac{13}{\sqrt{17}}, \cos \alpha = \frac{1}{\sqrt{17}}, \sin \alpha = \frac{4}{\sqrt{17}}$$

$$\tan \alpha = \frac{4}{1}$$

Therefore, the required equation is

$$x \cos \alpha + y \sin \alpha = \frac{13}{\sqrt{17}}$$

$$\text{and } \alpha = \tan^{-1}(4)$$

45. Equation of line passing through the point of intersection of L_1 and L_2 is

$$L_1 + \lambda L_2 = 0$$

$$(2x + 3y + 6) + \lambda(3x - y - 13) = 0$$

$$(2 + 3\lambda)x + (3 - \lambda)y + 6 - 13\lambda = 0$$

This is parallel to the line $3x - 4y + 5 = 0$

On comparing the coefficient of two equations,

$$\frac{2 + 3\lambda}{3} = \frac{3 - \lambda}{-4} = \frac{6 - 13\lambda}{5}$$

$$-8 - 12\lambda = 9 - 3\lambda$$

$$\lambda = \frac{17}{9}$$

Substitute the value of the λ .

$$\left(2 - \frac{51}{9}\right)x + \left(3 + \frac{17}{9}\right)y + 6 + \frac{221}{9} = 0$$

$$-33x + 44y + 275 = 0$$

$$3x - 4y - 25 = 0$$

46. The vertices of the triangle are $A(2, 3)$, $B(2, 4)$, $C(4, 3)$

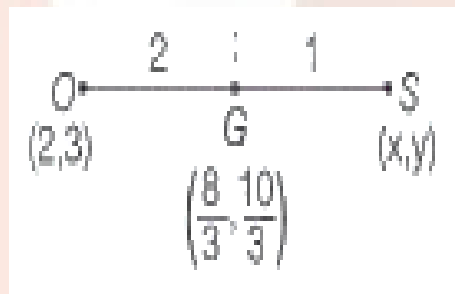
The length of the sides

$$AB = 1, BC = \sqrt{5}, CA = 2$$

Coordinate of the orthocentre is

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right) = \left(\frac{8}{3}, \frac{10}{3}\right)$$

It is given that the centroid of the triangle divides the line joining point O and S in ratio 2 : 1.



$$\frac{8}{3} = \frac{2x + 2}{3} \Rightarrow x = 3$$

$$\frac{10}{3} = \frac{2y + 3}{3} \Rightarrow y = \frac{7}{2}$$

The point $S\left(3, \frac{7}{2}\right)$.

Therefore, the value of

$$= AO^2 + 9BG^2 + 4CS^2$$

$$= 0 + 9 \times \left[\left(2 - \frac{8}{3} \right)^2 + \left(4 - \frac{10}{3} \right)^2 \right] + 4 \left[(4-3)^2 + \left(3 - \frac{7}{2} \right)^2 \right]$$

$$= 13$$

47. Given equation of pair is

$$3x^2 - 5xy + 2y^2 = 0$$

$$(3x - 2y)(x - y) = 0$$

The equation of the two sides of the triangle is

$$3x - 2y = 0$$

$$x - y = 0$$

The line $2x + 3y + k = 0$ passes through the point $(2, 1)$.

$$4 + 3 + k = 0$$

$$k = -7$$

Therefore,

$$2x + 3y - 7 = 0$$

The point of intersection of line $y - x = 0$ and $2x + 3y - 7 = 0$ is

$$x = \frac{7}{5}, y = \frac{7}{5}$$

The line $x + y + k = 0$ passes through the point $(2, 1)$.

$$2 + 1 + k = 0$$

$$k = -3$$

The point of intersection of line $x + y - 3$ and $3x - 2y = 0$ is

$$x = \frac{6}{5}, y = \frac{9}{5}$$

The equation of the third side of the triangle is

$$y - \frac{9}{5} = \frac{\frac{9}{5} - \frac{7}{5}}{\frac{6}{5} - \frac{7}{5}} \left(x - \frac{6}{5} \right)$$

$$y - \frac{9}{5} = \frac{2}{-1} \left(x - \frac{6}{5} \right)$$

$$\frac{5y - 9}{5} = \frac{-10x + 12}{5}$$

$$10x + 5y = 21$$

48. The equation of the common chord is given by

$$S_1 - S_2 = 0$$

$$(x^2 + y^2 - 4x + 8y + 5) - (x^2 + y^2 + 2x + 4y - 3) = 0$$

$$3x - 2y = 4$$

$$\frac{x}{4/3} + \frac{y}{-2} = 1$$

Homogenization of equation

$$x^2 + y^2 - 4x\left(\frac{x}{4/3} + \frac{y}{-2}\right) + 8y\left(\frac{x}{4/3} + \frac{y}{-2}\right) + 5\left(\frac{x}{4/3} + \frac{y}{-2}\right)^2 = 0$$

$$x^2 + y^2 - 3x^2 + 8xy - 4y^2 + 5\left(\frac{9}{16}x^2 + \frac{y^2}{4} - \frac{3xy}{4}\right) = 0$$

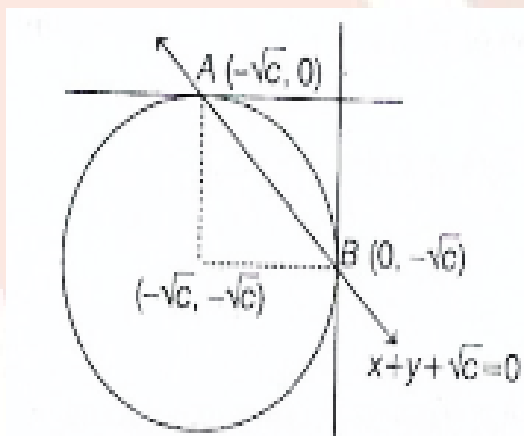
$$13x^2 - 28y^2 + 68xy = 0$$

49. The circle's equation is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Coordinate of centre = $(-g, -f)$

$$\text{radius} = \sqrt{g^2 + f^2 - c}$$



Since circle touches both the axes

$$g^2 = f^2$$

$$g = \pm\sqrt{c}$$

Therefore, the coordinate of the centre $(-\sqrt{c}, -\sqrt{c})$.

The equation of the given line is

$$x + y + \sqrt{c} = 0$$

Therefore, intercept

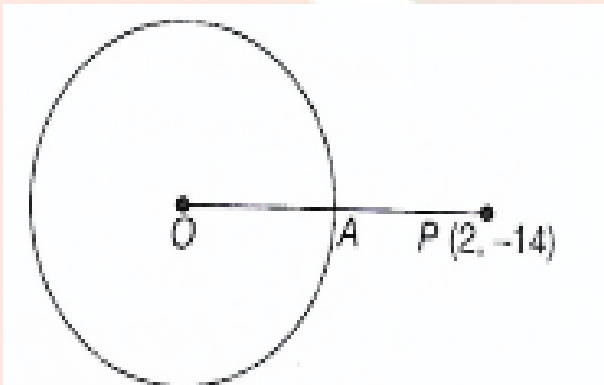
$$AB = \sqrt{(-\sqrt{c} - 0)^2 + (0 + \sqrt{c})^2} = \sqrt{2c}$$

50. The given equation of the circle is

$$x^2 + y^2 + 6x + 4y - 12 = 0$$

The centre $(-3, -2)$

$$\text{The radius} = \sqrt{9 + 4 + 12} = 5$$



The shortest distance between point P and the circle is

$$AP = OP - OA$$

$$\begin{aligned} AP &= \sqrt{(-3 - 2)^2 + (-2 + 14)^2} - 5 \\ &= 13 \end{aligned}$$

$$AP = 8$$

Length of tangent from point P is

$$= \sqrt{2^2 + (-14)^2 + 6 \cdot 2 + 4(-14) - 12}$$

$$= 12$$

The required length is $= \sqrt{8+12} = 2\sqrt{5}$

51. Given circle's equation is

$$x^2 + y^2 - 4x - 2y + k = 0$$

Centre $c_1(2,1)$ and $r_1 = 2$

For $x^2 + y^2 - 6x - 4y + l = 0$

Centre $c_2(3,2)$ and $r_2 = 3$

$$c_1c_2 = \sqrt{(2-3)^2 + (1-2)^2}$$

$$= \sqrt{2}$$

Since $c_1c_2 < r_1 + r_2$

Therefore, the given circle have two common tangents.

52. Let the equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Centre is $(-g, -f)$.

Since the circle passes through the point $(0,2)$.

$$4 + 0 + 4g + 0 + c = 0$$

$$4 + 4g + c = 0$$

And circle also touches the X-axis.

$$g^2 = c$$

Substitute the values

$$4 + 4g + g^2 = 0$$

$$(g + 2)^2 = 0$$

$$g = -2$$

$$\text{and } c = 4$$

This circle also cuts the circle $x^2 + y^2 - 2x - 2y - 2 = 0$ orthogonally

$$2g_1g_2 + 2f_1f_2 = c_1 + c_2$$

$$2(-2)(-1) + 2f(-1) = 4 - 2$$

$$f = 1$$

Hence the coordinate of the centre of the circle is $(2, -1)$.

53. Given equations of circle are

$$c_1 : x^2 + y^2 + 2gx + 2fy + c = 0$$

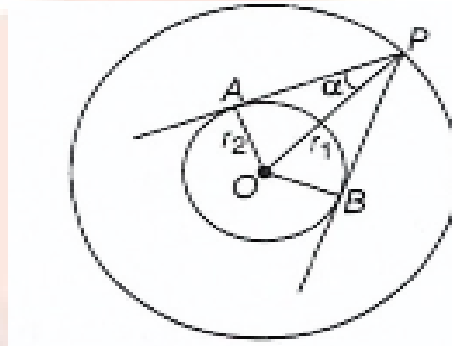
The centre is $O(-g, -f)$, $r_1 = \sqrt{g^2 + f^2 - c}$

$$c_2 : x^2 + y^2 + 2gx + 2fy + c \sin^2 \alpha + (g^2 + f^2) \cos^2 \alpha = 0$$

Centre $O(-g, -f)$

$$\begin{aligned}
 r_2 &= \sqrt{g^2 + f^2 - c \sin^2 \alpha - g^2 \cos^2 \alpha - f^2 \cos^2 \alpha} \\
 &= \sqrt{g^2 + f^2 - c} \times \sin \alpha
 \end{aligned}$$

Since, the two circles are concentric as shown in figure.



$$r_2 = r_1 \sin \alpha$$

$$\frac{r_2}{r_1} = \sin \alpha$$

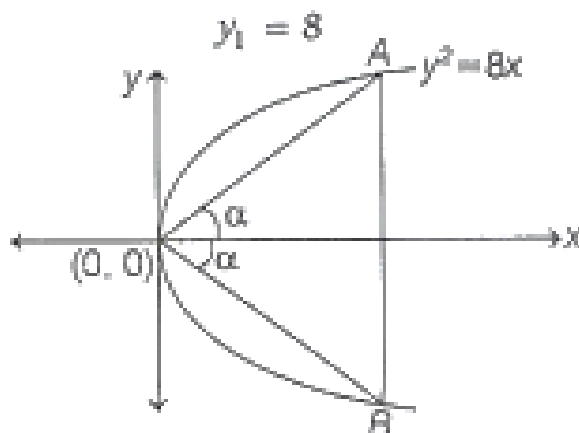
$$\sin(\angle APO) = \sin \alpha$$

$$\angle APO = \alpha$$

Therefore,

$$\angle APB = 2\angle APO = 2\alpha$$

54. The parabola is represented in the figure below



The coordinate of A is $(x_1, 8)$ and the coordinate of B is $(x_1, -8)$.

Put these coordinates in the equation of parabola.

$$x_1 = 8$$

$$\tan \alpha = \frac{8}{8} = 1$$

$$\alpha = \frac{\pi}{4}$$

The angle subtended at the vertex $= 2\alpha = \frac{\pi}{2}$

55. The intersection point of $y^2 = 32x$ and $2x^2 = 27y$.

The coordinate of point $P(18, 24)$.

Equation of tangent through the point $P(18, 24)$ on the parabola $y^2 = 32x$

$$y \cdot 24 = 16(x + 18)$$

$$3y = 2x + 36$$

The slope $m_1 = \frac{2}{3}$

The equation of tangent on $2x^2 = 27y$

$$2x \cdot 18 = 27 \frac{(y + 24)}{2}$$

$$3y = 8x - 72$$

Slope $m_2 = \frac{8}{3}$

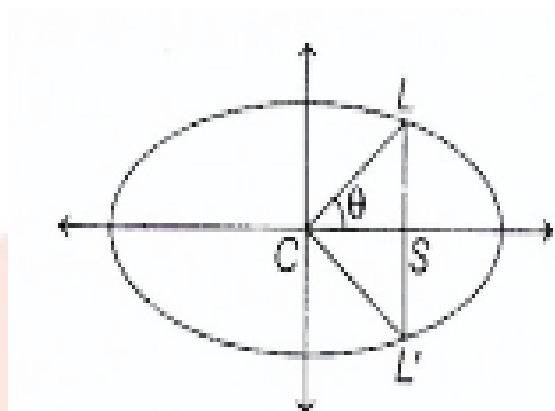
Angle between the tangents is

$$\begin{aligned}\tan \theta &= \frac{m_2 - m_1}{1 + m_1 m_2} \\ &= \frac{\frac{8}{3} - \frac{2}{3}}{1 + \frac{8}{3} \times \frac{2}{3}} = \frac{18}{25}\end{aligned}$$

The required value

$$5\sqrt{\tan \theta} = 5 \times \sqrt{\frac{18}{25}} = 3\sqrt{2}$$

56. The ellipse is shown in the figure below



The value of $\angle LCS = 45^\circ$

$$\tan 45^\circ = \frac{b^2}{ae}$$

$$1 = \frac{b^2}{a^2 e}$$

$$a^2 e = b^2 = a^2 (1 - e^2)$$

$$e^2 + e - 1 = 0$$

$$e = \frac{-1 \pm \sqrt{5}}{2}$$

The possible value of e is

$$= \frac{\sqrt{5} - 1}{2}$$

57. Equation of tangent at point $(1, 2)$ at the ellipse $3x^2 + 4y^2 = 19$ is

$$3x(1) + 4y(2) = 19$$

$$3x + 8y = 19$$

This line is also tangent to the parabola

$$y^2 - kx = 0$$

Substitute the value in the above equation.

$$y^2 - k\left(\frac{19 - 8y}{3}\right) = 0$$

$$3y^2 + 8ky - 19k = 0$$

The equation has equal roots.

$$(8k)^2 - 4(3)(-19k) = 0$$

$$64k^2 = -(12 \times 19)k$$

$$k = \frac{-57}{16}$$

58. The given equation of parabola is

$$14x^2 + 38xy + 20y^2 + x - 7y - 91 = 0$$

$$\text{Factorize } 14x^2 + 38xy + 20y^2 = 0$$

$$(7x + 5y)(2x + 4y) = 0$$

The first asymptote is $7x + 5y - 3 = 0$

The second asymptote is $2x + 4y + k = 0$

The combine equation of the asymptote is

$$(7x + 5y - 3)(2x + 4y + k) = 0$$

Compare the coefficient of x .

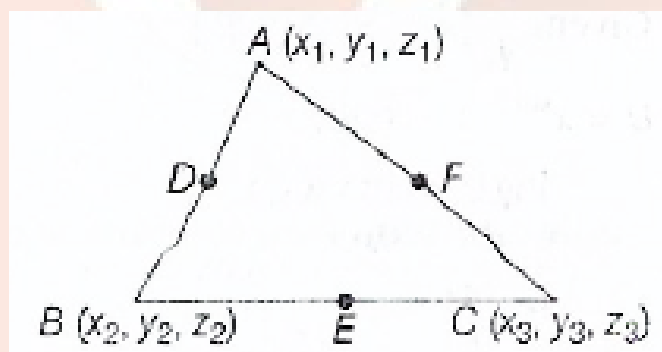
$$7k - 6 = 1$$

$$k = 1$$

Therefore, the equation of the above asymptote is $2x + 4y + 1 = 0$.

59. The coordinates of the mid-points are

$$D(1, 5, -1), E(0, 4, -2), F(2, 3, 4)$$



The relation between the coordinates of vertex of triangle and the mid points is

$$\frac{x_1 + x_2}{2} = 1, \frac{y_1 + y_2}{2} = 5, \frac{z_1 + z_2}{2} = -1$$

$$x_1 + x_2 = 2, y_1 + y_2 = 10, z_1 + z_2 = -2$$

For other two sides,

$$x_2 + x_3 = 0, y_2 + y_3 = 8, z_2 + z_3 = -4$$

$$x_1 + x_3 = 4, y_1 + y_3 = 6, z_1 + z_3 = 8$$

By using the above relations

$$2(x_1 + x_2 + x_3) = 6$$

$$2(y_1 + y_2 + y_3) = 24$$

$$2(z_1 + z_2 + z_3) = 2$$

$$x_3 = 1, y_3 = 2, z_3 = 3$$

The length of the median CD is

$$CD = \sqrt{(1-1)^2 + (2-5)^2 + (-1-3)^2} = 5$$

60. Direction cosine of the line which makes angle α with the x-axis

$$l = \cos \alpha, m = \cos \frac{\pi}{4}, n = \cos \frac{\pi}{3}$$

$$l^2 + m^2 + n^2 = 1$$

Substitute values in above expression.

$$l^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 = 1$$

$$l^2 = \frac{1}{4}$$

$$l = \pm \frac{1}{2}$$

$$\cos \alpha = -\frac{1}{2}$$

$$\alpha = \frac{2\pi}{3}$$

61. Mid-point of the line joining the points $(2, 0, 6)$ and $(-6, 2, 4)$ is $(-2, 1, 5)$.

Direction ratio of the line segment joining the points $(2, 0, 6)$ and $(-6, 2, 4)$

$$= (-6 - 2), (2 - 0), (4 - 6)$$

$$= -8, 2, -2 = a_1, b_1, c_1$$

Direction ratio of line joining (x, y, z) and $(-2, 1, 5)$ are

$$= (x + 2), (y - 1), (z - 5)$$

The relation between direction ratio is

$$a_1a_2 + b_1b_2 + c_1c_2 = 0$$

$$-8(x+2) + 2(y-1) - 2(z-5) = 0$$

$$-8x + 2y - 2z - 8 = 0$$

$$4x - y + z + 4 = 0$$

62. Given,

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x\sin x} - \sqrt{\cos x}}{\tan^2 2x}$$

Solve the above expression,

$$= \lim_{x \rightarrow 0} \frac{\sqrt{1+x\sin x} - \sqrt{\cos x}}{\tan^2 2x} \times \frac{\sqrt{1+x\sin x} + \sqrt{\cos x}}{\sqrt{1+x\sin x} + \sqrt{\cos x}}$$

$$= \lim_{x \rightarrow 0} \frac{1+x\sin x - \cos x}{\tan^2 2x} \times \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+x\sin x} + \sqrt{\cos x}}$$

$$= \lim_{x \rightarrow 0} \frac{(1-\cos x) + x\sin x}{\tan^2 2x} \times \frac{1}{2}$$

Solve further,

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} + 2x \sin \frac{x}{2} \cos \frac{x}{2}}{\tan^2 2x} \times \frac{1}{2}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2} \left[1 + \frac{x}{\tan \frac{x}{2}} \right]}{\tan^2 2x} \times \frac{1}{2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{4} \left(\frac{\sin \frac{x}{2}}{\frac{x}{2}} \right) \left[1 + \frac{2 \frac{x}{2}}{\tan \frac{x}{2}} \right]}{4 \left[\frac{\tan 2x}{2x} \right]^2}$$

Apply limits,

$$= \frac{\frac{1}{4} \times (1)^2 [1 + 2 \times 1]}{4(1)^2} = \frac{3}{16}$$

63. Since $f(x)$ is continuous at R ,

$$\text{at } x = 0 \quad \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$0 + \alpha = \cos 0 = 1$$

$$\alpha = 1$$

$$\text{at } x = 2 \quad \lim_{x \rightarrow 2^-} f(x) = f(2)$$

$$6 + \alpha = 2\beta + 3$$

$$7 = 2\beta + 3$$

$$\beta = 2$$

Therefore, the required expression

$$\begin{aligned}\alpha^2 + \beta^2 &= 1 + 4 \\ &= 5\end{aligned}$$

$$64. \text{ Let } g(x) = f(f(x)) = \begin{cases} f(1+x) & 0 \leq x \leq 2 \\ f(3-x) & 2 \leq x \leq 3 \end{cases}$$

So,

$$g(x) = \begin{cases} f(1+x) & ; \text{ for } 0 \leq x \leq 1 \Rightarrow 1 \leq x+1 \leq 2 \\ f(1+x) & ; \text{ for } 1 \leq x \leq 2 \Rightarrow 2 \leq x+1 \leq 3 \\ f(3-x) & ; \text{ for } 2 \leq x \leq 3 \Rightarrow 0 \leq 3-x \leq 1 \end{cases}$$

If $(1+x) \in [1, 2]$ then,

$$f(1+x) = 1 + (1+x) = 2+x$$

Similarly If $((1+x) \in (2, 3))$ then,

$$f(1+x) = 1 + (1+x) = 2+x$$

Similarly, if $(3-x) \in (0, 1)$ then,

$$f(3-x) = 1 + (3-x) = 4-x$$

$$g(x) = \begin{cases} 2+x & ; \quad 0 \leq x < 1 \\ 2-x & ; \quad 1 < x \leq 2 \\ 4-x & ; \quad 2 < x \leq 3 \end{cases}$$

At $x = 1$

$$\text{LHL} = \lim_{x \rightarrow 1^-} g(x) = \lim_{x \rightarrow 1^-} (2 + x) = 3$$

$$\text{RHL} = \lim_{x \rightarrow 1^+} g(x) = \lim_{x \rightarrow 1^+} (2 - x) = 1$$

Here, $g(x)$ is not continuous at $x = 1$.

At $x = 2$

$$\text{LHL} = \lim_{x \rightarrow 2^-} g(x) = \lim_{x \rightarrow 2^-} (2 - x) = 0$$

$$\text{RHL} = \lim_{x \rightarrow 2^+} g(x) = \lim_{x \rightarrow 2^+} (4 - x) = 2$$

Here, $g(x)$ is not continuous at $x = 2$.

Therefore, the function $g(x)$ is continuous for all $x \in [0, 1) \cup (1, 2)$.

$$65. \text{ Let } x = 4, y = 0 \Rightarrow f(4 + 0) = f(4) \cdot f(0)$$

$$f(4) = f(4) \cdot f(0) \Rightarrow f(0) = 1$$

$$\begin{aligned} f'(4) &= \lim_{h \rightarrow 0} \frac{f(4 + h) - f(4)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(4)f(h) - f(4)}{h} = \lim_{h \rightarrow 0} \left(\frac{f(h) - 1}{h} \right) f(4) \end{aligned}$$

$$f'(4) = f(4) \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

$$f'(4) = f(4) \cdot f'(0)$$

$$= f(4) \cdot f'(0)$$

$$f(4) = \frac{24}{3}$$

$$= 8$$

66. Let $U = x^{\sin x}$ and $V = (\sin x)^x$

$$\log U = \sin x \log x$$

$$\frac{1}{U} \frac{dU}{dx} = \frac{\sin x}{x} + \cos x \log x$$

$$\frac{dU}{dx} = x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right]$$

Similarly,

$$\log V = x \log \sin x$$

$$\frac{1}{V} \frac{dV}{dx} = \log \sin x + x \cot x$$

$$\frac{dV}{dx} = (\sin x)^x [\log \sin x + x \cot x]$$

Therefore,

$$\begin{aligned} \frac{d}{dx} \left[x^{\sin x} + (\sin x)^x \right] &= \frac{dU}{dx} + \frac{dV}{dx} \\ &= x^{\sin x} \left[\frac{\sin x}{x} + \cos x \log x \right] + (\sin x)^x \left[\log \sin x + x \cot x \right] \end{aligned}$$

Thus, no option is correct.

67. The given parametric function are

$$x = a(t + \sin t) \text{ and } y = a(1 - \cos t)$$

$$\frac{dx}{dt} = a(1 + \cos t), \frac{dy}{dt} = a \sin t$$

$$\begin{aligned} \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{a \sin t}{a(1 + \cos t)} \\ &= \tan \frac{t}{2} \end{aligned}$$

$$\begin{aligned} \frac{d^2y}{dx^2} &= \sec^2 \left(\frac{t}{2} \right) \times \frac{1}{2} \times \frac{dt}{dx} \\ &= \frac{\sec^2 \frac{t}{2}}{2a(1 + \cos t)} \\ &= \frac{1}{4a \cos^4 \left(\frac{t}{2} \right)} \end{aligned}$$

68. Find point of intersection $x^2 + 4y = 0$ and $xy = 2$

$$x = -2$$

Slope of the first curve at $x = -2$

$$2x + 4 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{2} = 1$$

Slope of the second curve $xy = 2$

$$y = \frac{2}{x}$$

$$\frac{dy}{dx} = \frac{-2}{x^2} = \frac{-1}{2}$$

The angle between the curves is

$$\begin{aligned} \tan \theta &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{1 - \left(\frac{-1}{2} \right)}{1 + (1) \left(\frac{-1}{2} \right)} \right| \\ &= 3 \end{aligned}$$

69. The given function $f(x) = x^3 + 2px^2 + 27x + 16$

$$f'(x) = 3x^2 + 4px + 27$$

For $f(x)$ to be strictly increasing

$$f'(x) > 0$$

$$3x^2 + 4px + 27 > 0$$

For the above statement to be true

$$D < 0$$

$$(4p)^2 - 4(3)(27) < 0$$

$$P \in \left(-\frac{9}{2}, \frac{9}{2} \right)$$

70. Given,

$$f(x) = x^{1/2}$$

So,

$$f'(x) = \frac{1}{2}x^{-1/2} \Rightarrow f'(c) = \frac{1}{2}c^{-1/2}$$

$$g(x) = x^{-1/2}$$

$$g'(x) = \frac{-1}{2}x^{-3/2} \Rightarrow g'(c) = \frac{-1}{2}c^{-3/2}$$

$$\frac{f'(c)}{g'(c)} = \frac{f(12) - f(3)}{g(12) - g(3)}$$

$$\frac{\frac{1}{2}c^{-1/2}}{\frac{-1}{2}c^{-3/2}} = \frac{\sqrt{12} - \sqrt{3}}{\frac{1}{\sqrt{12}} - \frac{1}{\sqrt{3}}}$$

$$c = 6$$

71. The displacement is given as

$$S = t^5 - 40t^3 + 80t - 25$$

$$V = \frac{dS}{dt} = 5t^4 - 120t^2 + 80$$

$$a = \frac{d^2S}{dt^2} = 20t^3 - 240t + 60$$

Let,

$$a = f(t) = 20t^3 - 240t + 60$$

$$f'(t) = 60t^2 - 240$$

$$\text{Put } f'(t) = 0$$

$$60t^2 = 240$$

$$t = \pm 2$$

$$\text{At } t = 2 \quad f'(t) > 0$$

Therefore, the minimum value of acceleration is

$$\begin{aligned} a_{\min} &= f(2) = 20(2)^3 - 240(2) + 60 \\ &= -260 \end{aligned}$$

72. As,

$$g\left(\frac{t+1}{2t+1}\right) = t+1$$

Consider

$$\frac{t+1}{2t+1} = x$$

$$t = \frac{x-1}{1-2x}$$

Therefore,

$$\begin{aligned} g(x) &= \frac{x-1}{1-2x} + 1 \\ &= \frac{-x}{1-2x} \end{aligned}$$

Perform integration for $g(x)$.

$$\begin{aligned} \int g(x) dx &= \int \frac{-x}{1-2x} dx \\ &= \frac{1}{2} \int 1 dx - \frac{1}{2} \int \frac{1}{1-2x} dx \\ &= \frac{x}{2} + \frac{1}{4} \log_e |2x-1| + c \end{aligned}$$

73. The given integral is

$$\begin{aligned} I &= \int \frac{x^8 - 9x^2 + 18}{x^4 - 3x^2 + 3} dx \\ &= \int (x^4 + 3x^2 + 6) dx \\ &= \frac{x^5}{5} + \frac{3x^3}{3} + 6x + c \\ &= \frac{x^5}{5} + x^3 + 6x + c \end{aligned}$$

$$\begin{aligned} 74. \quad I &= \int \frac{x^4 + 1}{1 + x^6} dx = \int \frac{x^4 - x^2 + 1 + x^2}{1 + x^6} dx \\ &= \int \frac{x^4 - x^2 + 1}{1 + x^6} dx + \int \frac{x^2}{1 + x^6} dx \end{aligned}$$

Let,

$$I = I_1 + I_2$$

$$\begin{aligned} I_1 &= \int \frac{x^4 - x^2 + 1}{1 + (x^2)^3} dx \\ &= \int \frac{x^4 - x^2 + 1}{(1 + x^2)(x^4 - x^2 + 1)} dx \\ &= \int \frac{1}{1 + x^2} dx = \tan^{-1} x \end{aligned}$$

$$I_2 = \int \frac{x^2}{1 + x^6} dx$$

$$\text{Let } x^3 = t \Rightarrow x^2 dx = \frac{dt}{3}$$

$$\begin{aligned} I_2 &= \frac{1}{3} \int \frac{dt}{1 + t^2} = \frac{1}{3} \tan^{-1}(t) \\ &= \frac{1}{3} \tan^{-1}(x^3) \end{aligned}$$

Therefore,

$$I = \tan^{-1} x + \frac{1}{3} \tan^{-1}(x^3) + C$$

75. $\int (\log x)^3 x^5 dx$

Use integration by parts

$$\begin{aligned} &= (\log x)^3 \int x^5 dx - \int \left[\frac{d}{dx} (\log x)^3 \int x^5 dx \right] dx \\ &= (\log x)^3 \frac{x^6}{6} - \int \frac{3(\log x)^2}{x} \frac{x^6}{6} dx \\ &= \frac{x^6 (\log x)^3}{6} - \frac{1}{2} \left[(\log x)^2 \int x^5 dx - \int \left\{ \frac{d}{dx} (\log x)^2 \int x^5 dx \right\} dx \right] \\ &= \frac{x^6 (\log x)^3}{6} - \frac{x^6}{12} (\log x)^2 + \frac{1}{6} \int (\log x) x^5 dx \end{aligned}$$

Solve further,

$$\begin{aligned} &= \frac{x^6 (\log x)^3}{6} - \frac{x^6}{12} (\log x)^2 + \frac{1}{6} \left[\log x \int x^5 dx - \int \left\{ \frac{d}{dx} (\log x) \int x^5 dx \right\} dx \right] \\ &= \frac{x^6 (\log x)^3}{6} - \frac{x^6}{12} (\log x)^2 + \frac{x^6}{36} (\log x) - \frac{1}{36} \int \left(\frac{x^6}{6} \right) + C \\ &= x^6 \left[\frac{(\log x)^3}{6} - \frac{(\log x)^2}{12} + \frac{\log x}{36} - \frac{1}{216} \right] + C \end{aligned}$$

76. Consider

$$A = \lim_{n \rightarrow \infty} \left[\left(1 + \frac{1}{n^2}\right) \left(1 + \frac{2^2}{n^2}\right) \cdots \left(1 + \frac{n^2}{n^2}\right) \right]^{\frac{1}{n}}$$

Take log on both sides

$$\log A = \lim_{n \rightarrow \infty} \frac{1}{n} \left[\log \left(1 + \frac{1}{n^2}\right) + \log \left(1 + \frac{2^2}{n^2}\right) + \cdots + \log \left(1 + \frac{n^2}{n^2}\right) \right]$$

$$\log A = \lim_{n \rightarrow \infty} \sum_{r=1}^n \log \left(1 + \frac{r^2}{n^2}\right) \frac{1}{n} = \int_0^1 \log(1+x^2) dx$$

Using integration by parts

$$\log A = \log(1+x^2) \int 1 dx - \int \frac{d}{dx} (\log(1+x^2)) \int 1 dx$$

$$= x \log(1+x^2) - \int \frac{2x}{1+x^2} x dx$$

$$= x \log(1+x^2) - 2 \int \frac{x^2}{1+x^2} dx$$

$$= \left[x \log(1+x^2) - 2x + 2 \tan^{-1}(x) \right]_0^1$$

$$\log A = \left[\log 2 - 2 + 2 \tan^{-1}(1) \right]$$

$$A = e^{\log 2 - 2 + 2\left(\frac{\pi}{4}\right)} = 2e^{\frac{\pi-4}{2}}$$

77. Let,

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos x dx}{\sqrt{1 + \cos x \sin x}}$$

Use property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$I = \int_0^{\frac{\pi}{2}} \frac{\cos\left(\frac{\pi}{2} - x\right)}{1 + \cos\left(\frac{\pi}{2} - x\right)\sin\left(\frac{\pi}{2} - x\right)} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x}{1 + \sin x \cos x} dx$$

Then,

$$\begin{aligned} 2I &= \int_0^{\frac{\pi}{2}} \frac{\cos x + \sin x}{\sqrt{1 + \cos x \sin x}} \times \frac{\sqrt{2}}{\sqrt{2}} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sqrt{2}(\cos x + \sin x)}{\sqrt{2 + 2\cos x \sin x}} dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sqrt{2}(\cos x + \sin x)}{\sqrt{3 - (\sin x - \cos x)^2}} dx \end{aligned}$$

Put $\sin x - \cos x = t \Rightarrow (\cos x + \sin x) dx = dt$

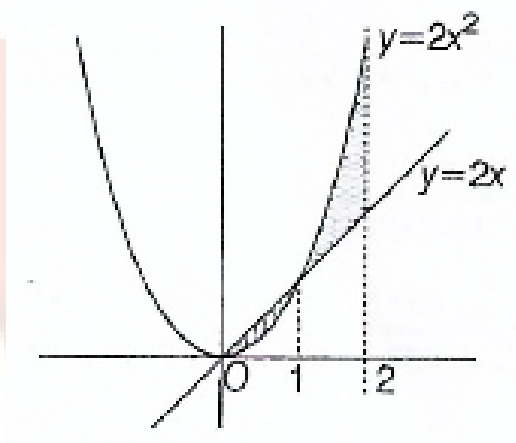
$$\begin{aligned} 2I &= \int_{-1}^1 \frac{\sqrt{2}}{\sqrt{3 - t^2}} dt \\ &= 2 \int_0^1 \frac{\sqrt{2}}{\sqrt{3 - t^2}} dt \\ I &= \sqrt{2} \left[\sin^{-1} \left(\frac{t}{\sqrt{3}} \right) \right]_{-1}^1 \\ &= \sqrt{2} \sin^{-1} \left(\frac{1}{\sqrt{3}} \right) \end{aligned}$$

78. $y = \max \{x - [x], x + |x|\}$

y can be expressed as

$$y = \begin{cases} 2x & x \geq 0 \\ \{x\} & x < 0 \end{cases}$$

The required region is represented in the figure below



Required area

$$\begin{aligned} &= \left| \int_0^1 (2x - x^2) dx \right| + \left| \int_1^2 (2x^2 - 2x) dx \right| \\ &= \left[x^2 - \frac{2x^3}{3} \right]_0^1 + \left[\frac{2x^3}{3} - x^2 \right]_1^2 \\ &= \frac{1}{3} + \frac{5}{3} = 2 \text{ sq units} \end{aligned}$$

79. The equation is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Differentiate the equation

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{y}{x} \frac{dy}{dx} = \frac{-b^2}{a^2}$$

Differentiate once again,

$$\frac{y}{x} \frac{d^2 y}{dx^2} + \frac{dy}{dx} \left[\frac{x \frac{dy}{dx} - y}{x^2} \right] = 0$$

$$xy \frac{d^2 y}{dx^2} + \frac{dy}{dx} \left[x \frac{dy}{dx} - y \right] = 0$$

$$xy \frac{d^2 y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \frac{dy}{dx} = 0$$

80. The differential equation is

$$\frac{dx}{dy} + \frac{x}{y} = x^2$$

$$\frac{1}{x^2} \frac{dx}{dy} + \left(\frac{1}{y} \right) \left(\frac{1}{x} \right) = 1$$

$$\text{Put, } \frac{1}{x} = t \Rightarrow \frac{1}{x^2} \frac{dx}{dy} = -\frac{dt}{dy}$$

$$-\frac{dt}{dy} + \frac{1}{y}(t) = 1$$

$$\frac{dt}{dy} + \left(-\frac{1}{y} \right)(t) = -1$$

The integrating factor

$$IF = e^{\int \frac{-1}{y} dy} = e^{\log \frac{1}{y}} = \frac{1}{y}$$

The solution of the equation is

$$t\left(\frac{1}{y}\right) = \int (-1)\left(\frac{1}{y}\right) dy + C$$

$$\frac{t}{y} = -\log y + C$$

$$t = -y \log y + Cy$$

$$\frac{1}{x} = cy - \log y$$

AP EAMCET-2018-Engineering-24 Apr-Shift-1

Solutions – Physics

81. The dimensional formula of Boltzmann's constant is,

$$A = \frac{J}{K} = \frac{ML^2T^{-2}}{K^1} = [ML^2T^{-2}K^{-1}]$$

The dimensional formula of Planck's constant is,

$$B = Js = [ML^2T^{-1}]$$

The dimensional formula of speed of light is,

$$C = [LT^{-1}]$$

The dimensional formula of $A^4B^{-3}C^{-2}$ is,

$$\begin{aligned}[A^4B^{-3}C^{-2}] &= [ML^2T^{-2}K^{-1}]^4 [ML^2T^{-1}]^{-3} [LT^{-1}]^{-2} \\ &= [M^1L^0T^{-3}K^{-4}]\end{aligned}$$

This is the dimensional formula of Stefan's constant.

82. Given expression is,

$$x = (2t - 3)^2$$

$$x = 4t^2 + 9 - 12t$$

The velocity is,

$$v = \frac{dx}{dt}$$

$$v = \frac{d}{dt}(4t^2 + 9 - 12t)$$

$$v = 8t - 12$$

The acceleration is,

$$a = \frac{dv}{dt}$$

$$a = \frac{d(8t - 12)}{dt}$$

$$a = 8 \text{ m/s}^2$$

83. Given, the expression of the velocity is,

$$v = x\hat{i} + yt\hat{j}$$

The magnitude of the velocity is,

$$|v| = \sqrt{x^2 + y^2 t^2}$$

The magnitude of the tangential acceleration is,

$$a_t = \frac{d|v|}{dt}$$

$$a_t = \frac{d\sqrt{x^2 + y^2 t^2}}{dt}$$

$$a_t = \frac{ty^2}{\sqrt{x^2 + y^2 t^2}}$$

The tangential acceleration at $t = \frac{x\sqrt{3}}{y}$ is,

$$\begin{aligned} a_t &= \frac{\left(\frac{x\sqrt{3}}{y}\right)y^2}{\sqrt{x^2 + y^2\left(\frac{x\sqrt{3}}{y}\right)^2}} \\ &= \frac{\sqrt{3}y}{2} \end{aligned}$$

The total acceleration is,

$$\begin{aligned} a &= \frac{dv}{dt} \\ a &= \frac{d}{dt}(x\hat{i} + yt\hat{j}) \\ a &= y\hat{j} \end{aligned}$$

The magnitude of the normal acceleration is,

$$\begin{aligned} |a_n| &= \sqrt{a^2 - a_t^2} \\ &= \sqrt{y^2 - \left(\frac{y^4 t^2}{x^2 + y^2 t^2}\right)} \\ &= \sqrt{\frac{x^2 y^2}{x^2 + y^2 t^2}} \\ &= \frac{xy}{\sqrt{x^2 + y^2 t^2}} \end{aligned}$$

The magnitude of normal acceleration at $t = \frac{x\sqrt{3}}{y}$ is,

$$|a_n| = \frac{xy}{\sqrt{x^2 + y^2} \frac{3x^2}{y^2}}$$

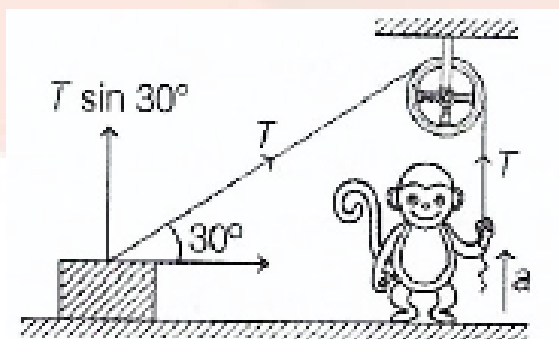
$$= \frac{y}{2}$$

84. In the uniform circular motion, the magnitude of velocity (speed) is constant.

In uniform circular motion, the direction of centripetal acceleration is always along the centre, which changes the direction of the motion of particle. It is a vector quantity which is not constant.

Therefore, assertion is correct and reason is incorrect.

85. Consider the figure,

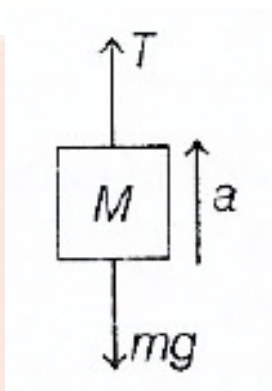


The force equilibrium equation is,

$$T \sin 30^\circ = 40$$

$$T = 80 \text{ N}$$

The free body diagram of monkey is shown below,



The force equilibrium equation is,

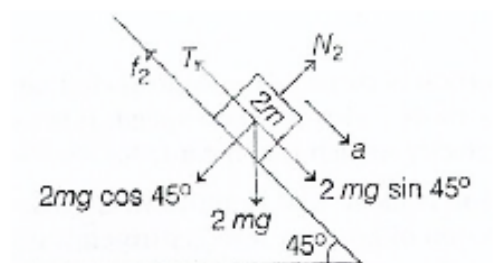
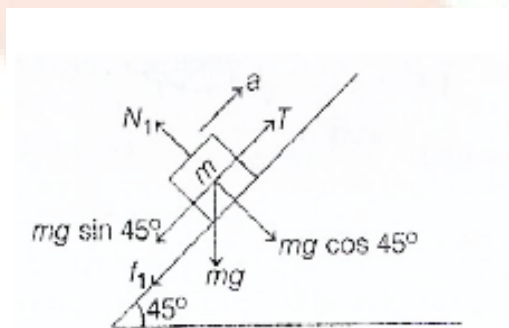
$$T - mg = ma$$

Substitute values.

$$80 - (5)(10) = (5)a$$

$$a = 6 \text{ m/s}^2$$

86. The free body diagram of the two masses is shown below.



The friction on mass 'm' is,

$$f_1 = \mu N_1$$

$$f_1 = \frac{2}{3}mg \cos 45^\circ$$

$$f_1 = \frac{\sqrt{2}}{3}mg$$

The friction on mass '2m' is,

$$f_2 = \mu N_2$$

$$f_2 = \frac{2}{3}(2m)g \cos 45^\circ$$

$$f_2 = \frac{2\sqrt{2}}{3}mg$$

The second law of motion for m mass,

$$T - f_1 - mg \cos 45^\circ = ma$$

$$T - \frac{\sqrt{2}}{3}mg - \frac{mg}{\sqrt{2}} = ma \quad \text{..... (I)}$$

The second law of motion for $2m$ mass,

$$2mg \cos 45^\circ - T - f_2 = 2ma$$

$$\sqrt{2}mg - T - \frac{2\sqrt{2}}{3}mg = 2ma \quad \text{..... (II)}$$

Solve equation (I) and (II),

$$3T - 2\sqrt{2}mg = 0$$

$$T = \frac{2\sqrt{2}mg}{3}$$

87. The tension is maximum at the bottom of the vertical circle.
The expression of the maximum tension is,

$$T_1 = \frac{mv^2}{r} + mg$$

The tension is minimum at the top of the vertical circle. The expression of the minimum tension is,

$$T_2 = \frac{mu^2}{r} - 5mg$$

Given, $\frac{T_1}{T_2} = 4$

Thus,

$$21mg = 3 \frac{mu^2}{r}$$

$$u^2 = 7rg$$

The velocity at the top is,

$$v = \sqrt{u^2 - 4rg}$$

$$v = \sqrt{7rg - 4rg}$$

$$v = \sqrt{3rg}$$

$$v = \sqrt{3 \left(\frac{5}{3} \right) (10)}$$

$$v = \sqrt{50} \text{ ms}^{-1}$$

88. The time and speed of the ball to reach on the ground is,

$$t_0 = \sqrt{\frac{2h}{g}} \text{ and } v_0 = \sqrt{2gh}$$

After the first collision, the speed of the ball is,

$$v_1 = ev_0 = e\sqrt{2gh}$$

The time when ball stops the motion is,

$$v = u + at$$

$$0 = v_1 - gt_1$$

$$t_1 = \frac{v_1}{g}$$

The time taken between first and second collision is $2t_1$.

Similarly the time taken between second and third collision

is, $2t_2 = \frac{2v_2}{g}$

Thus the total time before it ceases to rebound is,

$$T = t_0 + 2t_1 + 2t_2 + \dots$$

$$T = t_0 + \frac{2v_1}{g} + \frac{2v_2}{g} + \dots$$

$$T = t_0 + \frac{2ev_0}{g} + \frac{2e^2v_0}{g} + \dots \quad \left(\text{As, } v_2 = ev_1 = e^2v_0 \right)$$

$$T = \sqrt{\frac{2h}{g}} \left[1 + 2e(1 + e + e^2 + e^3 + \dots) \right] \quad \left(\text{As, } v_0 = \sqrt{\frac{2h}{g}} \right)$$

Solve further,

$$T = \sqrt{\frac{2h}{g}} \left[1 + 2e \left(\frac{1}{1-e} \right) \right]$$

$$T = \sqrt{\frac{2h}{g}} \left[\frac{1+e}{1-e} \right]$$

$$T = \sqrt{\frac{2(180)}{10}} \left(\frac{1+0.5}{1-0.5} \right)$$

$$T = 18 \text{ s}$$

Similarly the total distance in the entire motion of ball is,

$$H = h_0 + 2h_1 + 2h_2 + \dots$$

$$H = h + 2e^2h + 2e^2h + \dots$$

$$H = h \left[1 + 2e^2(1 + e^2 + e^4 + \dots) \right]$$

$$H = h \left(\frac{1+e^2}{1-e^2} \right)$$

Substitute values,

$$H = 180 \left(\frac{1 + \frac{1}{4}}{1 - \frac{1}{4}} \right)$$

$$H = 300 \text{ m}$$

The average speed is,

$$|v| = \frac{H}{T}$$

$$|v| = \frac{300}{18}$$

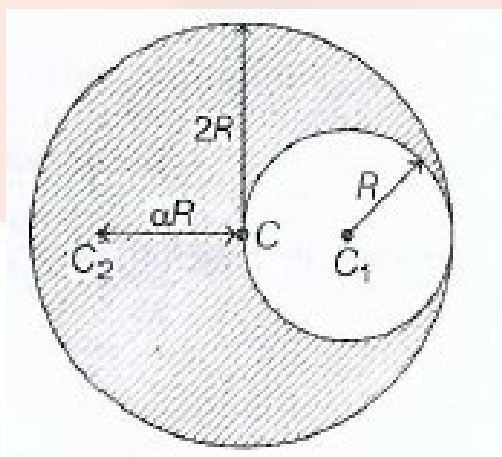
$$|v| = \frac{50}{3} \text{ m/s}$$

The average velocity is,

$$v = \frac{180}{18}$$

$$v = 10 \text{ m/s}$$

89. Consider the figure,



Let the mass of the entire disc is M .

$$\text{The mass per unit area is, } = \frac{M}{\pi(2R)^2} = \frac{M}{4\pi R^2}$$

The mass of the removed disc of radius R is,

$$M_1 = \frac{M}{4\pi R^2} \times (\pi R^2)$$

$$M_1 = \frac{M}{4}$$

The remaining mass of the disc is,

$$M_2 = M - M_1$$

$$M_2 = M - \frac{M}{4}$$

$$M_2 = \frac{3M}{4}$$

The centre of mass of combination of M_1 and M_2 will be $C(0, 0)$. So,

$$\frac{M_1 x_1 - M_2 x_2}{M_1 - M_2} = 0$$

$$M_1 x_1 = M_2 x_2$$

$$\frac{M}{4} R = \frac{3M}{4} (\alpha R)$$

$$\alpha = \frac{1}{3}$$

90. The torque produced in hollow cylinder is,

$$\tau = I\alpha$$

$$TR = (MR^2) \left(\frac{a}{R} \right)$$

$$TR = MRa$$

$$T = Ma$$

The force equilibrium equation is,

$$Mg - T = Ma$$

Substitute values,

$$Mg - Ma = Ma$$

$$a = \frac{g}{2}$$

91. Let initially, both the oscillating particles are at mean position i.e., ($x = 0$ at $t = 0$).

The first particle is in same phase at given time instances. So,
 $t = 0$ s, 3 s, 6 s, 9 s, 12 s, 15 s,...etc.

The time period of second particle will be selected from the given options which give same phase of the both particles at
 $t = 15$ s, 30 s, 45 s,...etc.

From option (1),

$t = 0$ s, 1 s, 2 s, 3 s, 4 s, 5 s,...etc.

From option (2),

$t = 0 \text{ s}, 1.5 \text{ s}, 3 \text{ s}, 4.5 \text{ s}, 6 \text{ s}, \dots \text{etc.}$

From option (3),

$t = 0 \text{ s}, 2 \text{ s}, 4 \text{ s}, 6 \text{ s}, \dots \text{etc.}$

From option (4),

$t = 0 \text{ s}, 2.5 \text{ s}, 5 \text{ s}, 7.5 \text{ s}, 10 \text{ s}, 12.5 \text{ s}, 15 \text{ s}, \dots \text{etc.}$

So, the both particles are in same phase again at $t = 15 \text{ s}, 30 \text{ s}, 45 \text{ s}, \dots \text{etc}$ only. Thus the time period must be 2.5 s .

92. From the conservation of mechanical energy,

$$\frac{1}{2}mv^2 + \left(-\frac{GMm}{r}\right) = \frac{1}{2}mv_e^2 + \left(-\frac{GMm}{R}\right)$$

$$v^2 = v_e^2 - \frac{2GM}{R} + \frac{2GM}{r}$$

$$v^2 = 2gR + 2gR\left(\frac{R}{r} - 1\right) \quad \left(\text{As, } v_e = \sqrt{2gR}, g = \frac{GM}{R^2}\right)$$

$$v = \frac{dr}{dt} = \sqrt{\frac{2gR^2}{r}}$$

Solve further,

$$\frac{dr}{dt} = R\sqrt{\frac{2g}{r}}$$

Integrate above equation,

$$\int_0^t dt = \frac{1}{R\sqrt{2g}} \int_R^{R+h} r^{1/2} dr$$

$$t = \frac{2}{3} \frac{1}{R\sqrt{2g}} \left[(R+h)^{3/2} - R^{3/2} \right]$$

$$t = \frac{1}{3} \sqrt{\frac{2R}{g}} \left[\left(1 + \frac{h}{R} \right)^{3/2} - 1 \right]$$

$$t = \frac{1}{3} \sqrt{\frac{2R}{g}} \left[\left(1 + \frac{3R}{R} \right)^{3/2} - 1 \right] \quad (\text{As, } h = 3R)$$

$$t = \frac{1}{3} \sqrt{\frac{2R}{g}} \times 7$$

Substitute values in above expression,

$$t = \frac{1}{3} \sqrt{\frac{2 \times 6400 \times 10^3}{9.8}} (7)$$

$$t = 2666.65 \text{ s}$$

$$t = 44.44 \text{ min}$$

93. The expression of longitudinal strain is given by,

$$\varepsilon = \frac{F}{YA}$$

$$\frac{\Delta l}{l} = \frac{F}{YA}$$

Substitute values,

$$\frac{\Delta l}{l} = \frac{22}{1.1 \times 10^{11} \times 10^{-6}}$$

$$\frac{\Delta l}{l} = 2 \times 10^{-4}$$

The expression of the Poisson's ratio is,

$$\mu = \frac{\Delta d / d}{\Delta l / l}$$

Substitute values,

$$0.32 = \frac{\Delta d / d}{2 \times 10^{-4}}$$

$$\frac{\Delta d}{d} = 6.4 \times 10^{-5}$$

So, the change in radius is,

$$\frac{\Delta r}{r} = 6.4 \times 10^{-5}$$

The fractional change in area is,

$$\frac{\Delta A}{A} = 2 \frac{\Delta r}{r}$$

$$\frac{\Delta A}{0.01} = 2(6.4 \times 10^{-5})$$

$$\Delta A = 1.28 \times 10^{-6} \text{ cm}^2$$

94. The expression of the excess pressure inside soap bubble due to the surface tension is,

$$p = \frac{4S}{R}$$

The expression of electrostatic pressure is,

$$p_e = \frac{\sigma^2}{2\epsilon_0}$$

Given condition, $p = p_e$

So,

$$\frac{4S}{R} = \frac{\sigma^2}{2\epsilon_0}$$

$$\frac{4S}{0.2 \times 10^{-3}} = \frac{(20 \times 10^{-6})^2}{2(8.85 \times 10^{-12})}$$

$$S = 11.3 \times 10^{-4} \text{ N/m}$$

95. The expression of the power radiating from a black body is,

$$P = \sigma AT^4$$

$$P = \sigma (4\pi R^2) \left(\frac{k}{R} \right)^4 \quad \left(\text{As, } T \propto \frac{1}{R} \Rightarrow T = \frac{k}{R} \right)$$

$$P \propto \frac{1}{R^2}$$

From above expression, when radius is doubled, the radiating power of the sphere becomes one fourth.

96. The expression of the apparent weight of object in water is,

$$w \propto \frac{\rho_{\text{sphere}}}{\rho_{\text{water}}}$$

The ratio of density of sphere and water will be greater at 50°C than 0°C because the cubical expansion (Y) of the metal (sphere) is less than water.

So,

$$\left(\frac{\rho_{\text{sphere}}}{\rho_{\text{water}}} \right)_{50^{\circ}\text{C}} > \left(\frac{\rho_{\text{sphere}}}{\rho_{\text{water}}} \right)_{0^{\circ}\text{C}}$$

$$w_2 > w_1$$

97. Given, $W = \frac{Q}{4}$

The expression of the efficiency is,

$$\eta = \frac{W}{Q}$$

$$1 - \frac{T_2}{T_1} = \frac{W}{Q}$$

Substitute values,

$$1 - \frac{T_2}{T_1} = \frac{Q/4}{Q}$$

$$1 - \frac{T_2}{T_1} = \frac{1}{4}$$

..... (I)

When the temperature is reduced by 50 K, efficiency becomes

$\frac{100}{3}\%$. Thus,

$$\eta' = \frac{100}{3} \times \frac{1}{100}$$

$$1 - \frac{T_2 - 50}{T_1} = \frac{1}{3}$$

$$1 - \frac{T_2}{T_1} + \frac{50}{T_1} = \frac{1}{3} \quad \text{.....(II)}$$

From equation (I) and (II),

$$\frac{50}{T_1} = \frac{1}{3} - \frac{1}{4}$$

$$T_1 = 600 \text{ K}$$

So,

$$T_2 = \frac{3}{4} T_1$$

$$T_2 = \frac{3}{4} (600)$$

$$T_2 = 450 \text{ K}$$

98. The expression of the heat added is,

$$\Delta Q_{ABC} = \Delta U_{ABC} + \Delta W_{ABC}$$

$$\Delta Q_{ABC} = nC_V(T_C - T_A) + 3p_0(2V_0 - V_0)$$

$$\Delta Q_{ABC} = n \frac{3R}{2}(T_C - T_A) + 3p_0V_0$$

$$\Delta Q_{ABC} = \frac{3}{2}(p_C V_C - p_A V_A) + 3p_0V_0$$

Solve further,

$$\Delta Q_{ABC} = \frac{3}{2}(3p_0 2V_0 - p_0 V_0) + 3p_0V_0$$

$$\Delta Q_{ABC} = \frac{3}{2}5p_0V_0 + 3p_0V_0$$

$$\Delta Q_{ABC} = 10.5p_0V_0$$

99. The expression of speed of sound in gas is,

$$v = \sqrt{\frac{\gamma RT_1}{M}}$$

$$v = \sqrt{\frac{(5/3)RT_1}{M}} \quad \left(\text{For monoatomic gas, } \gamma = \frac{5}{3} \right) \dots\dots (I)$$

The expression of speed of molecules is,

$$c = \sqrt{\frac{3RT_2}{M}} \dots\dots (II)$$

From equation (I) and (II),

$$\frac{v}{c} = \sqrt{\frac{(5/3)T_1}{3T_2}}$$

$$\frac{v}{c} = \sqrt{\frac{(5/3)(300)}{3(400)}}$$

$$\frac{v}{c} = \sqrt{\frac{5}{12}}$$

100. The mass per unit length is,

$$\dot{m} = \frac{\pi r^2 l d}{l}$$

$$\dot{m} = \pi r^2 d$$

The expression of frequency in stretched string is,

$$f = \frac{n}{2l} v$$

$$f = \frac{n}{2l} \sqrt{\frac{T}{\dot{m}}}$$

Given condition, $f_A(\text{At}, n=2) = f_B(\text{At}, n=3)$

So,

$$\frac{2}{2l_A} \sqrt{\frac{T}{\dot{m}_A}} = \frac{3}{2l_B} \sqrt{\frac{T}{\dot{m}_B}}$$
$$\frac{1}{l_A \sqrt{\pi r_A^2 d}} = \frac{3}{2l_B \sqrt{\pi r_B^2 d}}$$
$$\frac{l_A}{l_B} = \frac{2}{3} \left(\frac{r_B}{r_A} \right)$$

Substitute values.

$$\frac{l_A}{l_B} = \frac{2}{3} \left(\frac{1}{2} \right)$$
$$\frac{l_A}{l_B} = \frac{1}{3}$$

101. The wavelength of n^{th} overtone for open organ pipe is,

$$\lambda_n = \frac{2l}{n+1}$$

For $n = 3$

$$\lambda = \frac{2l}{3+1}$$

$$\lambda = \frac{l}{2}$$

102. From the expression of the magnifying power,

$$m = \frac{f_0}{f_e}$$

$$10 = \frac{f_0}{f_e}$$

$$f_0 = 10f_e \quad \text{..... (I)}$$

The length of tube is,

$$f_0 + f_e = 110 \quad \text{..... (II)}$$

From equation (I) and (II),

$$10f_e + f_e = 110$$

$$f_e = 10 \text{ cm}$$

$$f_0 = 100 \text{ cm}$$

The expression of magnifying power, when image is formed at near point is,

$$m = \frac{f_0}{f_e} \left(1 + \frac{f_e}{D} \right)$$

Substitute values,

$$m = \frac{100}{10} \left(1 + \frac{10}{25} \right)$$

$$m = 10 \times \frac{35}{25}$$

$$m = 14$$

103. Consider the equation,

$$n_1\lambda_1 = n_2\lambda_2$$

For minimum distance to be coincide,

$$n_1 = n + 1 \text{ and } n_2 = n$$

Thus,

$$(n + 1)(4200) = n(5040)$$

$$4200n + 4200 = 5040n$$

$$n = 5$$

The expression of the required distance is,

$$x = \frac{(n + 1)\lambda_1 D}{d}$$

Substitute values.

$$x = \frac{(5 + 1)(4200 \times 10^{-10})(2)}{2.4 \times 10^{-3}}$$

$$x = 2.1 \times 10^{-3} \text{ m}$$

$$x = 2.1 \text{ mm}$$

104. The expression of the electric flux is,

$$\phi = \frac{q}{\epsilon_m}$$

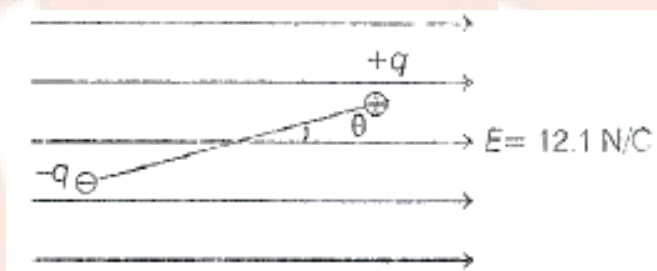
$$\phi = \frac{q}{K\epsilon_0}$$

Substitute values.

$$\phi = \frac{8}{4\epsilon_0}$$

$$\phi = \frac{2}{\epsilon_0}$$

105. Consider the figure,



The expression of the time period is,

$$T = 2\pi \sqrt{\frac{I}{PE}}$$

$$T = 2\pi \sqrt{\frac{2mr^2}{q(2l)E}}$$

$$T = 2\pi \sqrt{\frac{mr^2}{qlE}}$$

Substitute values.

$$T = 2\pi \sqrt{\frac{(9.8 \times 10^{-3})(25 \times 10^{-2})^2}{(20 \times 10^{-6})(25 \times 10^{-2})(121)}}$$

$$T = 2\pi(3.181)$$

$$T = 19.98$$

$$T \approx 20 \text{ s}$$

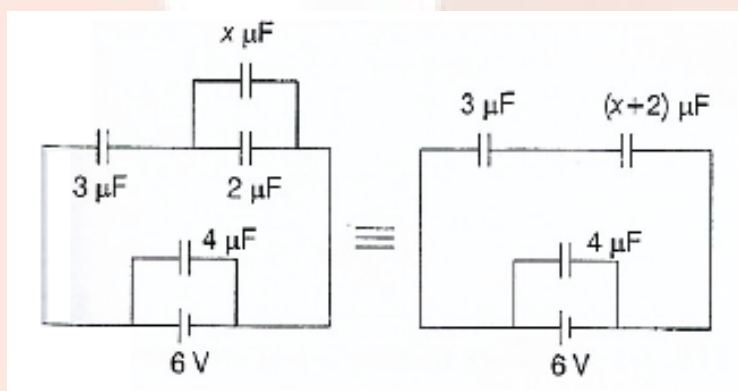
The time taken by the rod to become parallel is calculated as,

$$T' = \frac{T}{4}$$

$$T' = \frac{20}{4}$$

$$T' = 5 \text{ s}$$

106. Consider the circuit,



The charge on $4 \mu\text{F}$ capacitor is,

$$q_4 = C_4 V$$

$$q_4 = (4 \mu\text{F})(6)$$

$$q_4 = 24 \mu\text{C}$$

Given condition,

$$\frac{q_x}{q_4} = \frac{3}{8}$$

$$\frac{q_x}{24} = \frac{3}{8}$$

$$q_x = 9 \mu\text{C}$$

The potential difference on $x \mu\text{F}$ capacitor is,

$$V_1 = \frac{q_x}{C}$$

$$V_1 = \frac{9}{x}$$

Thus, the voltage across $2 \mu\text{F}$ will also be $\frac{9}{x}$.

The voltage across $3 \mu\text{F}$ will be $\left(6 - \frac{9}{x}\right)$ volt

The capacitors $3 \mu\text{F}$ and $(x+2) \mu\text{F}$ is in series, so there will be same charge on $3 \mu\text{F}$ and $(x+2) \mu\text{F}$.

So,

$$q_3 = q_{(x+2)}$$

$$3\left(6 - \frac{9}{x}\right) = (x+2)\frac{9}{x}$$

$$6x - 9 = 3x + 6$$

$$x = 5$$

107. The potential difference before inserting insulator is,

$$V = \frac{\sigma d}{\epsilon_0}$$

$$100 = \frac{\sigma d}{\epsilon_0}$$

The expression of the potential after inserting 2 mm of insulator is,

$$V_1 = E_1(2) + E_2(d - 2)$$

$$V_1 = \frac{\sigma}{\epsilon_0 k}(2) + \frac{\sigma}{\epsilon_0}(d - 2)$$

$$V_1 = \frac{\sigma}{\epsilon_0} \left(\frac{2}{k} + d - 2 \right)$$

To maintain potential difference, same distance is increased by 1.6 mm. So the expression of extra potential difference is,

$$V_2 = \frac{\sigma}{\epsilon_0}(1.6)$$

So,

$$V_1 + V_2 = V$$

Substitute values.

$$\frac{\sigma}{\epsilon_0} \left(\frac{2}{k} + d - 2 \right) + \frac{\sigma}{\epsilon_0} (1.6) = \frac{\sigma d}{\epsilon_0}$$

$$\frac{\sigma d}{\epsilon_0} - \frac{2\sigma}{\epsilon_0 k} - \frac{\sigma}{\epsilon_0} (d - 2) = \frac{\sigma}{\epsilon_0} (1.6)$$

$$\frac{2\sigma}{\epsilon_0} - \frac{2\sigma}{\epsilon_0 k} = \frac{\sigma}{\epsilon_0} (1.6)$$

$$k = 5$$

108. If the given resistances forms a Wheatstone bridge, then current will independent of resistance R_6 .

So,

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$R_1 R_4 = R_2 R_3$$

109. The expression of relation between current rating of fuse wire and radius of fuse wire is,

$$I^2 \propto r^3$$

$$I \propto r^{3/2}$$

$$\frac{I_2}{I_1} = \left(\frac{r_2}{r_1} \right)^{3/2}$$

Substitute values,

$$\frac{I_2}{5} = \left(\frac{0.3}{0.2} \right)^{3/2}$$

$$I_2 = 5 \sqrt{\frac{27}{8}} \text{ A}$$

110. The net magnetic field when the current is clockwise is,

$$B_1 = \frac{\mu_0 i}{2r} - \frac{\mu_0 i}{2\pi r}$$

$$B_1 = \frac{\mu_0 i}{2r} \left(1 - \frac{1}{\pi} \right)$$

The net magnetic field when the current is anticlockwise is,

$$B_2 = \frac{\mu_0 i}{2r} \left(1 + \frac{1}{\pi} \right)$$

So,

$$\frac{B_1}{B_2} = \frac{\frac{\pi - 1}{\pi}}{\frac{\pi + 1}{\pi}}$$

$$\frac{B_1}{B_2} = \frac{15}{29}$$

Thus, no option is correct.

111. The expression of the magnetic field intensity due to small loop at location of larger loop is,

$$B_1 = \frac{\mu_0 I r^2}{2x^3}$$

Substitute values,

$$B_1 = \frac{(4\pi \times 10^{-7})(2)(0.3 \times 10^{-2})^2}{2(15 \times 10^{-2})^3}$$

$$B_1 = \frac{36\pi}{15^3} \times 10^{-9} \text{ T}$$

The expression of the flux linked with larger loop is,

$$\phi_2 = B_1 A_2$$

Substitute values,

$$\phi_2 = \frac{86\pi}{15^3} \times 10^{-19} \times \pi (20 \times 10^{-2})^2$$

$$\phi_2 = 0.42 \times 10^{-11} \text{ Wb}$$

Thus no option is correct.

112. The horizontal component of earth's magnetic field is calculated as,

$$B_H = \frac{B_V}{\tan \delta}$$

$$B_H = \frac{0.3464}{1/\sqrt{3}}$$

$$B_H = 0.3464 \times 1.73$$

$$B_H = 0.6 \text{ G}$$

113. The expression of the induced emf is,

$$\varepsilon = -\frac{d\phi}{dt}$$

$$\varepsilon = -\frac{d(B\pi r^2)}{dt}$$

$$\varepsilon = -B\pi \frac{dr^2}{dt}$$

$$\varepsilon = -B\pi \left(2r \frac{dr}{dt} \right)$$

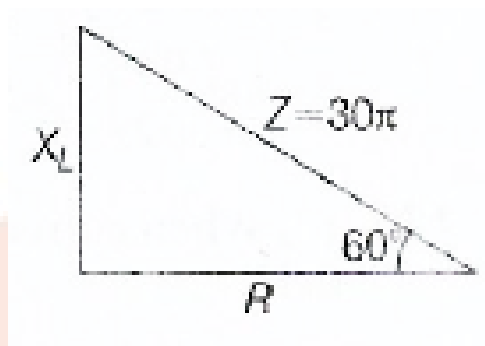
Substitute values.

$$\varepsilon = -10^{-4} \times \pi (2 \times 20 \times 10^{-2}) (-2 \times 10^{-3})$$

$$\varepsilon = 8\pi \times 10^{-8}$$

$$\varepsilon = 0.08\pi \text{ } \mu\text{V}$$

114. Consider the figure,



The phase difference between voltage and current is,

$$\phi = \pi - \frac{2\pi}{3}$$

$$\phi = \frac{\pi}{3}$$

$$\phi = 60^\circ$$

The expression of average power in R - L circuit is,

$$P_{avg} = V_{rms} I_{rms} \cos \phi$$

$$P_{avg} = \frac{V_0}{\sqrt{2}} \frac{I_0}{\sqrt{2}} \cos \phi$$

$$P_{avg} = \frac{V_0 I_0}{2} \cos \phi$$

Substitute values,

$$P_{avg} = \frac{(150)(5)}{2} \cos 60^\circ$$

$$P_{avg} = 187.5 \text{ W}$$

The impedance is calculated as,

$$Z = \frac{V_0}{I_0}$$

$$Z = \frac{150}{5}$$

$$Z = 30 \Omega$$

From the above figure,

$$\cos \phi = \frac{R}{Z}$$

$$\cos 60^\circ = \frac{R}{30}$$

$$R = 15 \Omega$$

115. The magnitude of magnetic field is calculated as,

$$B = \frac{E}{c}$$

$$B = \frac{750}{3 \times 10^8}$$

$$B = 2.5 \times 10^{-6} \text{ T}$$

In an electromagnetic wave, the magnetic field, electric field and direction of propagation are mutually perpendicular.

Thus,

$$\vec{B} = 2.5 \times 10^{-6} \hat{k} \text{ T}$$

116. The work function is proportional to the y -intercept on V verse ν graph. In the graph of option (4), the line of A cuts larger intercept, so it is the correct option.

117. The expression of the radius of the electron of hydrogen in n^{th} orbit is,

$$r_n = n^2 (0.53 \text{ \AA})$$

Substitute values.

$$530 \times 10^{-9} = n^2 (0.53 \times 10^{-10})$$

$$n^2 = \frac{5300}{0.53}$$

$$n = 100$$

118. The radioactive decay equation is,

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{\frac{t}{T}}$$

Substitute values.

$$\frac{X}{5X} = \left(\frac{1}{2}\right)^{\frac{t}{2}}$$

$$\left(\frac{1}{2}\right)^{\frac{t}{2}} = \frac{1}{5}$$

As,

$$\frac{1}{2^2} < \frac{1}{5} < \frac{1}{2^3}$$

$$\frac{1}{2^2} < \frac{1}{2^{t/2}} < \frac{1}{2^3}$$

$$2 < \frac{t}{2} < 3$$

$$4 < t < 6$$

119. For Small Scale Integration, logic gates should be less than 10.

For Medium Scale Integration, logic gates should be less than 100.

For Large Scale Integration, logic gates should be less than 1000.

For Very Large Scale Integration, logic gates should be greater than 100.

120. The expression of the modulation index is,

$$\mu = \frac{A_m}{A_c}$$

To keep signal without distortion, the relation between amplitude of modulating wave and carrier wave is, $A_m \leq A_c$

Thus, $\mu \leq 1$



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121. The energy is calculated as:

$$\begin{aligned} E &= h \frac{c}{\lambda} \\ &= 6.6 \times 10^{-34} \text{ Js} \times \frac{3 \times 10^8 \text{ m/s}}{450 \times 10^{-9} \text{ m}} \\ &= 4.4 \times 10^{-19} \text{ J} \end{aligned}$$

The energy in eV is derived as:

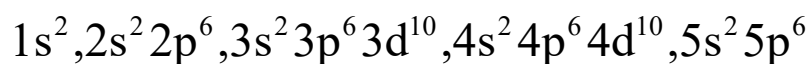
$$1 \text{ eV} = 1.602 \times 10^{-19}$$

$$\begin{aligned} \text{Energy} &= \frac{4.4 \times 10^{-19}}{1.602 \times 10^{-19}} \\ &= 2.74 \text{ eV} \end{aligned}$$

The metals which are having less value of work function than the energy calculated, they can undergo photoelectric effect. Therefore, metals are K, Na and Li.

122. The atomic number of Xenon is 54.

Its electronic configuration is as follows:



Therefore, in xenon atom, the number of completely filled orbital is 27.

123. Let's assume that the number of protons in X is equal to 'a'.

Thus, number of neutrons becomes $0.166a + a = 1.166a$

Since,

Mass number = Number of protons + Number of neutron

$$52 = a + 1.166a$$

$$52 = 2.166a$$

$$a = 24$$

'24' protons signify that the element X is ${}_{24}\text{Cr}$

Now, let's assume that the number of protons in Y is equal to 'b'.

Thus, number of neutrons becomes $0.273b + b = 1.273b$

Since,

Mass number = Number of protons + Number of neutron

$$75 = b + 1.273b$$

$$75 = 2.273b$$

$$b = 33$$

'33' protons signify that the element Y is ${}_{33}\text{As}$.

124. It is given that a molecule has electrons in 1s, 2s, 2p, 3s, 3p, 3d and 4s orbitals.

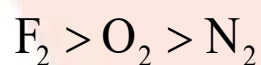
This means that the outermost shell is made up of 3d orbitals which can produce inner and outer orbital complex. The types of hybridization are:



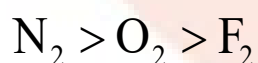
125. Since we know that, more is the bond order of any species, shorter will be its bond length and hence, stronger will be the bond.

	N_2	>	O_2	>	F_2
Bond order	3		2		1

As bond length is reverse of bond order, therefore, the correct order of bond length will be:



The order of dissociation energy is as follows:



126. The root mean square velocity can be calculated by using the relation shown below.

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

Substitute the values.

$$3 \times 10^3 \text{ m/s} = \sqrt{\frac{3RT}{28}}$$

$$9 \times 10^6 \text{ m}^2/\text{s}^2 = \frac{3RT}{28}$$

$$RT = 28 \times 3 \times 10^3 \text{ J/mol}$$

Since,

$$\frac{\text{m}^2}{\text{s}^2} = \frac{\text{J}}{\text{kg}}$$

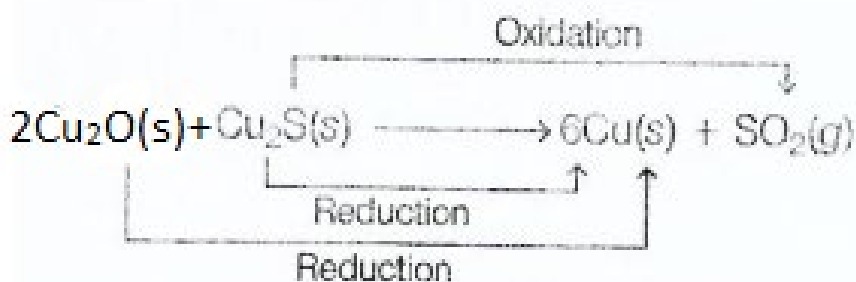
$$1 \text{ kg} = 1000\text{g}$$

Therefore, the kinetic energy for 1 mole of an ideal gas is calculated as:

$$\begin{aligned} K.E &= \frac{3}{2} RT \\ &= \frac{3}{2} \times \frac{28 \times 3 \times 10^3}{1000} \text{ kJ} \\ &= 126 \text{ kJ} \end{aligned}$$

127. In the given reaction,

Cu (I) of $\text{Cu}_2\text{O}_{(s)}$ and $\text{Cu}_2\text{S}_{(s)}$ shows reduction and form $\text{Cu}(s)$ while sulphide of $\text{Cu}_2\text{S}_{(s)}$ shows oxidation and form $\text{SO}_{2(g)}$ as shown below.



Therefore, Cu (I) of $\text{Cu}_2\text{O}_{(s)}$ and $\text{Cu}_2\text{S}_{(s)}$ acts as oxidant (oxidizing agent) and sulphide of $\text{Cu}_2\text{S}_{(s)}$ acts as reductant (reducing agent)

128. We know that $pV = nRT$

Therefore, the workdone is calculated as:

$$\begin{aligned}W &= -nRT \ln \left(\frac{V_2}{V_1} \right) \\&= -PV \ln \left(\frac{V_2}{V_1} \right) \\&= -2.303 \times 20 \times 1.5 \times \log \left(\frac{15}{1.5} \right) \\&= -69.09 \text{ L atm}\end{aligned}$$

129. The expression of equilibrium constant for the given reaction can be written as:

$$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$$
$$[\text{HI}]^2 = K_c [\text{H}_2][\text{I}_2]$$

Now, substitute all the values in the above expression.

$$\begin{aligned}[\text{HI}]^2 &= 49 \times 16 \times 10^{-4} \\&= \sqrt{49 \times 16 \times 10^{-4}} \\&= 0.28 \text{ mol}\end{aligned}$$

130. The dissociation reaction for monobasic acid is as follows:

	$\text{HA}^+(\text{aq})$	$\rightleftharpoons$	$\text{H}^+(\text{aq})$	+	$\text{A}^-(\text{aq})$
Initial conc.	0.1		0		0
Equilibrium	$(0.1 - x)$		x		x

The concentration of hydrogen ion can be calculated as:

$$\text{pH} = -\log[\text{H}^+]$$

$$5 = -\log[\text{H}^+]$$

$$[\text{H}^+] = 10^{-5} \text{ M}$$

The dissociation constant is calculated as:

$$K_a = \frac{[10^{-5}][10^{-5}]}{[0.10]}$$

$$= 10^{-9}$$

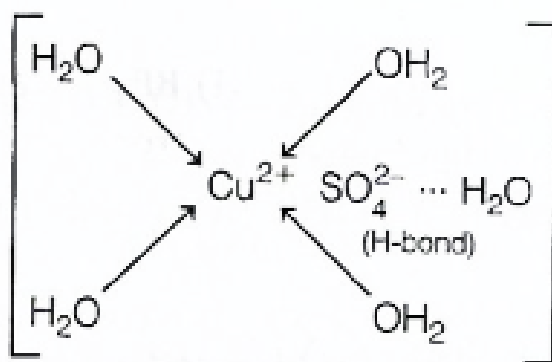
Now, the value of pK_a is derived as:

$$\text{pK}_a = -\log[\text{K}_a]$$

$$= -\log(10^{-9})$$

$$= 9$$

131. i. The number of hydrogen bonded water molecules in copper sulphate pentahydrate is one. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ has five molecules of water of crystallization. Four molecules of water are coordinated with Cu^{2+} and remaining one molecule of water coordinates through hydrogen bonding as shown below:



Structure of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

ii. Non-stoichiometric hydrides are formed by La and Zr.

Examples are LaH_2 , LaH_3 , ZrH_2

iii. In H_2O , each oxygen atom is linked to four hydrogen atoms.

Therefore, statements (i) and (ii) are correct.

132. The oxoacids salts of group II elements from the given list are CaCO_3 , MgCO_3 and $\text{Sr}(\text{NO}_3)_2$ as they all possess acid and base with oxygen atom.

CaCO_3 is a salt of H_2CO_3 and $\text{Ca}(\text{OH})_2$

MgCO_3 is a salt of H_2SO_4 and $\text{Mg}(\text{OH})_2$

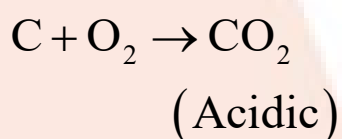
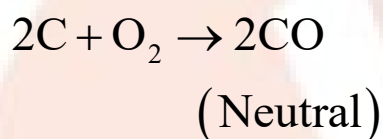
$\text{Sr}(\text{NO}_3)_2$ is a salt of HNO_3 and $\text{Sr}(\text{OH})_2$

133. Borax in aqueous solution is basic in nature due to hydrolysis.

Therefore, the pH range of the resultant solution is more than 7.

Its in the range of 7-14

134. When carbon (X) is burnt in the presence of oxygen, it produces carbon dioxide which is neutral and acidic in nature as shown below.



Therefore, X is C.

135. A) The condition which occurs by rising levels of globin inside the blood is called methenaglobinmia. This is due to the presence of 2000 ppm of nitrates in drinking water.

B) The presence of 1000 ppm of lead in potable water damages kidney.

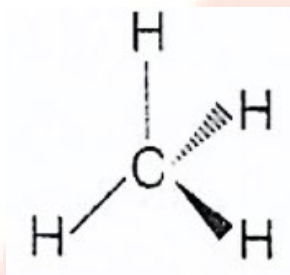
C) The presence of 50 ppm of fluoride in drinking water damage bones and teeth.

D) The growth of fish can be stopped if the presence of dissolved oxygen becomes less than 1 ppm.

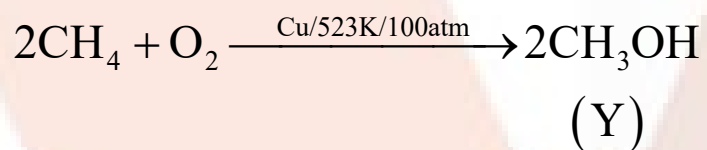
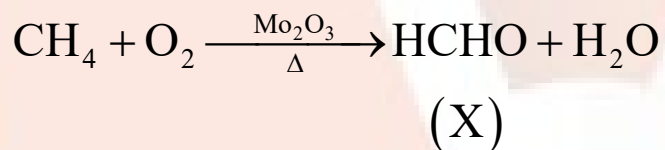
Therefore,

A	B	C	D
IV	II	V	I

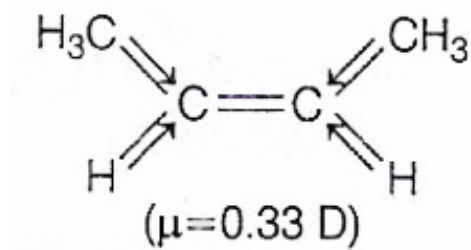
136. The bonds which are projecting out of the plane are W and Z.



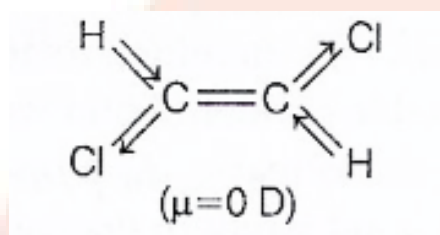
137. The products (X) and (Y) formed are as follows:.



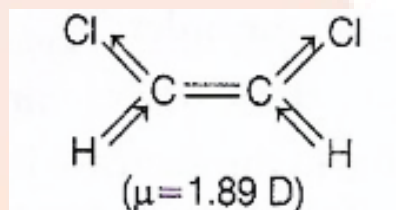
138. (1) The dipole moment of cis-But-2-ene is as follows:



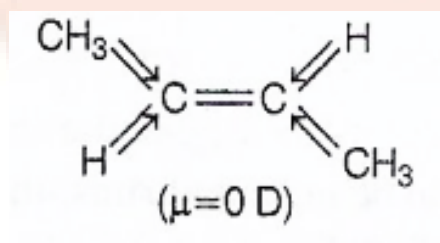
(2) The dipole moment of trans-1, 2-Dichloroethene is as follows:



(3) The dipole moment of cis-1, 2-Dichloroethene is as follows:



(4) The dipole moment of trans-But-2-ene is as follows:



Therefore, cis-1, 2-Dichloroethene has highest dipole moment.

139. The relation between side length (a) and radius (r) of a metal in face-centered unit cell is as follows:

$$r = \frac{a}{2\sqrt{2}}$$

Substitute the value of 'a' in the above expression.

$$\begin{aligned} r &= \frac{400}{2 \times 1.414} \\ &= 141.4 \text{ pm} \end{aligned}$$

140. The Henry's law equation is:

$$P_A = K_H x$$

Substitute the values.

$$1.67 \text{ bar} = 1.67 \times 1000 \text{ bar} \times x$$

$$\begin{aligned} x &= \frac{1.67}{1.67 \times 1000} \\ &= 0.001 \end{aligned}$$

The quantity of CO_2 can be derived as follows:

$$\begin{aligned} \text{Mole fraction}(x) &= \frac{m_{\text{CO}_2} / M_{\text{CO}_2}}{m_{\text{H}_2\text{O}} / M_{\text{H}_2\text{O}}} \\ 0.001 &= \frac{m_{\text{CO}_2} \times 18}{44 \times 1000} \\ m_{\text{CO}_2} &= \frac{0.001 \times 44 \times 1000}{18} \\ &= 2.44 \text{ g} \end{aligned}$$

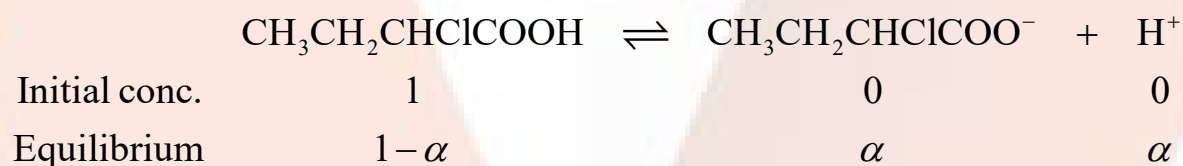
141. First, we will calculate molarity as follows:

$$\begin{aligned}\text{Molarity}(C) &= \frac{\text{Mass of solute}}{\text{Molar mass of solute} \times \text{Mass of solvent}} \times 1000 \\ &= \frac{12.25 \times 1000}{122.5 \times 250} \\ &= 0.40\end{aligned}$$

Now, the degree of ionization is calculated as:

$$\begin{aligned}\alpha &= \sqrt{\frac{K_a}{C}} \\ &= \sqrt{\frac{1.44 \times 10^{-3}}{0.4}} \\ &= 0.66\end{aligned}$$

Now, Van't Hoff factor (i) is calculated as:



Total number of moles is equal to $1 - \alpha + \alpha + \alpha = 1 + \alpha$

$$\begin{aligned}i &= \frac{\text{Moles after dissociation}}{\text{Moles before dissociation}} \\ &= \frac{1 + \alpha}{1} \\ &= 1 + \alpha\end{aligned}$$

Since, value of α is 0.06

Therefore,

$$i = 1 + 0.06$$

$$= 1.06$$

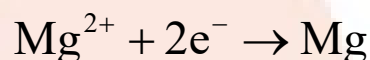
The change in freezing point is calculated as follows:

$$\Delta T_f = iK_f m$$

$$= 1.06 \times 0.4 \times 1.86$$

$$= 0.789^\circ\text{C}$$

142. The quantity of current required to deposit one mole of Mg from Mg^{2+} electrolyte is two faradays as follows:



The amount of electricity needed is equal to 2F.

From the above statement, we can conclude that one faraday is equal to the charge on one mole of electrons.

Therefore, both (A) and (R) are correct and (R) is the correct explanation of (A).

143. The activation energy can be calculated as:

$$\log \frac{k_1}{k_2} = -\frac{E_a}{2.303R} \left[\frac{1}{T_2} - \frac{1}{T_2} \right]$$

$$\text{Since, } k = \frac{0.693}{t_{1/2}},$$

Hence, we can write:

$$\log \frac{t_{1/2(1)}}{t_{1/2(2)}} = -\frac{E_a}{2.303R} \left[\frac{1}{T_2} - \frac{1}{T_1} \right]$$

Now, substitute all the values to calculate activation energy.

$$\log \frac{0.693}{20} = -\frac{E_a}{2.303 \times 8.314} \left[\frac{1}{350} - \frac{1}{300} \right]$$

$$\log \frac{20}{2} = -\frac{E_a}{2.303 \times 8.314} \left[\frac{-50}{350 \times 300} \right]$$

$$E_a = \frac{2.303 \times 8.314 \times 350 \times 300}{50}$$

$$= 40.29 \text{ kJ/mol}$$

144. Activated charcoal is present in gas mask as it is a great adsorbent.

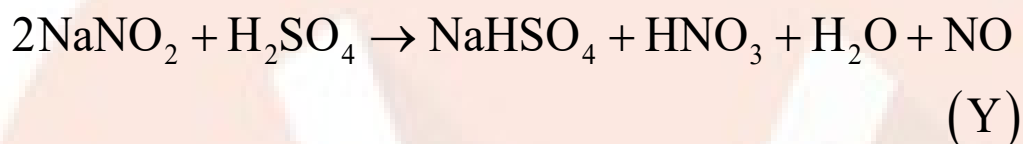
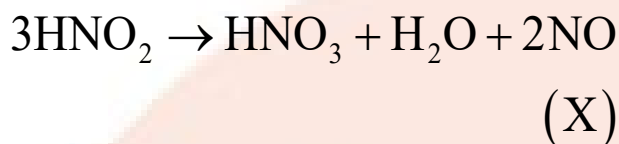
145. Bauxite is the most precious ore of Al which consists 30-60%

Al_2O_3 , the remaining is a mixture of SiO_2 , Fe_2O_3 and TiO_2 .

When powdered form of bauxite is reacted with concentrated NaOH, it mainly forms Al_2O_3 and SiO_2 whereas other oxides remain insoluble in the solution.

Therefore, the pair of oxides which can be leached out are Al_2O_3 and SiO_2 .

146. The reactions are shown below.



147. i. Oxygen shows $-2, -1, +1, +2$ oxidation states.

ii. As we move from top to bottom in a group, the atomic radius increases, thus, the potential to form stable covalent bond with hydrogen goes on decreasing. Therefore, the correct order is:



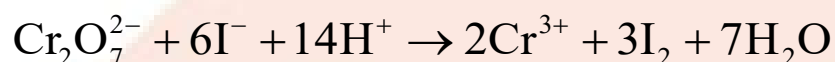
iii. Lesser the thermal stability of hydrides, the greater is reducing character, Therefore, the correct order is:



Hence, statements (i) and (iii) are correct.

148. The negative cell potential (non-spontaneous) is shown by the reaction of Br_2 with Cl^- . Therefore, this reaction is not possible.

149. In reaction (II), dichromate acts as oxidizing agent (which undergoes reduction) as shown below.



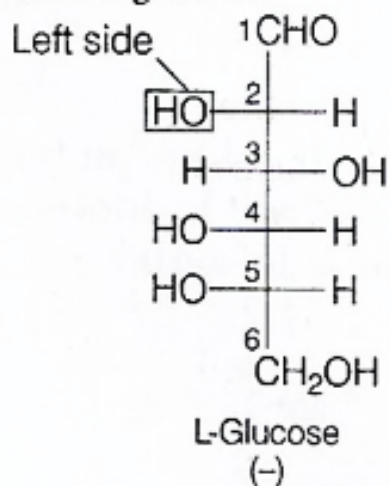
Cr^{6+} undergoes reduction and form Cr^{3+} .

I^- undergoes oxidation and form I_2 .

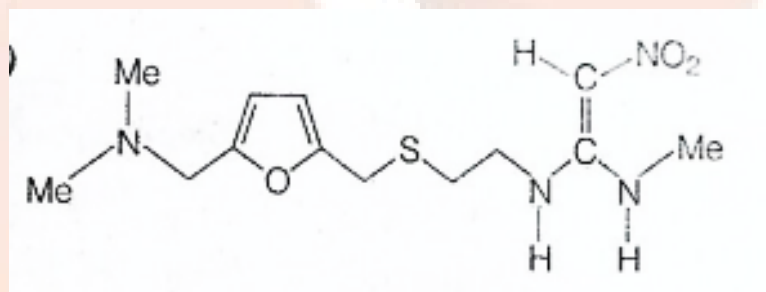
150. In $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$, H_2O acts as a weak ligand which does not cause pairing of unpaired electrons. Therefore, it is a colored complex.

151. Examples of synthetic and semi synthetic are polyethene and rayon respectively.

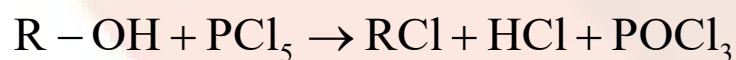
152. Fischer projection formula of L-(–)-glucose is as follows:



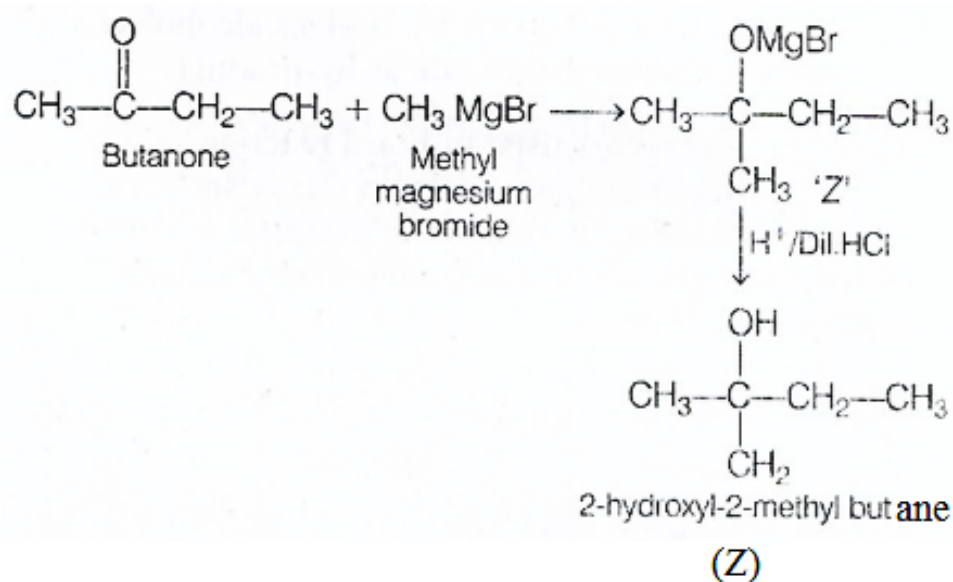
153. The structure of rantidine is shown below.



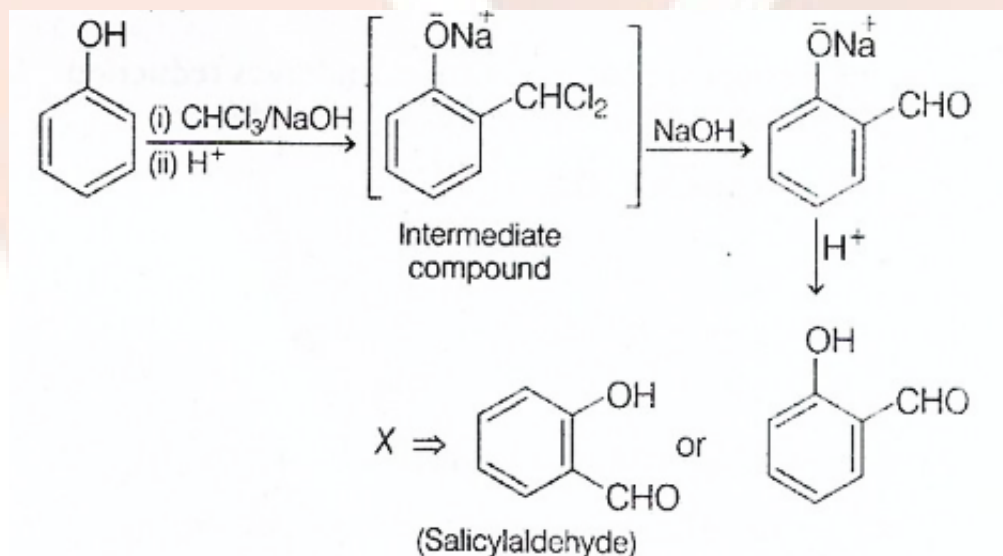
154. When alcohol undergoes a reaction with PCl_5 , it forms alkyl chloride, HCl and POCl_3 as shown below.



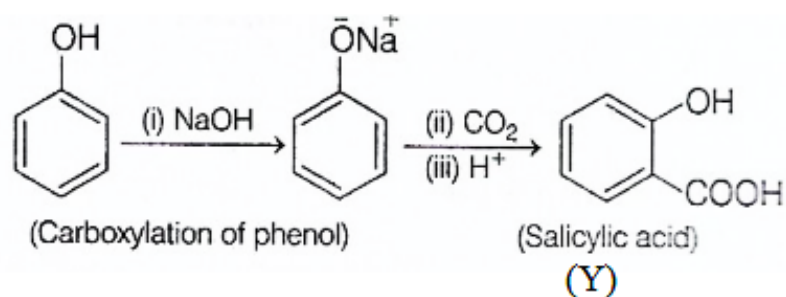
155. The reaction taking place is shown below.



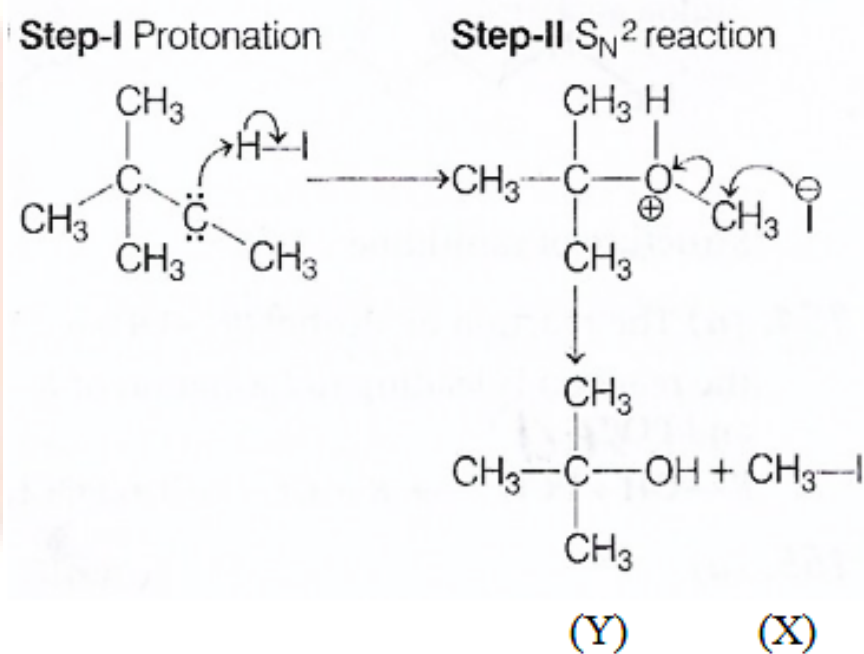
156. When phenol reacts with chloroform in the presence of NaOH, salicylaldehyde (X) is produced as shown below.



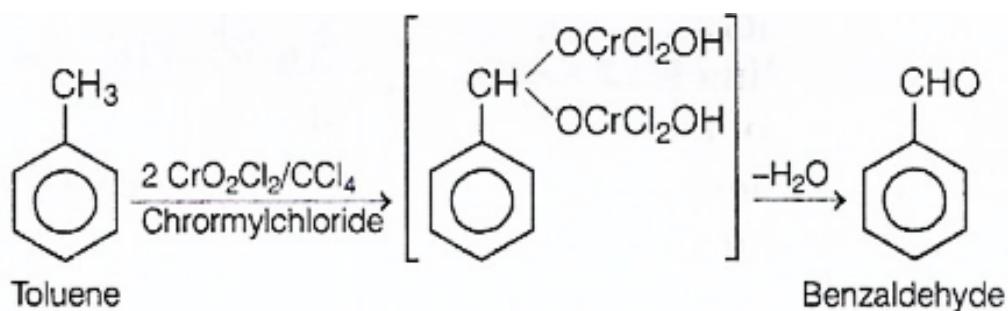
When phenol reacts with carbon dioxide and sodium hydroxide, salicylic acid (Y) is produced as shown below.



157. The products (X) and (Y) formed are shown below.

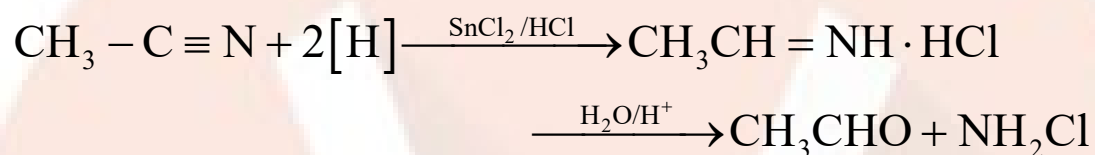


158. The reagent used in Etard reaction is CrO_2Cl_2 . Etard reaction is shown below.



The reagents used in Stephen's reaction is SnCl_2/HCl .

Stephen's reaction is shown below.



159. The products (X) and (Y) formed are as follows:



160. To decide the order of basic amines, inductive effect, solution effect and steric effect are used

On combining all the factors, the correct order is

